Quantum Groupoids from Thin Concurrent Games

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Abstract

Thin concurrent games provide a canonical source of quantum groupoids. Unique negative–positive factorization of configuration symmetries gives a vacant double groupoid whose boxes carry an involutive weak Hopf monoid in spans. We identify its separable Frobenius base with the configuration set and construct the associated Day–Street quantum groupoid, including its boundary coactions and inversion data. Tensor products are preserved, and game negation corresponds to compact transposition. Every distributor between configuration-symmetry groupoids yields an explicit quantum module; positive-witness distributors therefore realize thin strategies as quantum modules. We characterize the positive symmetry transport retained by these modules and show why full distributor data are needed for full-coend composition. Copycat agrees with the regular quantum identity exactly when negative symmetry is discrete. Retaining the generating distributors gives a symmetric monoidal double category and a canonical symmetric monoidal oplax realization of thin concurrent games.

Contents

1 Introduction

Day and Street [9] observed that the role of star-autonomous categories in linear logic suggests further interaction between computer science and quantum group theory. Their categorical formulation of quantum groupoids provides a setting for this connection. We construct a quantum groupoid from the polarized configuration symmetries of a thin concurrent game and extend the construction to strategies through their positive-witness distributors.

Thin concurrent games describe computation using causal dependencies, concurrent events, and selected symmetries between finite configurations [4]. Their negative and positive symmetry subgroupoids determine a unique factorization of every selected symmetry [7, Lemma 5.3]. This factorization is the input to our construction. It gives a unique commuting square with negative horizontal edges and positive vertical edges for each compatible corner. The resulting vacant double groupoid is denoted by $T(A)$.

The algebraic mechanism has two established antecedents. Andruskiewitsch and Natale [2] construct weak Hopf algebras from vacant double groupoids, using vertical composition and horizontal factorization of boxes. Pastro and Street [16] construct quantum groupoids from weak Hopf monoids with invertible antipode. We identify the required box structure in thin games and realize its operations in the category of spans. Spans retain the individual factorization witnesses, so the construction also applies when a game has infinitely many configurations or selected symmetries. Its structural equations are proved by bijections of witness sets.

For every thin concurrent game A, we construct an involutive weak Hopf monoid $\mathsf{WH}_{\mathrm{sp}}(A)$ and a Day–Street quantum groupoid $\mathsf{QG}_{\mathrm{sp}}(A)$. The configuration set $\mathsf{Conf}(A)$, with its canonical separable commutative Frobenius structure, is the object-of-objects. We exhibit both counital splittings, identify the cotensor of arrows as a boundary pullback, and give the boundary coactions, composition, unit, and inversion maps. The inversion map on the composition object is also described directly by factoring and pasting boxes. Tensor products act componentwise on this data, and game negation corresponds to compact transposition. A finite example with nontrivial symmetry of both polarities makes the incidence operations explicit.

The strategy construction starts with an ordinary distributor $F : \mathsf{Sym}(A)^{\mathrm{op}} \times \mathsf{Sym}(B) \to \mathbf{Set}$. Its endpoint-indexed carrier has a diagonal comonad, and positive endpoint transport gives a quantum-module action. Applying this construction to the positive-witness distributor $\|\sigma\|$ assigns a quantum module to each thin strategy σ. We prove the comonad and action equations, functoriality on natural transformations, and compatibility with external products.

Two comparison results describe the information retained by this realization. First, the quantum module depends exactly, at the level of its carrier and fixed-endpoint module maps, on the restriction of the distributor to positive symmetry (Proposition 5.7). Distributors with the same such restriction can have different full-coend composites (Proposition 6.3). Second, the quantum module of copycat is isomorphic to the regular quantum module exactly when the game's negative symmetry groupoid is discrete (Proposition 6.2). These results explain the role of the generating distributor in composition and identity.

Accordingly, our horizontal arrows are pairs (F, Q_F^{sp}), with composition induced by the full distributor coend. Functors between the symmetry groupoids supply representable adjunctions, and distributor squares supply the corresponding quantum-module maps. Adjoining these prescribed quantum components gives a symmetric monoidal double category canonically equivalent to the game-labelled distributor double category. The positive-witness collapse of [7, Theorem 5.15] then yields a symmetric monoidal oplax double functor

$$\mathfrak{Q} : \mathbf{TCG} \longrightarrow \mathbf{QMod}_{\mathrm{sp}}^{\mathrm{rep}}.$$

Its identity and tensor comparisons are invertible. A two-move interaction exhibits a proper composition comparison and identifies the causal compatibility that the full distributor coend does not impose.

Section 2 recalls the game-semantic input, and Section 3 fixes the quantum and span conventions. Section 4 constructs the game quantum groupoid. Section 5 treats distributor-induced modules, and Section 6 assembles the double-categorical realization and its comparison results.

Note on use of AI

Generative AI systems were used during the research and development of this manuscript. Responsibility for the mathematical content, arguments, and conclusions rests with the author.

2 Thin concurrent games and polarized symmetry

Concurrent games combine causal structure with symmetries between finite executions. In a thin concurrent game these symmetries are polarized: negative symmetries encode Opponent reindexing and positive symmetries encode Player reindexing. Their interaction is especially rigid, since every selected symmetry admits a unique negative–positive decomposition. Full definitions, including the strategy axioms, are given in [4, 7]; we use the polarized decomposition in the orientation of [7, Lemma 5.3]. We use the general consistency-family presentation of event structures from [4]. The constructions used below are polarized decomposition, symmetry lifting, and the positive-witness collapse of [7].

2.1 Event structures and configurations

Definition 2.1 (Event structure and configuration [4, Def. 2.1])**.** An **event structure** is a triple

$$E = (|E|, \leq_E, \mathsf{Con}_E)$$

where $|E|$ is a set of events, $\leq_E$ is a partial order with finite principal ideals

$$[e]_E := \{e' \mid e' \leq_E e\},$$

and Con_E is a nonempty family of finite subsets of $|E|$ satisfying:

1. **singleton consistency**: for every event $e \in |E|$,
$$\{e\} \in \mathsf{Con}_E;$$
2. **heredity**: if $X \in \mathsf{Con}_E$ and $Y \subseteq X$, then
$$Y \in \mathsf{Con}_E;$$
3. **causal closure**: if $X \in \mathsf{Con}_E$, $e \in X$, and $e' \leq_E e$, then
$$X \cup \{e'\} \in \mathsf{Con}_E.$$

A finite subset $X \subseteq |E|$ is **consistent** when $X \in \mathsf{Con}_E$. A finite **configuration** is a finite subset $x \subseteq |E|$ which is consistent and down-closed:

$$e \in x,\ e' \leq_E e \quad \Longrightarrow \quad e' \in x.$$

We write $\mathsf{Conf}(E)$ for the set of finite configurations.

The causal order records which events must already have occurred before a given event can occur, while consistency records which finite collections of events may coexist in one execution. A configuration is therefore a finite causal state of execution. Concurrent events are represented by incomparable, jointly consistent events, and their joint configuration records their coexistence directly.

For example, if E consists of two concurrent events e_1, e_2, then

$$\mathsf{Conf}(E) = \{\varnothing, \{e_1\}, \{e_2\}, \{e_1, e_2\}\}.$$

The configuration order is the diamond

$$\begin{tikzcd} & \varnothing \arrow[dl, hook] \arrow[dr, hook] & \\ \{e_1\} \arrow[dr, hook] & & \{e_2\} \arrow[dl, hook] \\ & \{e_1, e_2\} & \end{tikzcd}$$

whose two singleton configurations are incomparable one-event extensions of the empty configuration.

A **polarized event structure** carries a map

$$\mathrm{pol}_E : |E| \longrightarrow \{-, +\},$$

where $-$ denotes an Opponent/Environment move and $+$ denotes a Player/System move [4, Def. 2.3].

2.2 Configuration symmetries

Definition 2.2 (Isomorphism family and event structure with symmetry [4, Def. 3.1])**.** An **isomorphism family** on an event structure E is a groupoid $\mathsf{Sym}(E)$ whose object set is $\mathsf{Conf}(E)$ and whose arrows are selected bijections

$$\theta : x \xrightarrow{\sim} y, \qquad x, y \in \mathsf{Conf}(E),$$

satisfying:

1. **restriction**: if $\theta : x \xrightarrow{\sim} y$ is selected and $x' \in \mathsf{Conf}(E)$ with $x' \subseteq x$, then the restricted bijection
$$\theta|_{x'} : x' \xrightarrow{\sim} \theta(x')$$
is selected;
2. **extension**: if $\theta : x \xrightarrow{\sim} y$ is selected and $x \subseteq x' \in \mathsf{Conf}(E)$, then θ extends to some selected $\theta' : x' \xrightarrow{\sim} y'$.

An **event structure with symmetry** is an event structure equipped with an isomorphism family.

A **map of event structures** $f : E \to F$ is a function on events that sends configurations to configurations and is injective on each configuration. It is a **map of event structures with symmetry** when the pointwise image $f\theta : f(x) \xrightarrow{\sim} f(y)$ of every selected symmetry $\theta : x \xrightarrow{\sim} y$ is selected [4, Defs. 2.6 and 3.6].

2.3 Thin games and polarized decomposition

A thin concurrent game refines the configuration-symmetry groupoid by resolving selected reindexings according to polarity.

Definition 2.3 (Thin concurrent game [4, Def. 3.16]; [7, Def. 5.2])**.** A **thin concurrent game** A is a polarized event structure equipped with three isomorphism families

$$\mathsf{Sym}(A), \qquad \mathsf{Sym}^-(A), \qquad \mathsf{Sym}^+(A),$$

such that

$$\mathsf{Sym}^-(A), \mathsf{Sym}^+(A) \subseteq \mathsf{Sym}(A),$$

every selected symmetry preserves polarity, and the following polarized thinness conditions hold:

(i) **orthogonality**: if $\theta \in \mathsf{Sym}^-(A) \cap \mathsf{Sym}^+(A)$, then $\theta = \mathrm{id}_x$ for some $x \in \mathsf{Conf}(A)$;

(ii) $-$**-receptivity**: if $\theta \in \mathsf{Sym}^-(A)$ and $\theta \subseteq^- \theta' \in \mathsf{Sym}(A)$, then $\theta' \in \mathsf{Sym}^-(A)$;

(iii) $+$**-receptivity**: if $\theta \in \mathsf{Sym}^+(A)$ and $\theta \subseteq^+ \theta' \in \mathsf{Sym}(A)$, then $\theta' \in \mathsf{Sym}^+(A)$.

Here $\theta \subseteq^p \theta'$ means that θ' extends θ and all newly added pairs of events have polarity p.

We use *thin game* as shorthand for *thin concurrent game* throughout.
Since the three symmetry families are groupoids on the same configuration set and

$$\mathsf{Sym}^-(A), \mathsf{Sym}^+(A) \subseteq \mathsf{Sym}(A),$$

the groupoids $\mathsf{Sym}^-(A)$ and $\mathsf{Sym}^+(A)$ are wide subgroupoids of $\mathsf{Sym}(A)$, with common object set $\mathsf{Conf}(A)$. When the underlying arrow sets are needed explicitly, we write $\mathsf{Sym}_1(A)$, $\mathsf{Sym}_1^-(A)$, and $\mathsf{Sym}_1^+(A)$. Writing $\mathsf{Disc}(X)$ for the discrete groupoid on a set X, Condition (i) gives

$$\mathsf{Sym}^-(A) \cap \mathsf{Sym}^+(A) = \mathsf{Disc}(\mathsf{Conf}(A)). \tag{1}$$

The polarization records which side of the interaction carries a reindexing: positive symmetries are carried by Player moves, while negative symmetries are carried by Opponent moves [4].

The polarized subgroupoids jointly determine a unique normal form for every selected symmetry.

Theorem 2.4 (Polarized decomposition [7, Lemma 5.3])**.** *For every thin concurrent game A and every selected symmetry $\theta : x \xrightarrow{\sim} z$ in $\mathsf{Sym}(A)$, there are unique $y \in \mathsf{Conf}(A)$, $h : x \xrightarrow{\sim} y$ in $\mathsf{Sym}^-(A)$, and $v : y \xrightarrow{\sim} z$ in $\mathsf{Sym}^+(A)$ such that*

$$\theta = v \circ h. \tag{2}$$

Equivalently, every selected symmetry fits into a unique factorization triangle

$$\begin{tikzcd} x \arrow[rr, "\theta"] \arrow[dr, "h"'] & & z \\ & y \arrow[ur, "v"'] & \end{tikzcd}$$

with the first leg negative and the second positive.

Proof.

1. **Apply polarized decomposition.** The existence and uniqueness are [7, Lemma 5.3], with $h = \theta^-$ and $v = \theta^+$. The factorization equation gives the displayed triangle. □

Polarized decomposition expresses each selected symmetry as an Opponent reindexing followed by a Player reindexing, with the intermediate configuration uniquely determined.

2.4 Tensor product and duality

The dual $A^\perp$ has the same underlying event structure as A with reversed event polarity, while $A \otimes B$ is the tagged disjoint union of A and B, with causality and consistency inherited componentwise [4, Def. 2.8 and the following paragraph]. The configuration and polarized symmetry data are compatible with these operations as follows.

Proposition 2.5 (Tensor product and duality [4, Defs. 2.8, 3.5 and §§3.2.2–3.3.3])**.** *For thin concurrent games A, B, there are canonical identifications*

$$\mathsf{Conf}(A \otimes B) \cong \mathsf{Conf}(A) \times \mathsf{Conf}(B), \tag{3}$$

and

$$\begin{aligned} \mathsf{Sym}(A^\perp) &= \mathsf{Sym}(A), \\ \mathsf{Sym}^-(A^\perp) &= \mathsf{Sym}^+(A), \\ \mathsf{Sym}^+(A^\perp) &= \mathsf{Sym}^-(A), \end{aligned} \tag{4}$$

$$\begin{aligned} \mathsf{Sym}(A \otimes B) &\cong \mathsf{Sym}(A) \times \mathsf{Sym}(B), \\ \mathsf{Sym}^-(A \otimes B) &\cong \mathsf{Sym}^-(A) \times \mathsf{Sym}^-(B), \\ \mathsf{Sym}^+(A \otimes B) &\cong \mathsf{Sym}^+(A) \times \mathsf{Sym}^+(B). \end{aligned} \tag{5}$$

Proof.

1. **Decompose configurations and symmetries.** Configurations and selected symmetries of $A \otimes B$ decompose componentwise, giving the Cartesian-product identifications above [4, Defs. 2.8 and 3.5]. Tensor product therefore acts componentwise on the polarized symmetry families.

2. **Reverse polarity.** Duality reverses event polarity and exchanges the positive and negative symmetry families [4, §§3.2.2–3.3.3]. This gives (4). □

2.5 Strategies and positive-witness distributors

A strategy adds a causal realization of behaviour to the game structure introduced above. Its events record concrete occurrences of moves, together with the causal dependencies between them, while a display map records which move of the game each occurrence realizes. Different configurations of the strategy may therefore realize the same configuration of the game.

For a configuration $x \in \mathsf{Conf}(E)$ and an event $e \notin x$, write

$$x \vdash_E e$$

when $x \cup \{e\} \in \mathsf{Conf}(E)$. We also write $e_1 \prec_E e_2$ when $e_1 <_E e_2$ is an immediate causal dependency, meaning that no event lies strictly between them [7, §4.1.1].

Definition 2.6 (Thin strategy [7, Defs. 4.2 and 5.4])**.** Let A, B be thin concurrent games. A **thin strategy**

$$\sigma : A \longrightarrow B$$

consists of an event structure with symmetry σ together with a display map

$$\partial_\sigma : \sigma \longrightarrow A^\perp \otimes B$$

which is a map of event structures with symmetry. The polarity of an event $s \in |\sigma|$ is inherited from its image:

$$\mathrm{pol}_\sigma(s) := \mathrm{pol}_{A^\perp \otimes B}(\partial_\sigma(s)).$$

For $\theta : x \xrightarrow{\sim} y$ in $\mathsf{Sym}(\sigma)$, write

$$\partial_\sigma \theta : \partial_\sigma x \xrightarrow{\sim} \partial_\sigma y$$

for its pointwise image under the display map. The following conditions are required.

1. **receptivity**: if $x \in \mathsf{Conf}(\sigma)$ and a negative event a satisfies

$$\partial_\sigma x \vdash_{A^\perp \otimes B} a,$$

then there is a unique event s such that

$$x \vdash_\sigma s, \qquad \partial_\sigma(s) = a;$$

2. **courtesy**: if $s_1 \prec_\sigma s_2$ and either $\partial_\sigma(s_1)$ is positive or $\partial_\sigma(s_2)$ is negative, then

$$\partial_\sigma(s_1) \prec_{A^\perp \otimes B} \partial_\sigma(s_2);$$

3. **symmetry-receptivity**: let $\theta : x \xrightarrow{\sim} y$ be in $\mathsf{Sym}(\sigma)$. If a, b are negative events,

$$(a, b) \notin \partial_\sigma \theta,$$

and

$$\partial_\sigma \theta \cup \{(a, b)\}$$

is a selected symmetry of $A^\perp \otimes B$, then there are unique events s, t with

$$\partial_\sigma(s) = a, \qquad \partial_\sigma(t) = b,$$

such that

$$\theta \cup \{(s, t)\} : x \cup \{s\} \xrightarrow{\sim} y \cup \{t\}$$

belongs to $\mathsf{Sym}(\sigma)$;

4. **thinness**: if $\theta : x \xrightarrow{\sim} y$ belongs to $\mathsf{Sym}(\sigma)$ and

$$\partial_\sigma \theta \in \mathsf{Sym}^+(A^\perp \otimes B),$$

then

$$x = y, \qquad \theta = \mathrm{id}_x\,.$$

Receptivity and courtesy control the causal behaviour of the strategy relative to the game. Symmetry-receptivity performs the corresponding lifting for Opponent-carried reindexings, while thinness makes the remaining Player-carried symmetry rigid inside the strategy. Together these conditions give a canonical lifting property for arbitrary selected symmetries.

Lemma 2.7 (Symmetry lifting up to positive symmetry [7, Lemma 5.5])**.** *Let $\sigma : A \to B$ be a thin strategy, $x^\sigma \in \mathsf{Conf}(\sigma)$, and*

$$\psi : \partial_\sigma x^\sigma \xrightarrow{\sim} y$$

a selected symmetry of $A^\perp \otimes B$. There are unique

$$\varphi : x^\sigma \xrightarrow{\sim} z^\sigma \quad \textit{in } \mathsf{Sym}(\sigma)$$

and

$$\theta^+ : \partial_\sigma z^\sigma \xrightarrow{\sim} y \quad \textit{in } \mathsf{Sym}^+(A^\perp \otimes B)$$

such that

$$\theta^+ \circ \partial_\sigma \varphi = \psi. \tag{6}$$

Proof.

1. **Lift the selected endpoint symmetry.** Apply [7, Lemma 5.5] to the display map $\partial_\sigma : \sigma \to A^\perp \otimes B$. It supplies a strategy symmetry $\varphi : x^\sigma \to z^\sigma$ and a positive residual symmetry $\theta^+ : \partial_\sigma z^\sigma \to y$ satisfying $\theta^+ \circ \partial_\sigma \varphi = \psi$.

2. **Verify uniqueness.** For another such pair $\varphi', \theta^{+\prime}$,

$$\partial_\sigma(\varphi' \circ \varphi^{-1}) = (\theta^{+\prime})^{-1} \circ \theta^+ \in \mathsf{Sym}^+(A^\perp \otimes B).$$

Thinness makes $\varphi' \circ \varphi^{-1}$ an identity, hence $\varphi' = \varphi$ and $\theta^{+\prime} = \theta^+$. $\square$

For $x^\sigma \in \mathsf{Conf}(\sigma)$, the tensor-product identification of configurations gives unique

$$x^\sigma_A \in \mathsf{Conf}(A), \qquad x^\sigma_B \in \mathsf{Conf}(B)$$

such that

$$\partial_\sigma x^\sigma = x^\sigma_A \otimes x^\sigma_B.$$

If $\varphi : x^\sigma \xrightarrow{\sim} y^\sigma$ lies in $\mathsf{Sym}(\sigma)$, write

$$\partial_\sigma \varphi = \varphi^\sigma_A \otimes \varphi^\sigma_B$$

for its components under the tensor-product identification, so that $\varphi^\sigma_A : x^\sigma_A \xrightarrow{\sim} y^\sigma_A$ and $\varphi^\sigma_B : x^\sigma_B \xrightarrow{\sim} y^\sigma_B$.

Definition 2.8 (+-covered configuration [7, §4.1.4])**.** A configuration $x^\sigma \in \mathsf{Conf}(\sigma)$ is **+-covered** when every maximal event of x^σ is positive. Write

$$\mathsf{Conf}^+(\sigma) \subseteq \mathsf{Conf}(\sigma)$$

for the set of +-covered configurations.

Thus the display restricts to an endpoint map

$$\mathsf{Conf}^+(\sigma) \xrightarrow{x^\sigma \mapsto (x^\sigma_A, x^\sigma_B)} \mathsf{Conf}(A) \times \mathsf{Conf}(B).$$

By the tensor and duality identifications above,

$$\mathsf{Sym}^+(A^\perp \otimes B) \cong \mathsf{Sym}^-(A) \times \mathsf{Sym}^+(B).$$

Accordingly, the positive residual symmetry in Lemma 2.7 consists of a negative reindexing at the source and a positive reindexing at the target. This gives the endpoint-normalized configurations used below.

Definition 2.9 (Positive witness [7, §5.2])**.** Let $\sigma : A \to B$ be a thin strategy. A **positive witness** from $a \in \mathsf{Conf}(A)$ to $b \in \mathsf{Conf}(B)$ is a triple

$$(\theta^-_A, x^\sigma, \theta^+_B), \qquad \theta^-_A : a \longrightarrow x^\sigma_A \text{ in } \mathsf{Sym}^-(A), \qquad \theta^+_B : x^\sigma_B \longrightarrow b \text{ in } \mathsf{Sym}^+(B), \tag{7}$$

with $x^\sigma \in \mathsf{Conf}^+(\sigma)$. Write

$$\mathsf{Wit}^+_\sigma(a, b)$$

for the set of positive witnesses from a to b.

The polarized frames make these witnesses stable under arbitrary selected symmetries of their endpoints. The lifting lemma supplies the corresponding renormalization uniquely.

Proposition 2.10 (Endpoint transport [7, Proposition 5.6])**.** *Let*

$$w = (\theta^-_A, x^\sigma, \theta^+_B) \in \mathsf{Wit}^+_\sigma(a, b),$$

and let

$$\Omega_A : a' \longrightarrow a \quad \textit{in } \mathsf{Sym}(A), \qquad \Omega_B : b \longrightarrow b' \quad \textit{in } \mathsf{Sym}(B).$$

There are unique

$$\varphi : x^\sigma \xrightarrow{\sim} y^\sigma \quad \textit{in } \mathsf{Sym}(\sigma),$$

$$\vartheta^-_A : a' \longrightarrow y^\sigma_A \quad \textit{in } \mathsf{Sym}^-(A), \qquad \vartheta^+_B : y^\sigma_B \longrightarrow b' \quad \textit{in } \mathsf{Sym}^+(B),$$

for which the diagrams

$$
\begin{tikzcd}
a \arrow[r, "\theta_A^-"] & x_A^\sigma \arrow[d, "\varphi_A^\sigma"] \\
a' \arrow[u, "\Omega_A"] \arrow[r, "\vartheta_A^-"'] & y_A^\sigma
\end{tikzcd}
\qquad
\begin{tikzcd}
x_B^\sigma \arrow[r, "\theta_B^+"] \arrow[d, "\varphi_B^\sigma"'] & b \arrow[d, "\Omega_B"] \\
y_B^\sigma \arrow[r, "\vartheta_B^+"'] & b'
\end{tikzcd}
$$

commute. Hence endpoint transport is defined by

$$\Omega_{B*}\Omega_A^* w := (\vartheta_A^-, y^\sigma, \vartheta_B^+) \in \mathsf{Wit}_\sigma^+(a', b'). \tag{8}$$

Proof.

1. **Lift the endpoint transport.** Apply Lemma 2.7 to the selected endpoint symmetry determined by $\Omega_A, \Omega_B, \theta_A^-$, and θ_B^+.

2. **Identify the polarized residuals.** The tensor and duality formulas identify the positive residual symmetry with the displayed pair $\vartheta_A^-, \vartheta_B^+$, giving the two commuting diagrams. Uniqueness follows from the lifting lemma. This is the endpoint transport of [7, Proposition 5.6]. □

Proposition 2.11 (Positive-witness distributor [7, Proposition 5.7])**.** *For every thin strategy $\sigma : A \to B$, endpoint transport makes the positive-witness sets into a distributor*

$$\|\sigma\| : \mathsf{Sym}(A)^{\mathrm{op}} \times \mathsf{Sym}(B) \longrightarrow \mathbf{Set}, \qquad (a, b) \longmapsto \mathsf{Wit}_\sigma^+(a, b). \tag{9}$$

Proof.

1. **Verify identity transport.** Transport along identity symmetries returns the original witness by uniqueness in Proposition 2.10.

2. **Verify composition.** Successive endpoint transports and transport along the corresponding composite symmetries satisfy the same commuting diagrams. Uniqueness therefore makes them equal. These are the identity and composition laws for

$$\mathsf{Sym}(A)^{\mathrm{op}} \times \mathsf{Sym}(B) \longrightarrow \mathbf{Set}.$$ □

2.6 The double category of thin concurrent games

Throughout, double categories may be weak in horizontal composition: their horizontal arrows form bicategories.

Thin games, strategies, and their morphisms assemble into a two-dimensional structure. Besides strategies relating games horizontally, maps of games act vertically, while positive morphisms compare strategies compatibly with those maps.

A **map of thin concurrent games**

$$h : A \longrightarrow C$$

is a map of event structures with symmetry that is rigid, meaning that it preserves causal dependency, and that preserves event polarities and the polarized symmetry families, in the sense that [7, §5.3.1]

$$h(\mathsf{Sym}^-(A)) \subseteq \mathsf{Sym}^-(C), \qquad h(\mathsf{Sym}^+(A)) \subseteq \mathsf{Sym}^+(C).$$

For such a map, $h^\perp : A^\perp \to C^\perp$ denotes the same underlying map on events, regarded between the dual games.

Given thin strategies

$$\sigma : A \longrightarrow B, \qquad \tau : C \longrightarrow D$$

and maps of thin games

$$h : A \longrightarrow C, \qquad k : B \longrightarrow D,$$

a **positive morphism of strategies**

$$\begin{tikzcd} A \arrow[r, Rightarrow, "\sigma"] \arrow[d, "h"'] \arrow[dr, phantom, "f"] & B \arrow[d, "k"] \\ C \arrow[r, Rightarrow, "\tau"'] & D \end{tikzcd}$$

is a rigid map of event structures with symmetry

$$f : \sigma \longrightarrow \tau$$

for which the display square commutes up to positive symmetry:

$$\partial_\tau \circ f \simeq^+ (h^\perp \otimes k) \circ \partial_\sigma.$$

Here $\simeq^+$ means that for every $x \in \mathsf{Conf}(\sigma)$, the canonical pointwise bijection

$$\big\{((h^\perp \otimes k)(\partial_\sigma(s)), \partial_\tau(f(s))) \mid s \in x\big\}$$

belongs to $\mathsf{Sym}^+(C^\perp \otimes D)$ [7, Def. 5.8].

Proposition 2.12 (The double category of thin concurrent games [7, Prop. 5.11])**.** *There is a symmetric monoidal double category* **TCG** *whose*

(i) *objects are thin concurrent games;*

(ii) *vertical arrows are maps of thin concurrent games;*

(iii) *horizontal arrows are thin strategies;*

(iv) *squares are positive morphisms of strategies.*

Horizontal composition is interaction followed by hiding, horizontal identities are copycat strategies, and the symmetric monoidal product is induced by tensor product.

Proof.

1. **Describe interaction and hiding.** For composable strategies

$$\sigma : A \longrightarrow B, \qquad \tau : B \longrightarrow C,$$

interaction synchronizes configurations $x^\sigma \in \mathsf{Conf}^+(\sigma)$ and $x^\tau \in \mathsf{Conf}^+(\tau)$ whose displayed B-configurations agree and whose synchronization is causally compatible, meaning that the relation on the synchronization graph induced by the two causal orders is acyclic [7, §4.2.2]. Hiding the synchronized B-events then produces a configuration of the composite

$$\tau \odot \sigma : A \longrightarrow C.$$

More precisely, there is an order-isomorphism

$$\left\{ (x^\sigma, x^\tau) \in \mathsf{Conf}^+(\sigma) \times \mathsf{Conf}^+(\tau) \;\middle|\; \begin{array}{c} x^\sigma_B = x^\tau_B \\ \text{and the pair is causally compatible} \end{array} \right\} \xrightarrow{\cong} \mathsf{Conf}^+(\tau \odot \sigma),$$

and the resulting composite configuration has exterior boundary

$$(x^\sigma_A, x^\tau_C).$$

This is the interaction-and-hiding description of strategy composition [7, Prop. 4.7].

2. **Identify the horizontal identity.** The horizontal identity on A is the copycat strategy

$$\mathsf{cc}_A : A \longrightarrow A.$$

Its two copies of A are linked so that corresponding occurrences are causally related from the negative occurrence to the positive one. Its $+$-covered configurations are canonically indexed by configurations of A:

$$\mathsf{Conf}(A) \xrightarrow{\cong} \mathsf{Conf}^+(\mathsf{cc}_A), \qquad x \longmapsto \mathsf{cc}_A x,$$

with

$$\partial_{\mathsf{cc}_A}(\mathsf{cc}_A x) = x \otimes x.$$

See [7, Prop. 4.6].

3. **Assemble the double-category structure.** For thin strategies, synchronization also determines the selected symmetries of the composite, while synchronization up to symmetry gives horizontal composition of positive morphisms [7, Props. 5.9–5.10]. Together with componentwise tensor product, these constructions satisfy the symmetric monoidal double-category axioms [7, Prop. 5.11]. □

Here **Dist** denotes the double category of groupoids, functors, distributors, and natural transformations [7, Example 3.3].

Theorem 2.13 (Positive-witness collapse [7, Thm. 5.15]). *The assignments*

$$A \longmapsto \mathsf{Sym}(A), \qquad \sigma \longmapsto \|\sigma\|$$

extend to a symmetric monoidal oplax double functor

$$\|-\| : \mathbf{TCG} \longrightarrow \mathbf{Dist}.$$

On vertical arrows it sends

$$h : A \longrightarrow B$$

to the induced functor

$$\mathsf{Sym}(h) : \mathsf{Sym}(A) \longrightarrow \mathsf{Sym}(B),$$

and on positive morphisms of strategies it acts by the corresponding natural transformations of positive-witness distributors.

The horizontal identity comparison is the natural isomorphism

$$\|\mathsf{cc}_A\| \xrightarrow{\cong} \mathsf{Sym}(A)(-,-),$$

while for composable thin strategies

$$\sigma : A \longrightarrow B, \qquad \tau : B \longrightarrow C,$$

the compositor is the natural transformation

$$\gamma_{\sigma,\tau} : \|\tau \odot \sigma\| \longrightarrow \|\tau\| \circ \|\sigma\|. \tag{10}$$

Proof.

1. **Construct the assignments.** The action on objects and horizontal arrows is the positive-witness distributor construction above. Maps of thin games induce functors between their configuration-symmetry groupoids, and positive morphisms induce natural transformations by endpoint transport [7, §§5.4.1–5.4.2].

2. **Use the collapse comparisons.** The copycat unitor and strategy-composition compositor are constructed in [7, Props. 5.12–5.13]. Their coherence and compatibility with tensor give the symmetric monoidal oplax double functor [7, Thm. 5.15]. □

The one-dimensional organization of the same game-semantic operations is the compact closed category **Tcg** of [4, Thm. 3.37], whose tensor product is $\otimes$ and whose duality is $A \mapsto A^{\perp}$. The double-category presentation additionally retains maps of games, positive morphisms of strategies, and the oplax compositor above.

3 Quantum categories and quantum modules

Quantum groupoids have their roots in several problems in mathematical physics and operator algebra in which natural symmetry structures turned out to be more general than ordinary Hopf algebras. In low-dimensional quantum theory and conformal field theory, Mack and Schomerus were led to weak quasi-Hopf symmetries in order to accommodate truncated tensor products of physical sectors [12]. In solvable lattice models, Hayashi introduced face algebras to describe quantum-group symmetry of interaction-round-a-face models whose partition functions depend on boundary conditions, with applications to the Jones index problem [10, 11]. Closely related weak Hopf and finite quantum-groupoid structures arose in the study of finite-index and finite-depth subfactors, where they describe finite-index inclusions, intermediate subalgebras, bimodule categories, and principal-graph data [14, 15]. These developments were organized algebraically in the theory of weak Hopf algebras of Böhm–Nill–Szlachányi [3]: much of the familiar Hopf calculus survives, but the unit and counit laws are weakened so that nontrivial source and target bases can remain. Day–Street subsequently formulated quantum categories and quantum groupoids categorically, with a quantum groupoid intended as a “Hopf algebra with several objects” [9]. Pastro–Street related this formulation directly to weak Hopf monoids in braided monoidal categories [16], and Chikhladze developed the corresponding theory of quantum modules [5]. We develop the span and comodule formulations used in the construction.

3.1 Spans as coefficient-free incidence morphisms

Definition 3.1 (Span category)**.** Let $\mathcal{E}$ be a locally small category with finite limits, in a fixed universe in which the isomorphism classes of spans between fixed endpoints form sets. Define

$$\mathbf{Sp}(\mathcal{E}) := \mathbf{Span}_{\cong}(\mathcal{E})$$

to be the category whose objects are those of $\mathcal{E}$ and whose morphisms $X \to Y$ are isomorphism classes of spans

$$X \xleftarrow{\ell} S \xrightarrow{r} Y.$$

Composition is pullback. Cartesian product makes $\mathbf{Sp}(\mathcal{E})$ symmetric monoidal. For $\mathcal{E} = \mathbf{Set}$, write simply $\mathbf{Sp}$.

Thus if

$$\begin{tikzcd} & S \arrow[dl] \arrow[dr] & \\ X & & Y \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} & T \arrow[dl] \arrow[dr] & \\ Y & & Z \end{tikzcd}$$

are composed, the witness object is displayed by

$$\begin{tikzcd} & & S \times_Y T \arrow[dl] \arrow[dr] \arrow[dd, phantom, "\lrcorner" rotate=-45, very near start] & & \\ & S \arrow[dl] \arrow[dr] & & T \arrow[dl] \arrow[dr] & \\ X & & Y & & Z. \end{tikzcd}$$

3.2 Weak Hopf monoids and incidence

Definition 3.2 (Weak bimonoid [16, Def. 1.1])**.** Let

$$\mathcal{V} = (\mathcal{V}, \otimes, I, c)$$

be a braided monoidal category. A **weak bimonoid** H consists of the following data and compatibility conditions:

1. **monoid structure**: morphisms

$$m : H \otimes H \longrightarrow H, \qquad \eta : I \longrightarrow H,$$

   making (H, m, η) a monoid, so that m is associative and η is its two-sided unit;

2. **comonoid structure**: morphisms

$$\Delta : H \longrightarrow H \otimes H, \qquad \varepsilon : H \longrightarrow I,$$

   making (H, Δ, ε) a comonoid, so that Δ is coassociative and ε is its counit;

3. **weak bimonoid compatibility**: multiplication and comultiplication satisfy the braided multiplicativity law, namely the diagram

$$\begin{tikzcd} & H \otimes H \otimes H \otimes H \arrow[r, "1 \otimes c_{H,H} \otimes 1"] & H \otimes H \otimes H \otimes H \arrow[dr, "m \otimes m"] & \\ H \otimes H \arrow[ur, "\Delta \otimes \Delta"] \arrow[drr, "m"'] & & & H \otimes H \\ & & H \arrow[ur, "\Delta"'] & \end{tikzcd} \tag{11}$$

commutes, while the bimonoid compatibility of the unit and counit with the opposite structure is replaced by the weak unit and weak counit equations of [16, Def. 1.1].

The weak unit and weak counit equations determine canonical source and target counital idempotents

$$\varepsilon_s, \varepsilon_t : H \longrightarrow H.$$

In the symmetric case used below, the counital maps are the composites

$$\varepsilon_t = (\varepsilon m \otimes 1)(1 \otimes c)(\Delta\eta \otimes 1), \tag{12}$$

$$\varepsilon_s = (1 \otimes \varepsilon m)(c \otimes 1)(1 \otimes \Delta\eta). \tag{13}$$

These formulas fix the counital convention independently of the later antipode calculation.

The adjective "weak" refers specifically to the interaction of the unit and counit with the opposite structure: the weakened unit and counit laws allow the resulting object to retain nontrivial source and target bases.

Definition 3.3 (Weak Hopf monoid [16, Def. 2.1])**.** A **weak Hopf monoid** is a weak bimonoid H equipped with a morphism

$$S : H \longrightarrow H,$$

called the **antipode**, satisfying the following commutative diagrams:

$$\begin{tikzcd} & H \arrow[dl, "\varepsilon_t"'] \arrow[d, "\Delta"] \arrow[dr, "\varepsilon_s"] & \\ H & H \otimes H \arrow[l, "m(1\otimes S)"] \arrow[r, "m(S\otimes 1)"'] & H \end{tikzcd} \qquad \begin{tikzcd} H \arrow[r, "\Delta^{(3)}"] \arrow[d, "S"'] & H \otimes H \otimes H \arrow[d, "m^{(3)}(S\otimes 1\otimes S)"] \\ H \arrow[r, "1_H"'] & H. \end{tikzcd} \tag{14}$$

For an ordinary Hopf monoid the counital idempotents collapse to $\eta\varepsilon$. For a weak Hopf monoid, splitting the counital idempotents gives the source and target base objects, as in the Karoubi-completion construction below. These bases provide the object structure of the corresponding quantum category.

3.3 Separable Frobenius bases and boundary comodules

Definition 3.4 (Frobenius, commutative, and separable Frobenius monoids)**.** Following [16, §A.4 and Def. A.5] for the Frobenius and separability conventions, let $(\mathcal{V}, \otimes, I)$ be a monoidal category. A **Frobenius monoid** C consists of a monoid

$$(C, \mu, \eta)$$

and a comonoid

$$(C, \delta, \epsilon)$$

on the same object, satisfying the **Frobenius equations**

$$(\mu \otimes 1_C)(1_C \otimes \delta) = \delta\mu = (1_C \otimes \mu)(\delta \otimes 1_C). \tag{15}$$

Equivalently, suppressing coherence isomorphisms

$$\begin{tikzcd}
 & C \otimes C \arrow[dl, "\delta \otimes 1_C"'] \arrow[d, "\mu"] \arrow[dr, "1_C \otimes \delta"] & \\
C \otimes C \otimes C \arrow[dr, "1_C \otimes \mu"'] & C \arrow[d, "\delta"] & C \otimes C \otimes C \arrow[dl, "\mu \otimes 1_C"] \\
 & C \otimes C, &
\end{tikzcd}$$

commutes.

If $\mathcal{V}$ is braided, a Frobenius monoid is **commutative** when its multiplication and comultiplication satisfy

$$\begin{tikzcd}
C \otimes C \arrow[rr, "c_{C,C}"] \arrow[dr, "\mu"'] & & C \otimes C \arrow[dl, "\mu"] \\
 & C &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & C \arrow[dl, "\delta"'] \arrow[dr, "\delta"] & \\
C \otimes C \arrow[rr, "c_{C,C}"'] & & C \otimes C.
\end{tikzcd}$$

Thus $\mu c_{C,C} = \mu$ and $c_{C,C}\delta = \delta$.

A Frobenius monoid is **separable** when

$$\begin{tikzcd}
 & C \otimes C \arrow[dr, "\mu"] & \\
C \arrow[ur, "\delta"] \arrow[rr, "1_C"'] & & C
\end{tikzcd} \tag{16}$$

commutes, equivalently

$$\mu\delta = 1_C.$$

Proposition 3.5 (Canonical Frobenius structure in spans [8, Examples 3.7 and 4.8])**.** *For every set P, the spans*

$$\mu_P : \begin{tikzcd}
 & P \arrow[dl, "\Delta_P"'] \arrow[dr, "1_P"] & \\
P \times P & & P
\end{tikzcd}
\qquad
\eta_P : \begin{tikzcd}
 & P \arrow[dl] \arrow[dr, "1_P"] & \\
1 & & P
\end{tikzcd}$$

$$\delta_P : \begin{tikzcd}
 & P \arrow[dl, "1_P"'] \arrow[dr, "\Delta_P"] & \\
P & & P \times P
\end{tikzcd}
\qquad
\epsilon_P : \begin{tikzcd}
 & P \arrow[dl, "1_P"'] \arrow[dr] & \\
P & & 1
\end{tikzcd}$$

define a canonical separable commutative Frobenius monoid

$$C_P = (P, \mu_P, \eta_P, \delta_P, \epsilon_P)$$

in **Sp**.

Proof.

1. **Identify the Frobenius spans.** Specializing the constructions of [8, Examples 3.7 and 4.8] to the discrete groupoid on P gives the displayed Frobenius spans.

2. **Verify commutativity and separability.** Commutativity follows from the diagonal. The pullback computing $\mu_P \delta_P$ is canonically P, with both exterior legs the identity, so

$$\mu_P \delta_P = 1_P. \qquad \square$$

Remark 3.6 (Pastro–Street convention [16, §2.1, Def. 2.1 and Prop. 2.3(15)]). We use categorical composition order. Relative to the notation of [16], our counital maps are

$$r^{\mathrm{PS}} = \varepsilon_t, \qquad t^{\mathrm{PS}} = \varepsilon_s. \tag{17}$$

Indeed, [16, Def 2.1] gives

$$m(S \otimes 1)\Delta = t^{\mathrm{PS}}, \qquad m(1 \otimes S)\Delta = r^{\mathrm{PS}}.$$

Moreover $r^{\mathrm{PS}} = Ss^{\mathrm{PS}}$ by [16, Prop. 2.3(15)]; hence, when S is invertible,

$$s^{\mathrm{PS}} = S^{-1}\varepsilon_t.$$

3.4 Quantum categories and quantum groupoids

The bicategorical description of an ordinary category provides a useful model for the quantum notion. A small category with object set A_0 and arrow set A_1 is equivalently a monad on A_0 in the bicategory **Span**(**Set**) [13, Example 2.3]. Here **Span**(**Set**) denotes the bicategory of spans and maps of spans; **Sp** denotes the category of isomorphism classes of spans introduced above. The underlying endomorphism is the span

$$\begin{tikzcd} & A_1 \arrow[dl, "s"'] \arrow[dr, "t"] & \\ A_0 & & A_0, \end{tikzcd}$$

while the multiplication and unit of the monad are maps of spans

$$\begin{tikzcd} & A_1 \times_{A_0} A_1 \arrow[dl, "s\pi_1"'] \arrow[d, "m"] \arrow[dr, "t\pi_2"] & \\ A_0 & A_1 \arrow[l, "s"] \arrow[r, "t"'] & A_0 \end{tikzcd} \qquad \begin{tikzcd} & A_0 \arrow[dl, "1_{A_0}"'] \arrow[d, "u"] \arrow[dr, "1_{A_0}"] & \\ A_0 & A_1 \arrow[l, "s"] \arrow[r, "t"'] & A_0. \end{tikzcd}$$

Thus span composition forms the pullback of composable arrows, and the monad associativity and unit laws are precisely the ordinary category axioms.

In Chikhladze's general comodule setting, where the required coreflexive equalizers are available and preserved by tensoring, the corresponding span-like bicategory is

$$\mathbf{Comod}(\mathcal{V})$$

of comonoids, bicomodules, and comodule maps [6, §3]. A 1-cell

$$M : C \rightsquigarrow D$$

is a C–D-bicomodule. For composable bicomodules

$$M : C \rightsquigarrow D, \qquad N : D \rightsquigarrow E,$$

their composite is the cotensor product

$$M \boxtimes_D N,$$

defined by the equalizer

$$M \boxtimes_D N \xhookrightarrow{\ j\ } M \otimes N \underset{1_M \otimes \lambda_N}{\overset{\rho_M \otimes 1_N}{\rightrightarrows}} M \otimes D \otimes N. \tag{18}$$

Thus cotensor product plays in **Comod**$(\mathcal{V})$ the role played by pullback composition in **Span**(**Set**). For configuration bases, Theorem 5.1 constructs the cotensors explicitly as split absolute equalizers.

For $\mathcal{V} = \mathbf{Set}$, each set has its diagonal comonoid, and a C–D-bicomodule is equivalently a span

$$C \leftarrow M \rightarrow D.$$

Under this identification the equalizer above is the ordinary pullback, so that

$$\mathbf{Comod}(\mathbf{Set}) \simeq \mathbf{Span}(\mathbf{Set}).$$

Correspondingly, quantum categories in **Set** are precisely ordinary small categories [6, Def. 5.2, p. 13].

For a comonoid

$$A_0 = (A_0, \delta_0, \epsilon_0)$$

in the braided monoidal category $\mathcal{V}$, define its **opposite comonoid**

$$A_0^\circ := \big(A_0, c_{A_0,A_0}\delta_0, \epsilon_0\big).$$

Thus A_0° has the same underlying object and counit as A_0, but its comultiplication is reversed by the braiding. In **Comod**$(\mathcal{V})$, the comonoid A_0° is a right bidual of A_0 [6, §3]. The corresponding evaluation and coevaluation are 1-cells

$$e : A_0 \otimes A_0^\circ \rightsquigarrow I, \qquad n : I \rightsquigarrow A_0^\circ \otimes A_0,$$

satisfying the biduality triangle identities.

The resulting **enveloping pseudomonoid**

$$A_0^e := A_0^\circ \otimes A_0$$

has multiplication and unit

$$A_0^e \otimes A_0^e \xrightarrow{p_A^e = 1_{A_0^\circ} \otimes e \otimes 1_{A_0}} A_0^e \xleftarrow{\ j_A^e = n\ } I.$$

Definition 3.7 (Quantum category [9, §12]; [16, §5.2]; [6, Def. 5.2])**.** A **quantum category** (A_0, A_1) in $\mathcal{V}$ consists of a comonoid A_0, called the **object-of-objects**, together with a monoidal comonad

$$A_1 : A_0^e \rightsquigarrow A_0^e$$

on the enveloping pseudomonoid

$$A_0^e = A_0^\circ \otimes A_0$$

in **Comod**$(\mathcal{V})$. We call A_1 the **arrow object**.

Unpacked, the underlying comonad determines an arrow comonoid A_1 with source and target maps

$$\begin{tikzcd} & A_1 \arrow[dl, "s"'] \arrow[dr, "t"] & \\ A_0^\circ & & A_0, \end{tikzcd}$$

while the monoidal structure on the comonad supplies a composition morphism and unit

$$A_1 \boxtimes_{A_0} A_1 \xrightarrow{\ \mu\ } A_1 \xleftarrow{\ u\ } A_0.$$

The quantum-category axioms are the corresponding monoidal-comonad compatibility conditions. In this form the analogy with an ordinary category is direct: the object set A_0 is replaced by the object comonoid A_0, the arrow set A_1 by the arrow comonoid A_1, spans by bicomodules, and pullback of composable arrows by cotensor product.

The weak-bimonoid formulation introduced above is related to this bicategorical description by the following correspondence.

Theorem 3.8 (Weak-bimonoid/quantum-category correspondence [16, Thm. 6.1])**.** *In the setting of [16, §5], where tensoring preserves its coreflexive equalizers, weak bimonoids in the Cauchy (Karoubi) completion* $\mathrm{Kar}(\mathcal{V})$ *correspond to quantum categories whose object-of-objects is a separable Frobenius monoid. An invertible antipode induces the corresponding quantum-groupoid structure.*

The passage to $\mathrm{Kar}(\mathcal{V})$ allows the source and target counital idempotents of a weak bimonoid to split. Recall that an object of $\mathrm{Kar}(\mathcal{V})$ is a pair (X, e) with

$$e : X \longrightarrow X, \qquad e^2 = e,$$

and that a splitting

$$Y \xrightarrow{\ i\ } X \xrightarrow{\ r\ } Y \qquad ri = 1_Y, \qquad ir = e$$

identifies the idempotent with its image Y. In the forward Pastro–Street construction, splitting the target counital idempotent t^{PS} produces the separable Frobenius object-of-objects A_0, while the corresponding idempotent on $A_1 \otimes A_1$ selects the object of composable arrows.

For the inversion data we also need the composition comodules. Write Δ_1, ε_1 for the arrow comonoid structure and put

$$\gamma_l = c^{-1}_{A_0, A_1}(1 \otimes s)\Delta_1, \qquad \gamma_r = (1 \otimes t)\Delta_1.$$

The opposite-comonoid notation is understood in these underlying morphisms of $\mathcal{V}$. Let

$$P = A_1 \boxtimes_{A_0} A_1, \qquad \iota_P : P \longrightarrow A_1 \otimes A_1$$

be the cotensor and its equalizer inclusion. We use "cotensor" for this equalizer; it is the tensor product over the base in the terminology of [16].

The quantum-category structure induces commuting coactions

$$\lambda_P : P \longrightarrow A_1 \otimes A_1 \otimes P, \qquad \rho_P : P \longrightarrow P \otimes A_1,$$

characterized by

$$(1 \otimes 1 \otimes \iota_P)\lambda_P = (1 \otimes c_{A_1,A_1} \otimes 1)(\Delta_1 \otimes \Delta_1)\iota_P, \tag{19}$$
$$(\iota_P \otimes 1)\rho_P = (1 \otimes 1 \otimes \mu)\lambda_P. \tag{20}$$

The equalizer universal property gives the first map; the quantum-category compatibility equations give the factorization in the second. Their combined coaction is

$$\chi_P = (\lambda_P \otimes 1)\rho_P = (1 \otimes 1 \otimes \rho_P)\lambda_P : P \longrightarrow A_1 \otimes A_1 \otimes P \otimes A_1. \tag{21}$$

These are the composition comodules of [16, §§5.1–5.3].

Definition 3.9 (Quantum groupoid [16, §5.3])**.** A **quantum groupoid** is a quantum category $A = (A_0, A_1)$ equipped with comonoid isomorphisms

$$\upsilon : A_0^{\circ\circ} \xrightarrow{\cong} A_0, \qquad \nu : A_1^{\circ} \xrightarrow{\cong} A_1,$$

and an isomorphism $\theta : P_l \xrightarrow{\cong} P_r$ of left $A_1^{\otimes 3}$-comodules. Here P_l and P_r have the common carrier P and the respective coactions

$$\lambda_{P_l} = (1 \otimes 1 \otimes c_{P,A_1})(1 \otimes 1 \otimes 1_P \otimes \nu)\chi_P, \tag{22}$$
$$\lambda_{P_r} = c^{-1}_{A_1 \otimes A_1 \otimes P, A_1}(1 \otimes 1 \otimes 1_P \otimes \nu^{-1})\chi_P. \tag{23}$$

These data satisfy:

(G1) $s\nu = t$;

(G2) $t\nu = \upsilon s$;

(G3)
$$(1_{A_0} \otimes 1_{A_0} \otimes \upsilon)c_{A_0, A_0 \otimes A_0}\, \varsigma = \varsigma\theta,$$

where
$$\varsigma = (s \otimes 1_{A_0} \otimes t)(\gamma_r \otimes 1_{A_1})\iota_P = (s \otimes 1_{A_0} \otimes t)(1_{A_1} \otimes \gamma_l)\iota_P.$$

3.5 Quantum modules and multitensors

Quantum modules play for quantum categories the role that distributors play for ordinary categories [5]. In the module calculus distinguish the formal Chikhladze orientation

$$\widehat{E}_{A,B} := B_0^{\circ} \otimes A_0$$

from its right-mate presentation

$$E_{A,B} := A_0^{\circ} \otimes B_0, \qquad E_A := E_{A,A}.$$

The formal module comonad is on $\widehat{E}_{A,B}$; all explicit boundary, action, and multitensor calculations below use the equivalent right-mate presentation on $E_{A,B}$, made explicit for configuration bases in Proposition 5.2. The canonical action of the enveloping pseudomonoids is

$$\mathsf{act}_{A,B} := 1_{A_0^{\circ}} \otimes e_{A_0} \otimes e_{B_0} \otimes 1_{B_0} : E_A \otimes E_{A,B} \otimes E_B \rightsquigarrow E_{A,B}. \tag{24}$$

Its associativity and unit constraints are induced by the biduality identities.

Definition 3.10 (Quantum module [5, Defs. 3 and 7])**.** Let $A = (A_0, A_1)$ and $B = (B_0, B_1)$ be quantum categories. Formally, a **quantum module** $M : A \nrightarrow B$ has its module comonad on $\widehat{E}_{A,B}$. In the right-mate presentation used below it is represented by a comonad M on $E_{A,B}$ and a 2-cell

$$\kappa_M : \mathsf{act}_{A,B}(A_1 \otimes M \otimes B_1) \Longrightarrow M\mathsf{act}_{A,B}. \tag{25}$$

Put $D = A_1 \otimes M \otimes B_1$. The pair $(\mathsf{act}_{A,B}, \kappa_M)$ is required to be a map of comonads, meaning

$$(\epsilon_M \mathsf{act}_{A,B})\kappa_M = \mathsf{act}_{A,B}\epsilon_D, \tag{26}$$

$$(\delta_M \mathsf{act}_{A,B})\kappa_M = (M\kappa_M)(\kappa_M D)(\mathsf{act}_{A,B}\delta_D). \tag{27}$$

Write p^e_A, j^e_A and p^e_B, j^e_B for the multiplications and units of the two enveloping pseudomonoids, and write

$$\phi_A : p^e_A(A_1 \otimes A_1) \Rightarrow A_1 p^e_A, \quad \phi^0_A : j^e_A \Rightarrow A_1 j^e_A,$$
$$\phi_B : p^e_B(B_1 \otimes B_1) \Rightarrow B_1 p^e_B, \quad \phi^0_B : j^e_B \Rightarrow B_1 j^e_B$$

for the monoidal structure cells. Put $\mathsf{act} = \mathsf{act}_{A,B}$ and

$$\pi = p^e_A \otimes 1_{E_{A,B}} \otimes p^e_B, \qquad \lambda = 1_{E_A} \otimes \mathsf{act} \otimes 1_{E_B}, \qquad \jmath = j^e_A \otimes 1_{E_{A,B}} \otimes j^e_B.$$

The canonical action constraints identify $\mathsf{act}\pi \cong \mathsf{act}\lambda$ and $\mathsf{act}\jmath \cong 1_{E_{A,B}}$. Under these identifications the action equations are

$$(\kappa_M \pi)\mathsf{act}(\phi_A \otimes 1_M \otimes \phi_B) = (\kappa_M \lambda)\mathsf{act}(1_{A_1} \otimes \kappa_M \otimes 1_{B_1}), \tag{28}$$

$$(\kappa_M \jmath)\mathsf{act}(\phi^0_A \otimes 1_M \otimes \phi^0_B) = 1_M. \tag{29}$$

The first equation compares multiplication followed by action with two successive actions; the second identifies the action of the units.

A map of quantum modules is a comonad transformation intertwining the actions.

The direct action has a Kan-extension presentation whose inputs are endocomodules with source and target boundary maps.

Definition 3.11 (Multitensors [5, §3, Eqs. (4)–(5)])**.** Let $C_0, \dots, C_n$ be comonoids and let

$$X_i : C^\circ_{i-1} \otimes C_i \rightsquigarrow C^\circ_{i-1} \otimes C_i \qquad (1 \le i \le n)$$

be endocomodules. For $n \ge 2$, let q_n contract the consecutive middle boundaries by the evaluations e_{C_i}. When the indicated left Kan extension exists, set

$$T_n(X_1, \dots, X_n) := \mathrm{Lan}_{q_n}\big(q_n(X_1 \otimes \cdots \otimes X_n)\big).$$

Also set $T_1(X) = X$ and $T_0(C) = \mathrm{Lan}_{n_C}\, n_C$, where $n_C : I \rightsquigarrow C^\circ \otimes C$ is coevaluation.

For comonad inputs, the Kan-extension universal property induces a comonad on T_n. Under this correspondence, the direct action (25) is equivalently a map of comonads

$$\alpha_M : T_3(A_1, M, B_1) \longrightarrow M \tag{30}$$

satisfying the induced action equations. Section 5 constructs the direct action and verifies these equations by identifying the corresponding spans.

Proposition 3.12 (Span multitensors)**.** *Let X_i be an endocomodule on $C_{P_{i-1}\times P_i}$, with source and target boundary maps*

$$d_i^\diamond(x_i) = (a_i^\diamond(x_i), b_i^\diamond(x_i)), \qquad \diamond \in \{L, R\}.$$

For $n \geq 2$, the carrier of $T_n(X_1, \dots, X_n)$ is canonically

$$K_n = \left\{ (x_1, \dots, x_n) \;\middle|\; \begin{array}{l} b_i^L(x_i) = a_{i+1}^L(x_{i+1}), \\ b_i^R(x_i) = a_{i+1}^R(x_{i+1}) \end{array} \text{ for } 1 \leq i < n \right\}, \tag{31}$$

with boundary maps

$$d^\diamond(x_1, \dots, x_n) = (a_1^\diamond(x_1), b_n^\diamond(x_n)).$$

The carrier of $T_0(C_P)$ is $P \times P$, with source (p, p) and target (q, q) at (p, q). Its comultiplication has witnesses

$$(p, q) \longmapsto ((p, z), (z, q)) \quad (z \in P), \tag{32}$$

and its counit is supported on $p = q$.

Proof.

1. **Express coactions by boundary maps.** For a coaction over a canonical diagonal comonoid, the counit equation identifies the coaction apex with its carrier. Thus coactions are graphs of boundary functions, and comodule maps are spans preserving these functions.

2. **Construct the Kan extension.** The composite $q_n(X_1 \otimes \cdots \otimes X_n)$ imposes the target-boundary equations. A 2-cell from this composite to Zq_n has support where the source labels also satisfy the source-boundary equations. On that locus, the source labels uniquely determine the contraction witness in Zq_n. Forgetting or inserting this witness gives inverse operations

$$\begin{gathered} \{\text{boundary spans } q_n(X_1 \otimes \cdots \otimes X_n) \Rightarrow Zq_n\} \\ \cong \{\text{boundary spans } K_n \Rightarrow Z\}, \end{gathered}$$

   natural in Z. This is the left Kan-extension universal property.

3. **Construct the nullary carrier.** A 2-cell $n_P \Rightarrow Zn_P$ is supported at elements of Z with source (p, p) and target (q, q). Inserting or forgetting those two labels identifies it with a boundary-preserving span from the displayed P^2-carrier to Z.

4. **Compute the nullary comonad.** The induced comultiplication splits (p, q) at an intermediate label z, giving (32). Its counit selects $p = q$. The two iterated comultiplications have witnesses (p, z_1, z_2, q), and either counit removes the corresponding repeated endpoint. This is the matrix coalgebra in spans. $\square$

For quantum modules $M : A \nrightarrow B$ and $N : B \nrightarrow C$, Chikhladze's intrinsic composition involves the balancing pair

$$T_3(M, B_1, N) \underset{d_l}{\overset{d_r}{\rightrightarrows}} T_2(M, N) \longrightarrow N \bullet_B M. \tag{33}$$

Chikhladze's Theorem 11 constructs this composition on classes of modules closed under the indicated coequalizers and compatible multitensors [5]. The generated composition of Section 6 is defined on pairs consisting of a distributor and its associated quantum module.

Proposition 3.13 (Classical specialization [5, §4])**.** *For $\mathcal{V} = \mathbf{Set}$, quantum categories are ordinary small categories, quantum modules are distributors, and* (33) *is the ordinary coend composition*

$$(N \circ M)(a, c) = \int^{b \in B} M(a, b) \times N(b, c).$$

Proof.

1. **Identify the module data.** In the cartesian setting the comonad and action data reduce to a boundary-indexed set and its commuting endpoint actions. These are the data of a distributor.

2. **Identify composition.** The balancing pair identifies the two ways of applying a middle arrow. This is exactly the defining relation of the displayed coend. □

4 Quantum symmetries of thin concurrent games

We now begin the main construction. Let A be a thin concurrent game. We use the polarized decomposition of its configuration symmetries to extract the structure from which the quantum groupoid associated to A will be built.

The construction proceeds as

$$A \longmapsto T(A) \longmapsto \mathsf{WH}_{\mathrm{sp}}(A) \longmapsto \mathsf{QG}_{\mathrm{sp}}(A),$$

from polarized decomposition to vacant boxes, then incidence weak Hopf monoid, and finally the Pastro–Street quantum groupoid.

4.1 Polarized boxes of a thin game

A **polarized box** of A is a commuting square of selected symmetries

$$\begin{tikzcd} p \arrow[r, "h_t"] \arrow[d, "v_l"'] & q \arrow[d, "v_r"] \\ r \arrow[r, "h_b"'] & s, \end{tikzcd} \qquad h_b \circ v_l = v_r \circ h_t, \tag{34}$$

where

$$h_t, h_b \in \mathsf{Sym}^-(A), \qquad v_l, v_r \in \mathsf{Sym}^+(A).$$

For a box X displayed as in (34), set

$$\mathrm{nw}(X) = p, \quad \mathrm{ne}(X) = q, \quad \mathrm{sw}(X) = r, \quad \mathrm{se}(X) = s.$$

We keep these corner names separate from the boundary functions ℓ_X, r_X of an arbitrary bicomodule. Horizontal and vertical pasting are the evident pastings of commuting squares.

Definition 4.1 (Double and vacant double groupoids [2, §1.4, Def. 2.1 and Prop. 2.2])**.** A **double groupoid** consists of a set P of objects, horizontal and vertical groupoids

$$H \rightrightarrows P, \qquad V \rightrightarrows P,$$

and a set B of boxes

$$\begin{tikzcd} p \arrow[r, "h_t"] \arrow[d, "v_l"'] & q \arrow[d, "v_r"] \\ r \arrow[r, "h_b"'] & s, \end{tikzcd}$$

such that the boxes form groupoids under both horizontal and vertical composition, with objects V and H, respectively, and the two compositions satisfy the interchange law.

A double groupoid is **vacant** when every compatible pair of adjacent horizontal and vertical edges determines a unique box. Equivalently, every compatible southwest corner

$$\begin{tikzcd} p \arrow[r, dashed] \arrow[d, "v"'] & q \arrow[d, dashed] \\ r \arrow[r, "h"'] & s \end{tikzcd}$$

with $v \in V$ and $h \in H$ has a unique filler.

Proposition 4.2 (Polarized box double groupoid)**.** *The polarized boxes of A form a vacant double groupoid, denoted*

$$T(A). \tag{35}$$

Its horizontal edge groupoid is $\mathsf{Sym}^-(A)$, its vertical edge groupoid is $\mathsf{Sym}^+(A)$, and its object set is $\mathsf{Conf}(A)$.

Proof.

1. **Construct the two box compositions.** Pasting polarized boxes preserves the commuting-square equation. The horizontal edges compose in $\mathsf{Sym}^-(A)$ and the vertical edges in $\mathsf{Sym}^+(A)$, so the pasted squares remain polarized.

2. **Identify identities and inverses.** Horizontal and vertical identities are the evident identity squares. Horizontal and vertical inverses are obtained by reversing the corresponding direction, and remain polarized because $\mathsf{Sym}^-(A)$ and $\mathsf{Sym}^+(A)$ are groupoids.

3. **Verify the double-groupoid laws.** Associativity and the identity and inverse laws in each direction are inherited from composition in $\mathsf{Sym}(A)$. The interchange law follows from the equality of the two ways of composing the boundary arrows of a pasted 2×2 array of commuting squares. Hence the polarized boxes form a double groupoid.

4. **Form the symmetry around a southwest corner.** Let

$$v : p \longrightarrow r \quad \text{in } \mathsf{Sym}^+(A), \qquad h : r \longrightarrow s \quad \text{in } \mathsf{Sym}^-(A)$$

   be a compatible southwest corner. Their composite

$$h \circ v : p \longrightarrow s$$

   is a selected symmetry of A.

5. **Apply polarized decomposition.** By Theorem 2.4, there are unique $q \in \mathsf{Conf}(A)$,

$$h' : p \longrightarrow q \quad \text{in } \mathsf{Sym}^-(A), \qquad v' : q \longrightarrow s \quad \text{in } \mathsf{Sym}^+(A),$$

   such that

$$h \circ v = v' \circ h'.$$

6. **Identify the unique box filler.** The preceding factorization is precisely the commuting square

$$\begin{tikzcd} p \arrow[r, "h'"] \arrow[d, "v"'] & q \arrow[d, "v'"] \\ r \arrow[r, "h"'] & s. \end{tikzcd}$$

   Thus every compatible southwest corner has a unique polarized box filler. Therefore the double groupoid $T(A)$ is vacant. ☐

Lemma 4.3 (Top-right determination)**.** *Let*

$$h : p \longrightarrow q \quad \textit{in } \mathsf{Sym}^-(A), \qquad v : q \longrightarrow s \quad \textit{in } \mathsf{Sym}^+(A).$$

There is a unique polarized box whose top horizontal edge is h and whose right vertical edge is v.

Proof.

1. **Construct the filler.** Apply polarized decomposition to $(v \circ h)^{-1}$, say

$$(v \circ h)^{-1} = u \circ k$$

   with $k : s \to r$ negative and $u : r \to p$ positive. Inverting gives $v \circ h = k^{-1} \circ u^{-1}$, hence the required filler.

2. **Verify uniqueness.** Any other filler gives, after inversion, a negative–positive decomposition of $(v \circ h)^{-1}$. Uniqueness follows from Theorem 2.4. ☐

Write Boxes_A for the set of polarized boxes. For compatible $h \in \mathsf{Sym}^-(A)$ and $v \in \mathsf{Sym}^+(A)$, write

$$h \urcorner v$$

for the unique box of Lemma 4.3. Its left and bottom edges are denoted

$$\mathsf{L}(h, v), \qquad \mathsf{B}(h, v),$$

respectively.

For a polarized box X, write X^{-1} for its total inverse, obtained by inverting in both double-groupoid directions; interchange makes the two orders equal.

Lemma 4.4 (Polarized box calculus)**.** *The following identities and factorization properties hold.*

1. ***Horizontal pasting.***

$$(h_2 \urcorner v) \circ_{\mathrm{h}} (h_1 \urcorner \mathsf{L}(h_2, v)) = (h_2 \circ h_1) \urcorner v. \tag{36}$$

2. ***Vertical pasting.***

$$\big(\mathsf{B}(h,v_1) \urcorner v_2\big) \circ_{\mathrm{v}} (h \urcorner v_1) = h \urcorner (v_2 \circ v_1). \tag{37}$$

3. ***Vertical identities.***

$$\mathrm{id}^{\mathrm{v}}_h = h \urcorner \mathrm{id}_{t(h)} . \tag{38}$$

4. ***Horizontal identities.***

$$\mathrm{id}^{\mathrm{h}}_v = \mathrm{id}_{s(v)} \urcorner v. \tag{39}$$

5. ***Total inversion.***

$$\big(h \urcorner v\big)^{-1} = \big(\mathsf{B}(h,v)\big)^{-1} \urcorner \big(\mathsf{L}(h,v)\big)^{-1}. \tag{40}$$

6. ***Horizontal factorizations.*** *A horizontal factorization of $h \urcorner v$ is uniquely determined by a factorization*

$$h = h_2 \circ h_1$$

in $\mathsf{Sym}^{-}(A)$, and its two factors are

$$h_1 \urcorner \mathsf{L}(h_2, v), \qquad h_2 \urcorner v. \tag{41}$$

7. ***Vertical factorizations.*** *A vertical factorization of $h \urcorner v$ is uniquely determined by a factorization*

$$v = v_2 \circ v_1$$

in $\mathsf{Sym}^{+}(A)$, and its two factors are

$$h \urcorner v_1, \qquad \mathsf{B}(h, v_1) \urcorner v_2. \tag{42}$$

8. ***Factors of identity boxes.*** *Every horizontal factor of a vertical identity is a vertical identity, and every vertical factor of a horizontal identity is a horizontal identity.*

Proof.

1. **Paste horizontally.** The two boxes

$$h_1 \urcorner \mathsf{L}(h_2, v), \qquad h_2 \urcorner v$$

are horizontally composable by definition of $\mathsf{L}(h_2, v)$. Their paste has top edge $h_2 \circ h_1$ and right edge v. By Lemma 4.3, it is therefore

$$(h_2 \circ h_1) \urcorner v.$$

This proves (36).

2. **Paste vertically.** The boxes

$$h \urcorner v_1, \qquad \mathsf{B}(h, v_1) \urcorner v_2$$

are vertically composable by definition of $\mathsf{B}(h, v_1)$. Their paste has top edge h and right edge $v_2 \circ v_1$. Top-right determination therefore gives (37).

3. **Identify the two kinds of identity box.** The vertical identity on h has top edge h and right edge $\mathrm{id}_{t(h)}$, so top-right determination gives

$$\mathrm{id}^{\mathrm{v}}_{h} = h \urcorner \mathrm{id}_{t(h)} .$$

Likewise, the horizontal identity on v has top edge $\mathrm{id}_{s(v)}$ and right edge v, hence

$$\mathrm{id}^{\mathrm{h}}_{v} = \mathrm{id}_{s(v)} \urcorner v.$$

4. **Compute total inversion.** Total inversion exchanges the top edge with the inverse of the bottom edge and the right edge with the inverse of the left edge. Thus the top and right edges of $(h \urcorner v)^{-1}$ are

$$\big(\mathsf{B}(h,v)\big)^{-1}, \qquad \big(\mathsf{L}(h,v)\big)^{-1}.$$

Top-right determination gives (40).

5. **Classify horizontal factorizations.** Suppose

$$h \urcorner v = X_2 \circ_{\mathrm{h}} X_1.$$

If h_1, h_2 are the top edges of X_1, X_2, respectively, then

$$h = h_2 \circ h_1.$$

Since X_2 has top edge h_2 and right edge v, top-right determination forces

$$X_2 = h_2 \urcorner v.$$

Its left edge is therefore $\mathsf{L}(h_2, v)$, which is the right edge of X_1. Hence

$$X_1 = h_1 \urcorner \mathsf{L}(h_2, v).$$

Conversely, Step 1 shows that these two boxes paste to $h \urcorner v$. This proves the horizontal factorization statement and its uniqueness.

6. **Classify vertical factorizations.** Suppose

$$h \urcorner v = X_2 \circ_{\mathrm{v}} X_1.$$

If v_1, v_2 are the right edges of X_1, X_2, respectively, then

$$v = v_2 \circ v_1.$$

The upper factor has top edge h and right edge v_1, so

$$X_1 = h \urcorner v_1.$$

Its bottom edge is $\mathsf{B}(h, v_1)$, which is the top edge of X_2. Hence

$$X_2 = \mathsf{B}(h, v_1) \urcorner v_2.$$

Conversely, Step 2 gives the required paste. This proves the vertical factorization statement and its uniqueness.

7. **Classify factors of identity boxes.** Let

$$\mathrm{id}^{\mathrm{v}}_{h} = h \urcorner \mathrm{id}_{t(h)}$$

be a vertical identity, and suppose

$$h = h_2 \circ h_1.$$

By the horizontal factorization classification, its right horizontal factor is

$$h_2 \urcorner \mathrm{id}_{t(h)}\,.$$

Since $t(h_2) = t(h)$, the identity formula of Step 3 gives

$$h_2 \urcorner \mathrm{id}_{t(h)} = h_2 \urcorner \mathrm{id}_{t(h_2)} = \mathrm{id}^{\mathrm{v}}_{h_2}\,.$$

The left edge of this box is therefore

$$\mathrm{id}_{s(h_2)} = \mathrm{id}_{t(h_1)}\,.$$

Hence the left horizontal factor is

$$h_1 \urcorner \mathrm{id}_{t(h_1)} = \mathrm{id}^{\mathrm{v}}_{h_1}\,.$$

Thus every horizontal factor of a vertical identity is a vertical identity.

Dually, let

$$\mathrm{id}^{\mathrm{h}}_{v} = \mathrm{id}_{s(v)} \urcorner v$$

be a horizontal identity and suppose

$$v = v_2 \circ v_1.$$

The vertical factorization classification gives the upper factor

$$\mathrm{id}_{s(v)} \urcorner v_1.$$

Since $s(v_1) = s(v)$, this is

$$\mathrm{id}_{s(v_1)} \urcorner v_1 = \mathrm{id}^{\mathrm{h}}_{v_1}\,.$$

Its bottom edge is $\mathrm{id}_{t(v_1)} = \mathrm{id}_{s(v_2)}$, so the lower factor is

$$\mathrm{id}_{s(v_2)} \urcorner v_2 = \mathrm{id}^{\mathrm{h}}_{v_2}\,.$$

Thus every vertical factor of a horizontal identity is a horizontal identity. □

Proposition 4.5 (Tensor product of polarized boxes)**.** *For thin concurrent games A, B,*

$$T(A \otimes B) \cong T(A) \times T(B).$$

Proof.

1. **Factor the box data.** Equations (3) and (5) make polarized decomposition and every box operation componentwise. The resulting bijection preserves both groupoid structures. □

Proposition 4.6 (Negation as transposition)**.** *For every thin concurrent game A,*

$$T(A^{\perp}) \cong T(A)^{\mathsf{t}}, \tag{43}$$

where transposition exchanges the horizontal and vertical edge groupoids and the two box compositions.

Proof.

1. **Transpose the box data.** Equation (4) exchanges $\mathsf{Sym}^{-}(A)$ and $\mathsf{Sym}^{+}(A)$. Transposition preserves the commuting-square equation and exchanges the two box compositions. □

4.2 Box incidence and the weak Hopf monoid

The two box directions now give transverse incidence operations. Put

$$M_{\mathrm{v}}(A) = \{(X,Y) \mid Y \circ_{\mathrm{v}} X \text{ is defined}\}, \qquad M_{\mathrm{h}}(A) = \{(L,R) \mid R \circ_{\mathrm{h}} L \text{ is defined}\}.$$

Define spans in **Sp**

$$\begin{aligned}
m_A:&\quad \mathsf{Boxes}_A^2 \xleftarrow{(X,Y)\mapsto(X,Y)} M_{\mathrm{v}}(A) \xrightarrow{(X,Y)\mapsto Y\circ_{\mathrm{v}} X} \mathsf{Boxes}_A,\\
\eta_A:&\quad 1 \longleftarrow \mathsf{Sym}_1^-(A) \xrightarrow{h\mapsto \mathrm{id}_h^{\mathrm{v}}} \mathsf{Boxes}_A,\\
\Delta_A:&\quad \mathsf{Boxes}_A \xleftarrow{(L,R)\mapsto R\circ_{\mathrm{h}} L} M_{\mathrm{h}}(A) \xrightarrow{(L,R)\mapsto(L,R)} \mathsf{Boxes}_A^2,\\
\varepsilon_A:&\quad \mathsf{Boxes}_A \xleftarrow{v\mapsto \mathrm{id}_v^{\mathrm{h}}} \mathsf{Sym}_1^+(A) \longrightarrow 1,\\
S_A:&\quad \mathsf{Boxes}_A \xrightarrow{X\mapsto X^{-1}} \mathsf{Boxes}_A.
\end{aligned} \tag{44}$$

where the last map is regarded as its graph span.

For reference, in the symmetric category **Sp** the weak unit and counit equations used below are

$$\begin{aligned}
(\Delta_A\otimes 1)\Delta_A\eta_A &= (1\otimes m_A\otimes 1)(\Delta_A\eta_A\otimes\Delta_A\eta_A)\\
&= (1\otimes m_A\otimes 1)(1\otimes c\otimes 1)(\Delta_A\eta_A\otimes\Delta_A\eta_A),
\end{aligned} \tag{45}$$

$$\begin{aligned}
\varepsilon_A m_A(m_A\otimes 1) &= (\varepsilon_A m_A\otimes\varepsilon_A m_A)(1\otimes\Delta_A\otimes 1)\\
&= (\varepsilon_A m_A\otimes\varepsilon_A m_A)(1\otimes c\Delta_A\otimes 1).
\end{aligned} \tag{46}$$

Lemma 4.7 (Polarized rectangular decomposition)**.** *Let $X, Y \in \mathsf{Boxes}_A$ be vertically composable, and suppose their vertical paste admits a horizontal factorization*

$$Y \circ_{\mathrm{v}} X = R \circ_{\mathrm{h}} L.$$

Then there is a unique 2×2 grid of polarized boxes

$$\begin{matrix} X_1 & X_2 \\ Y_1 & Y_2 \end{matrix}$$

such that

$$X = X_2 \circ_{\mathrm{h}} X_1, \qquad Y = Y_2 \circ_{\mathrm{h}} Y_1,$$

the two columns are vertically composable, and

$$L = Y_1 \circ_{\mathrm{v}} X_1, \qquad R = Y_2 \circ_{\mathrm{v}} X_2.$$

Proof.

1. **Fix the common paste.** Put

$$P := Y \circ_{\mathrm{v}} X = R \circ_{\mathrm{h}} L.$$

2. **Factor the right column.** Since
$$P = Y \circ_{\mathrm{v}} X,$$
the right edge of P factors as the right edge of X followed by the right edge of Y. Since
$$P = R \circ_{\mathrm{h}} L,$$
the right edge of P is the right edge of R. By the vertical factorization statement of Lemma 4.4, there are therefore unique vertically composable boxes X_2, Y_2 such that
$$R = Y_2 \circ_{\mathrm{v}} X_2,$$
with the right edges of X_2 and Y_2 equal respectively to the right edges of X and Y.

3. **Factor the left column.** Since L and R are horizontally composable, the right edge of L is the left edge of R. From
$$R = Y_2 \circ_{\mathrm{v}} X_2,$$
that left edge factors as the left edge of X_2 followed by the left edge of Y_2. Applying the vertical factorization statement again gives unique vertically composable boxes X_1, Y_1 such that
$$L = Y_1 \circ_{\mathrm{v}} X_1,$$
where the right edge of X_1 is the left edge of X_2, and the right edge of Y_1 is the left edge of Y_2. Hence the two rows
$$X_1,\ X_2 \qquad \text{and} \qquad Y_1,\ Y_2$$
are horizontally composable.

4. **Identify the upper row with X.** The horizontal paste
$$X_2 \circ_{\mathrm{h}} X_1$$
has right edge equal to the right edge of X. Its top edge is the horizontal composite of the top edges of L and R, hence is the top edge of
$$R \circ_{\mathrm{h}} L = P.$$
Since $P = Y \circ_{\mathrm{v}} X$, this is the top edge of X. Thus $X_2 \circ_{\mathrm{h}} X_1$ and X have the same top and right edges. By Lemma 4.3,
$$X = X_2 \circ_{\mathrm{h}} X_1.$$

5. **Identify the lower row with Y.** By interchange,
$$\begin{aligned} P &= R \circ_{\mathrm{h}} L \\ &= \big(Y_2 \circ_{\mathrm{v}} X_2\big) \circ_{\mathrm{h}} \big(Y_1 \circ_{\mathrm{v}} X_1\big) \\ &= \big(Y_2 \circ_{\mathrm{h}} Y_1\big) \circ_{\mathrm{v}} \big(X_2 \circ_{\mathrm{h}} X_1\big) \\ &= \big(Y_2 \circ_{\mathrm{h}} Y_1\big) \circ_{\mathrm{v}} X. \end{aligned}$$
But also
$$P = Y \circ_{\mathrm{v}} X.$$
Since the boxes form a groupoid under vertical composition, cancellation of the common upper factor X gives
$$Y = Y_2 \circ_{\mathrm{h}} Y_1.$$
By construction,
$$L = Y_1 \circ_{\mathrm{v}} X_1, \qquad R = Y_2 \circ_{\mathrm{v}} X_2.$$

6. **Prove uniqueness.** The right-edge factorization of R determined by the given vertical factorization
$$P = Y \circ_{\mathrm{v}} X$$
uniquely determines X_2, Y_2 by the vertical factorization classification. Their left edges then uniquely determine the vertical factorization of L, hence uniquely determine X_1, Y_1. Therefore the entire 2×2 grid is unique. □

Lemma 4.8 (Incidence weak bimonoid)**.** *The spans*
$$m_A, \eta_A, \Delta_A, \varepsilon_A$$
make Boxes_A *a weak bimonoid in* **Sp**.

Its canonical target and source counital maps are represented respectively by the spans
$$\mathsf{Boxes}_A \xleftarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{h}}_v} \big\{(v,h) \in \mathsf{Sym}^{+}_1(A) \times \mathsf{Sym}^{-}_1(A) \mid s(v) = s(h)\big\} \xrightarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{v}}_h} \mathsf{Boxes}_A \tag{47}$$
and
$$\mathsf{Boxes}_A \xleftarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{h}}_v} \big\{(v,h) \in \mathsf{Sym}^{+}_1(A) \times \mathsf{Sym}^{-}_1(A) \mid t(v) = t(h)\big\} \xrightarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{v}}_h} \mathsf{Boxes}_A. \tag{48}$$

Proof.

1. **Verify the monoid laws.** The multiplication span records vertical composition of boxes, and η_A records vertical identity boxes. The apex of either parenthesization of the triple product is the set of vertically composable triples
$$(X_1, X_2, X_3),$$
and both composites send such a triple to its total vertical paste. Associativity follows from associativity in the vertical box groupoid.

   For the unit laws, pulling back against η_A inserts the appropriate vertical identity box. The two resulting composites are therefore the identity span on Boxes_A. Thus (m_A, η_A) is a monoid.

2. **Verify the comonoid laws.** Dually, Δ_A records horizontal factorizations and ε_A records horizontal identity boxes. Both iterated coproducts have as apex the set of horizontally composable triples
$$(X_1, X_2, X_3)$$
together with their common total horizontal paste. Associativity in the horizontal box groupoid identifies the two apexes canonically, with the same exterior legs.

   Pullback against ε_A forces the corresponding factor to be a horizontal identity, and the horizontal identity laws then identify either counit composite with the identity span on Boxes_A. Hence $(\Delta_A, \varepsilon_A)$ is a comonoid.

3. **Prove multiplicativity.** A witness for
$$\Delta_A m_A$$
consists of a vertically composable pair X, Y, together with a horizontal factorization
$$Y \circ_{\mathrm{v}} X = R \circ_{\mathrm{h}} L.$$

By Lemma 4.7, this is uniquely equivalent to a 2×2 grid

$$\begin{matrix} X_1 & X_2 \\ Y_1 & Y_2 \end{matrix}$$

whose rows and columns are composable. Such a grid is exactly a witness for

$$(m_A \otimes m_A)(1 \otimes c \otimes 1)(\Delta_A \otimes \Delta_A).$$

Under this correspondence the four exterior boxes are unchanged. Hence the two spans are equal.

4. **Verify the weak unit equations.** Put

$$U_n^- := \left\{(h_1, \ldots, h_n) \in \mathsf{Sym}_1^-(A)^n \;\middle|\; h_n \circ \cdots \circ h_1 \text{ is defined}\right\}.$$

By the factor-of-identity statement in Lemma 4.4, a horizontal factorization of a vertical identity

$$\mathrm{id}_h^{\mathrm{v}}$$

consists entirely of vertical identities. More precisely, horizontal factorizations of $\mathrm{id}_h^{\mathrm{v}}$ are in bijection with factorizations

$$h = h_2 \circ h_1,$$

and the two factors are

$$\mathrm{id}_{h_1}^{\mathrm{v}}, \qquad \mathrm{id}_{h_2}^{\mathrm{v}}.$$

Consequently $\Delta_A \eta_A$ is represented by the span whose witness set is U_2^-, with output

$$(h_1, h_2) \longmapsto \left(\mathrm{id}_{h_1}^{\mathrm{v}}, \mathrm{id}_{h_2}^{\mathrm{v}}\right).$$

Iterating the coproduct shows that either coassociative threefold factorization of the unit has witness set U_3^-, with output

$$(h_1, h_2, h_3) \longmapsto \left(\mathrm{id}_{h_1}^{\mathrm{v}}, \mathrm{id}_{h_2}^{\mathrm{v}}, \mathrm{id}_{h_3}^{\mathrm{v}}\right).$$

Consider now either product-side composite occurring in the two weak-unit equations of [16, Def. 1.1]. A witness consists of two horizontal factorizations of vertical identity boxes together with a vertical multiplication of the two middle factors. Every factor is a vertical identity by the preceding paragraph, and two vertical identity boxes are vertically composable precisely when their underlying horizontal edges agree. Thus the two middle witnesses are forced to carry the same horizontal edge.

Hence in either weak-unit composite the data reduce uniquely to a composable triple

$$(h_1, h_2, h_3) \in U_3^-,$$

and the exterior output is again

$$\left(\mathrm{id}_{h_1}^{\mathrm{v}}, \mathrm{id}_{h_2}^{\mathrm{v}}, \mathrm{id}_{h_3}^{\mathrm{v}}\right).$$

The symmetry appearing in the second weak-unit equation only exchanges the two middle witness coordinates before the same equality is imposed. Since **Sp** is symmetric, the resulting witness set and exterior legs are the same. Thus both weak-unit equations hold.

5. **Verify the weak counit equations.** Put

$$U_n^+ := \left\{(v_1, \dots, v_n) \in \mathsf{Sym}_1^+(A)^n \;\middle|\; v_n \circ \cdots \circ v_1 \text{ is defined}\right\}.$$

A witness for the counit of a total vertical paste is a vertically composable family of boxes whose total paste is a horizontal identity. By the factor-of-identity statement in Lemma 4.4, every vertical factor of such a paste is itself a horizontal identity. Thus a threefold witness is uniquely of the form

$$\left(\mathrm{id}^{\mathrm{h}}_{v_1}, \mathrm{id}^{\mathrm{h}}_{v_2}, \mathrm{id}^{\mathrm{h}}_{v_3}\right)$$

for a composable triple

$$(v_1, v_2, v_3) \in U_3^+.$$

On either product-side composite in the weak-counit equations of [16, Def. 1.1], the middle box is first horizontally factorized and the two resulting vertical pastes are then required by the two counits to be horizontal identities. Each factor in those vertical pastes is therefore a horizontal identity. Since two horizontally composable horizontal identity boxes must be identities on the same vertical edge, the inserted horizontal factorization is uniquely the identity factorization on that common edge.

Hence both product-side composites again have witness set U_3^+, with the same exterior legs as the total-paste side. The second weak-counit equation differs only by the symmetry interchanging the two middle coproduct coordinates, and gives the same witness set in the symmetric category **Sp**. Therefore both weak-counit equations hold.

6. **Compute the target counital map.** Expand the canonical target counital composite of the weak bimonoid. By the identity calculations above, a witness reduces uniquely to a pair

$$(v, h) \in \mathsf{Sym}_1^+(A) \times \mathsf{Sym}_1^-(A)$$

satisfying

$$s(v) = s(h).$$

Its input box is

$$\mathrm{id}^{\mathrm{h}}_v,$$

and its output box is

$$\mathrm{id}^{\mathrm{v}}_h.$$

Hence the target counital map is represented by

$$\mathsf{Boxes}_A \xleftarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{h}}_v} \left\{(v, h) \mid s(v) = s(h)\right\} \xrightarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{v}}_h} \mathsf{Boxes}_A,$$

which is (47).

7. **Compute the source counital map.** Reflecting the preceding identity calculation exchanges sources with targets. The surviving witnesses are therefore the pairs

$$(v, h) \in \mathsf{Sym}_1^+(A) \times \mathsf{Sym}_1^-(A)$$

satisfying

$$t(v) = t(h),$$

again with input $\mathrm{id}^{\mathrm{h}}_v$ and output $\mathrm{id}^{\mathrm{v}}_h$. This is precisely (48). □

Theorem 4.9 (Quantum symmetry weak Hopf monoid)**.** *For every thin concurrent game A, the polarized box object carries a canonical involutive weak Hopf monoid structure*

$$\mathsf{WH}_{\mathrm{sp}}(A) := \mathsf{Boxes}_A \tag{49}$$

in **Sp***, with structure maps $m_A, \eta_A, \Delta_A, \varepsilon_A, S_A$ above and*

$$S_A^2 = 1_{\mathsf{Boxes}_A}.$$

Proof.

1. **Use the incidence weak bimonoid.** By Lemma 4.8,

$$(\mathsf{Boxes}_A, m_A, \eta_A, \Delta_A, \varepsilon_A)$$

is a weak bimonoid. It remains to verify the three antipode equations for total box inversion.

2. **Set up the target antipode convolution.** Write an input box as

$$X = h \urcorner v.$$

By Lemma 4.4, a horizontal factorization of X is uniquely determined by

$$h = h_2 \circ h_1$$

and has factors

$$Z = h_1 \urcorner \mathsf{L}(h_2, v), \qquad Y = h_2 \urcorner v.$$

A witness for

$$m_A(1 \otimes S_A)\Delta_A$$

requires Z and $S_A(Y)$ to be vertically composable. Since the top edge of $S_A(Y)$ is

$$\big(\mathsf{B}(h_2, v)\big)^{-1},$$

this condition is

$$\mathsf{B}(h_1, \mathsf{L}(h_2, v)) = \big(\mathsf{B}(h_2, v)\big)^{-1}.$$

3. **Force the input to be a horizontal identity.** The bottom edge of the horizontal paste $X = Y \circ_{\mathrm{h}} Z$ is

$$\mathsf{B}(h_2, v) \circ \mathsf{B}(h_1, \mathsf{L}(h_2, v)).$$

Hence the preceding composability condition is equivalent to

$$\mathsf{B}(h, v) = \mathrm{id}\,.$$

The commuting-square equation for X then gives

$$\mathsf{L}(h, v) = v \circ h.$$

The left-hand side lies in $\mathsf{Sym}^+(A)$. Comparing the polarized decomposition

$$v \circ h$$

with the purely positive decomposition

$$\mathsf{L}(h, v) \circ \mathrm{id}$$

and using uniqueness in Theorem 2.4 forces

$$h = \mathrm{id}, \qquad \mathsf{L}(h, v) = v.$$

Thus the convolution is supported precisely on horizontal identity boxes.

4. **Identify the target counital map.** If $X = \mathrm{id}^{\mathrm{h}}_{v}$, then a factorization

$$\mathrm{id} = h_2 \circ h_1$$

has $h_2 = h_1^{-1}$, and the corresponding antipode witness exists uniquely. Its vertical paste is $\mathrm{id}^{\mathrm{v}}_{h_1}$. Therefore

$$m_A(1 \otimes S_A)\Delta_A$$

is exactly the target counital span (47).

5. **Identify the source counital map.** Reflecting the preceding calculation gives

$$m_A(S_A \otimes 1)\Delta_A$$

equal to the source counital span (48).

6. **Set up the third antipode equation.** A threefold horizontal factorization of

$$X = h \urcorner v$$

is uniquely determined by

$$h = h_3 \circ h_2 \circ h_1$$

and has factors

$$A_1 = h_1 \urcorner \mathsf{L}(h_2, \mathsf{L}(h_3, v)), \qquad A_2 = h_2 \urcorner \mathsf{L}(h_3, v), \qquad A_3 = h_3 \urcorner v.$$

A witness for

$$m_A^{(3)}(S_A \otimes 1 \otimes S_A)\Delta_A^{(3)}$$

requires $S_A(A_1), A_2, S_A(A_3)$ to be vertically composable.

7. **Determine the unique threefold witness.** The first composability condition gives

$$h_2 = h_1^{-1}.$$

The second gives

$$\mathsf{B}(h_2, \mathsf{L}(h_3, v)) = \big(\mathsf{B}(h_3, v)\big)^{-1}.$$

Hence the horizontal paste

$$A_3 \circ_{\mathrm{h}} A_2 = (h_3 \circ h_2) \urcorner v$$

has identity bottom edge. By Step 3,

$$h_3 \circ h_2 = \mathrm{id}\,.$$

Therefore

$$h_2 = h_1^{-1}, \qquad h_3 = h_1, \qquad h = h_1.$$

Moreover $A_3 = X$, and the left vertical edge of the horizontal identity $A_3 \circ_{\mathrm{h}} A_2$ is v; hence the right edge of A_1 is v, so $A_1 = X$ as well. Thus there is exactly one such threefold witness over each input box.

8. **Compute the threefold output.** The total vertical paste has top edge

$$\big(\mathsf{B}(h,v)\big)^{-1}$$

and right edge

$$\big(\mathsf{L}(h,v)\big)^{-1} \circ \mathsf{L}(h,v) \circ \big(\mathsf{L}(h,v)\big)^{-1} = \big(\mathsf{L}(h,v)\big)^{-1}.$$

By top-right determination, this box is precisely

$$\big(h \urcorner v\big)^{-1} = S_A(X).$$

Hence

$$m_A^{(3)}(S_A \otimes 1 \otimes S_A)\Delta_A^{(3)} = S_A.$$

9. **Conclude involutivity.** Total inversion in a double groupoid is involutive, so

$$S_A^2 = 1_{\mathsf{Boxes}_A}.$$

The three antipode equations therefore make $\mathsf{WH}_{\mathrm{sp}}(A)$ an involutive weak Hopf monoid. □

When A is fixed, we suppress the subscript A on the weak Hopf structure maps $m, \eta, \Delta, \varepsilon, S$.

Example 4.10 (A finite game with symmetry of both polarities)**.** Let A have four concurrent events n_1, n_2, p_1, p_2, with the n_i negative and the p_i positive, all finite subsets consistent, and all polarity-preserving bijections selected. Negative symmetries fix the positive events pointwise, and positive symmetries fix the negative events pointwise. At the full configuration $x = \{n_1, n_2, p_1, p_2\}$, let h exchange the two negative events and let v exchange the two positive events. Write $X_{ij} = h^i \urcorner v^j$, with indices in $\mathbb{Z}/2\mathbb{Z}$. The four boxes at this configuration satisfy

$$m_A(X_{ij}, X_{k\ell}) = X_{i,j+\ell} \quad \text{exactly when } i = k,$$

and the horizontal factorization witnesses of X_{ij} are

$$(X_{aj}, X_{bj}) \quad \text{with } a + b = i.$$

In particular, $m_A(X_{11}, X_{11}) = X_{10}$, while $\Delta_A(X_{11})$ has the two witnesses (X_{01}, X_{11}) and (X_{11}, X_{01}).

Proof.

1. **Check the polarized game structure.** Restriction and extension hold by extending the bijections separately on each polarity. The polarized families intersect in the identities and satisfy their respective receptivity conditions, so A is a thin concurrent game.

2. **Display the nontrivial polarized box.** The permutations h and v commute because they act on disjoint sets of events. Thus

$$\begin{tikzcd} x \arrow[r, "h"] \arrow[d, "v"'] & x \arrow[d, "v"] \\ x \arrow[r, "h"'] & x \end{tikzcd} \qquad X_{11} = h \urcorner v.$$

More generally, X_{ij} has top and bottom edges h^i and left and right edges v^j. Every symmetry out of the full configuration has that same full configuration as target, so its factorization witnesses remain in this component.

3. **Compute multiplication and comultiplication.** Vertical pasting requires equality of the intermediate horizontal edges, hence $i = k$, and composes the positive edges to $v^{j+\ell}$. Horizontal pasting requires equal positive edges and multiplies the negative edges; consequently the factorizations are exactly the displayed pairs with $a + b = i$. This proves the formulas and the two particular calculations.

4. **Identify the remaining operations.** In this component the unit witnesses are X_{00}, X_{10}, the counit is supported on X_{00}, X_{01}, and total inversion fixes every X_{ij}. The unit and counit therefore select different box directions even in this finite example. $\square$

4.3 The counital base

For reference, the relation between the two counital conventions and the base-valued maps is

Pastro–Street notation	configuration-span presentation
r_A^{PS}	$\varepsilon_t = i_t r_t$
t_A^{PS}	$\varepsilon_s = i_s r_s$
s_A^{PS}	$i_s r_t$
$\bar{s}_A, \bar{t}_A$	r_t, r_s

The spans below construct these presentations and prove the displayed identities.

Let Θ_a denote the double identity at $a \in \mathsf{Conf}(A)$. Define the spans

$$
\begin{aligned}
i_t : \mathsf{Conf}(A) \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A), &\qquad
\begin{tikzcd} & \mathsf{Sym}_1^-(A) \arrow[dl, "s"'] \arrow[dr, "h \mapsto \mathrm{id}_h^{\mathrm{v}}"] & \\ \mathsf{Conf}(A) & & \mathsf{Boxes}_A \end{tikzcd} \\
i_s : \mathsf{Conf}(A) \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A), &\qquad
\begin{tikzcd} & \mathsf{Sym}_1^-(A) \arrow[dl, "t"'] \arrow[dr, "h \mapsto \mathrm{id}_h^{\mathrm{v}}"] & \\ \mathsf{Conf}(A) & & \mathsf{Boxes}_A \end{tikzcd} \\
r_t : \mathsf{WH}_{\mathrm{sp}}(A) \longrightarrow \mathsf{Conf}(A), &\qquad
\begin{tikzcd} & \mathsf{Sym}_1^+(A) \arrow[dl, "v \mapsto \mathrm{id}_v^{\mathrm{h}}"'] \arrow[dr, "s"] & \\ \mathsf{Boxes}_A & & \mathsf{Conf}(A) \end{tikzcd} \\
r_s : \mathsf{WH}_{\mathrm{sp}}(A) \longrightarrow \mathsf{Conf}(A), &\qquad
\begin{tikzcd} & \mathsf{Sym}_1^+(A) \arrow[dl, "v \mapsto \mathrm{id}_v^{\mathrm{h}}"'] \arrow[dr, "t"] & \\ \mathsf{Boxes}_A & & \mathsf{Conf}(A). \end{tikzcd}
\end{aligned}
\tag{50}
$$

Lemma 4.11 (Counital retract spans)**.** *The composites of the configuration retract spans are*

$$
i_t r_t = \left[\mathsf{Boxes}_A \xleftarrow{(v,h) \mapsto \mathrm{id}_v^{\mathrm{h}}} \big\{ (v,h) \in \mathsf{Sym}_1^+(A) \times \mathsf{Sym}_1^-(A) \mid s(v) = s(h) \big\} \xrightarrow{(v,h) \mapsto \mathrm{id}_h^{\mathrm{v}}} \mathsf{Boxes}_A \right],
\tag{51}
$$

and

$$i_s r_s = \left[\mathsf{Boxes}_A \xleftarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{h}}_v} \big\{(v,h) \in \mathsf{Sym}^+_1(A) \times \mathsf{Sym}^-_1(A) \mid t(v) = t(h)\big\} \xrightarrow{(v,h)\mapsto \mathrm{id}^{\mathrm{v}}_h} \mathsf{Boxes}_A\right]. \quad (52)$$

Hence $i_t r_t$ and $i_s r_s$ are respectively the target and source counital maps of the incidence weak bimonoid.

Proof.

1. **Compose r_t with i_t.** The composite

$$\mathsf{Boxes}_A \xrightarrow{r_t} \mathsf{Conf}(A) \xrightarrow{i_t} \mathsf{Boxes}_A$$

   is formed by pulling back the two apex maps

$$s : \mathsf{Sym}^+_1(A) \longrightarrow \mathsf{Conf}(A), \qquad s : \mathsf{Sym}^-_1(A) \longrightarrow \mathsf{Conf}(A).$$

   Its apex is therefore

$$\big\{(v,h) \in \mathsf{Sym}^+_1(A) \times \mathsf{Sym}^-_1(A) \mid s(v) = s(h)\big\},$$

   with exterior legs

$$(v,h) \longmapsto \mathrm{id}^{\mathrm{h}}_v, \qquad (v,h) \longmapsto \mathrm{id}^{\mathrm{v}}_h\,.$$

   This gives (51).

2. **Compose r_s with i_s.** Likewise, the pullback defining

$$\mathsf{Boxes}_A \xrightarrow{r_s} \mathsf{Conf}(A) \xrightarrow{i_s} \mathsf{Boxes}_A$$

   imposes

$$t(v) = t(h),$$

   giving (52).

3. **Identify the counital maps.** These are exactly the target and source counital spans computed in Lemma 4.8. □

Proposition 4.12 (Explicit counital maps)**.** *For* $\mathsf{WH}_{\mathrm{sp}}(A)$,

$$m(1 \otimes S)\Delta = i_t r_t, \qquad m(S \otimes 1)\Delta = i_s r_s. \quad (53)$$

Moreover

$$r_t i_t = r_s i_s = r_s i_t = r_t i_s = 1_{\mathsf{Conf}(A)}, \quad (54)$$

$$S i_t = i_s, \qquad S i_s = i_t, \quad (55)$$

$$r_t S = r_s, \qquad r_s S = r_t. \quad (56)$$

Proof.

1. **Identify the two counital convolutions.** By Theorem 4.9, the antipode satisfies

$$m(1 \otimes S)\Delta = \varepsilon_t, \qquad m(S \otimes 1)\Delta = \varepsilon_s.$$

   Lemma 4.11 identifies these counital maps with

$$\varepsilon_t = i_t r_t, \qquad \varepsilon_s = i_s r_s.$$

   This proves (53).

2. **Compute the four configuration composites.** Consider, for example, $r_t i_t$. Its pullback apex consists of pairs
$$(h, v) \in \mathsf{Sym}_1^-(A) \times \mathsf{Sym}_1^+(A)$$
satisfying
$$\mathrm{id}_h^{\mathrm{v}} = \mathrm{id}_v^{\mathrm{h}}\,.$$
The same equation occurs for each of the other three composites, with the exterior configuration labels read using the appropriate source or target maps.

3. **Identify the surviving boxes.** A box which is simultaneously a vertical identity and a horizontal identity is necessarily a double identity
$$\Theta_a$$
at a unique $a \in \mathsf{Conf}(A)$. Conversely every Θ_a gives such a witness. In each of the four composites the input and output configuration are therefore both a, with a unique witness. Hence
$$r_t i_t = r_s i_s = r_s i_t = r_t i_s = 1_{\mathsf{Conf}(A)}.$$

4. **Track inversion on the inclusions.** Total box inversion sends a vertical identity to
$$S(\mathrm{id}_h^{\mathrm{v}}) = \mathrm{id}_{h^{-1}}^{\mathrm{v}}\,.$$
Since
$$s(h^{-1}) = t(h), \qquad t(h^{-1}) = s(h),$$
inversion exchanges the source- and target-labelled inclusions:
$$S i_t = i_s, \qquad S i_s = i_t.$$

5. **Track inversion on the retractions.** Likewise,
$$S(\mathrm{id}_v^{\mathrm{h}}) = \mathrm{id}_{v^{-1}}^{\mathrm{h}}\,.$$
On the common apex $\mathsf{Sym}_1^+(A)$,
$$r_t S(\mathrm{id}_v^{\mathrm{h}}) = s(v^{-1}) = t(v) = r_s(\mathrm{id}_v^{\mathrm{h}}),$$
so
$$r_t S = r_s.$$
Applying $S^2 = 1_{\mathsf{Boxes}_A}$ gives
$$r_s S = r_t.$$
□

By Proposition 3.5, the configuration set $\mathsf{Conf}(A)$ carries the canonical separable commutative Frobenius monoid
$$C_{\mathsf{Conf}(A)} := \big(\mathsf{Conf}(A), \mu_{\mathsf{Conf}(A)}, \eta_{\mathsf{Conf}(A)}, \delta_{\mathsf{Conf}(A)}, \epsilon_{\mathsf{Conf}(A)}\big)$$
in **Sp**.

Proposition 4.13 (Configuration Frobenius retract). *Transport the Pastro–Street target counital retract through*

$$\mathsf{Conf}(A) \xrightarrow{i_s} \big(\mathsf{WH}_{\mathrm{sp}}(A), t_A^{\mathrm{PS}}\big) \xrightarrow{r_s} \mathsf{Conf}(A).$$

The transported Frobenius structure is

$$\mu_A^C : \quad \mathsf{Conf}(A)^2 \xleftarrow{\Delta_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \xrightarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A),$$

$$\eta_A^C : \quad 1 \longleftarrow \mathsf{Conf}(A) \xrightarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A),$$

$$\delta_A^C : \quad \mathsf{Conf}(A) \xleftarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \xrightarrow{\Delta_{\mathsf{Conf}(A)}} \mathsf{Conf}(A)^2,$$

$$\epsilon_A^C : \quad \mathsf{Conf}(A) \xleftarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \longrightarrow 1.$$

Hence the transported separable commutative Frobenius monoid is precisely $C_{\mathsf{Conf}(A)}$.

Proof.

1. **Set up the transported Frobenius structure.** By [16, Proposition 1.4], transporting the target Frobenius structure along i_s, r_s gives the four composites

$$r_s m(i_s \otimes i_s), \qquad r_s\eta, \qquad (r_s \otimes r_s)\Delta i_s, \qquad \varepsilon i_s.$$

   We calculate them as spans.

2. **Calculate the transported multiplication.** The multiplication is

$$r_s m(i_s \otimes i_s).$$

   A multiplication witness begins with vertical identities $\mathrm{id}^{\mathrm{v}}_{h_1}$ and $\mathrm{id}^{\mathrm{v}}_{h_2}$. Vertical composability forces

$$h_1 = h_2 =: h,$$

   and their paste is $\mathrm{id}^{\mathrm{v}}_h$. The final application of r_s requires this paste also to be a horizontal identity. Thus

$$\mathrm{id}^{\mathrm{v}}_h = \mathrm{id}^{\mathrm{h}}_v$$

   for some $v \in \mathsf{Sym}^+(A)$, which forces the common box to be the double identity Θ_a for a unique $a \in \mathsf{Conf}(A)$.

   Hence the two input labels are both a, the output label is a, and the surviving witness is unique. Therefore

$$\mu_A^C = \left[\mathsf{Conf}(A) \times \mathsf{Conf}(A) \xleftarrow{\Delta_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \xrightarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A)\right].$$

3. **Calculate the transported unit.** The unit is

$$r_s\eta.$$

   Its pullback apex consists of pairs

$$(h, v) \in \mathsf{Sym}_1^-(A) \times \mathsf{Sym}_1^+(A)$$

satisfying

$$\mathrm{id}^{\mathrm{v}}_h = \mathrm{id}^{\mathrm{h}}_v \,.$$

Again the common box is uniquely a double identity Θ_a. Therefore

$$\eta^C_A = \left[1 \leftarrow \mathsf{Conf}(A) \xrightarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \right].$$

4. **Calculate the transported comultiplication.** The comultiplication is

$$(r_s \otimes r_s)\Delta i_s.$$

An i_s-witness over a is a vertical identity $\mathrm{id}^{\mathrm{v}}_h$ with $t(h) = a$. By Lemma 4.4, a horizontal factorization of $\mathrm{id}^{\mathrm{v}}_h$ is uniquely determined by a factorization

$$h = h_2 \circ h_1,$$

with factors

$$\mathrm{id}^{\mathrm{v}}_{h_1}, \qquad \mathrm{id}^{\mathrm{v}}_{h_2} \,.$$

For both factors to survive the two applications of r_s, each must also be a horizontal identity. Hence h_1, h_2, and therefore h, are identity arrows.

The unique surviving factorization is the double identity at a, and its two output labels are (a, a). Thus

$$\delta^C_A = \left[\mathsf{Conf}(A) \xleftarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \xrightarrow{\Delta_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \times \mathsf{Conf}(A) \right].$$

5. **Calculate the transported counit.** Finally, εi_s has the same double-identity pullback as the unit calculation, now with exterior legs $1_{\mathsf{Conf}(A)}$ and the unique map to 1. Hence

$$\epsilon^C_A = \left[\mathsf{Conf}(A) \xleftarrow{1_{\mathsf{Conf}(A)}} \mathsf{Conf}(A) \to 1 \right].$$

6. **Identify the canonical Frobenius structure.** The four transported spans are exactly the canonical Frobenius structure on $\mathsf{Conf}(A)$ from Proposition 3.5. Hence the transported Frobenius monoid is $C_{\mathsf{Conf}(A)}$. $\square$

Write

$$C_A := C_{\mathsf{Conf}(A)}.$$

With the Pastro–Street convention of (17), Proposition 4.12 gives

$$r^{\mathrm{PS}}_A = \varepsilon_t = i_t r_t, \qquad t^{\mathrm{PS}}_A = \varepsilon_s = i_s r_s.$$

Moreover,

$$s^{\mathrm{PS}}_A = S^{-1}\varepsilon_t.$$

Since $S^2 = 1$ and $S i_t = i_s$,

$$s^{\mathrm{PS}}_A = S i_t r_t = i_s r_t.$$

Thus

$$r_A^{\mathrm{PS}} = i_t r_t, \tag{57}$$
$$t_A^{\mathrm{PS}} = i_s r_s, \tag{58}$$
$$s_A^{\mathrm{PS}} = i_s r_t. \tag{59}$$

After splitting the target retract by $\mathsf{Conf}(A)$, the base-valued source and target maps are therefore

$$\bar{s}_A = r_t, \qquad \bar{t}_A = r_s. \tag{60}$$

4.4 Boundary coactions and composition

Define the Pastro–Street coactions

$$\gamma_l := c_{\mathsf{WH}_{\mathrm{sp}}(A),C_A}(1 \otimes \bar{s}_A)\Delta : \mathsf{WH}_{\mathrm{sp}}(A) \longrightarrow C_A \otimes \mathsf{WH}_{\mathrm{sp}}(A), \tag{61}$$
$$\gamma_r := (1 \otimes \bar{t}_A)\Delta : \mathsf{WH}_{\mathrm{sp}}(A) \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A) \otimes C_A. \tag{62}$$

For a box

$$\begin{tikzcd} p \arrow[r, "h_t"] \arrow[d, "v_l"'] & q \arrow[d, "v_r"] \\ r \arrow[r, "h_b"'] & s, \end{tikzcd}$$

put

$$\ell_A(X) := q, \qquad r_A(X) := s. \tag{63}$$

Proposition 4.14 (Boundary graph coactions)**.** *The left coaction is the graph of*

$$X \longmapsto (\ell_A(X), X),$$

and the right coaction is the graph of

$$X \longmapsto (X, r_A(X)).$$

Equivalently, the two graph coactions are

$$\mathsf{WH}_{\mathrm{sp}}(A) \xrightarrow{X \mapsto (\ell_A(X),X)} \mathsf{Conf}(A) \times \mathsf{WH}_{\mathrm{sp}}(A) \qquad \mathsf{WH}_{\mathrm{sp}}(A) \xrightarrow{X \mapsto (X,r_A(X))} \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{Conf}(A).$$

Proof.

1. **Compute the left boundary coaction.** In $(1 \otimes \bar{s}_A)\Delta(X)$, a horizontal factorization survives exactly when its right factor lies in the domain of r_t, hence is a horizontal identity. The unique such factorization has right factor $\mathrm{id}^{\mathrm{h}}_{v_r}$, and

$$r_t(\mathrm{id}^{\mathrm{h}}_{v_r}) = s(v_r) = q.$$

2. **Compute the right boundary coaction.** The right-coaction calculation is identical with r_s, giving $t(v_r) = s$. □

Proposition 4.15 (Right-mate boundary calculus in spans)**.** *Let P, Q be sets.*

1. *Every right C_P-coaction on a set X in* **Sp** *is represented by the graph coaction of a unique map*
$$r_X : X \longrightarrow P.$$
2. *Since C_P is commutative and*
$$C_P \otimes C_Q \cong C_{P\times Q},$$
every right $C_P^\circ \otimes C_Q$-coaction on a set X is represented by the graph coaction of a unique map
$$d_X : X \longrightarrow P \times Q.$$
3. *Writing*
$$d_X = (\ell_X, r_X),$$
the Chikhladze right-mate convention identifies this comodule with the C_P–C_Q-bicomodule having boundary presentation
$$P \xleftarrow{\quad \ell_X \quad} X \xrightarrow{\quad r_X \quad} Q. \tag{64}$$
Conversely, this boundary diagram reconstructs the original right $C_P^\circ \otimes C_Q$-coaction on X.

Proof.

1. **Normalize a right C_P-coaction.** Represent a right C_P-coaction on X by a span
$$X \xleftarrow{\ell} R \xrightarrow{(u,b)} X \times P.$$
Composing with the counit of C_P gives the span
$$X \xleftarrow{\ell} R \xrightarrow{u} X.$$
The comodule counit equation says that this morphism of **Sp** is the identity span on X. Hence there is an apex isomorphism
$$\varphi : R \xrightarrow{\cong} X$$
commuting with both exterior legs, so that
$$\ell = \varphi, \qquad u = \varphi.$$
Transporting the coaction along φ therefore puts it in graph form
$$X \xleftarrow{1_X} X \xrightarrow{(1_X, r_X)} X \times P,$$
where
$$r_X := b \circ \varphi^{-1}.$$
Coassociativity is then the equality of the two graph spans of
$$x \longmapsto \big(x, r_X(x), r_X(x)\big),$$
so every such map $r_X : X \to P$ indeed defines a right C_P-coaction.

To prove uniqueness, suppose that the graph coactions of

$$r_X, r'_X : X \longrightarrow P$$

represent the same morphism

$$X \longrightarrow X \times P$$

in **Sp**. An isomorphism between their apexes must commute with the two left legs 1_X, and is therefore 1_X. Commutation with the right legs then gives

$$(1_X, r_X) = (1_X, r'_X),$$

hence

$$r_X = r'_X.$$

2. **Apply the graph description to the product base.** The canonical Frobenius monoid C_P is commutative, so

$$C_P^\circ = C_P.$$

Moreover, taking Cartesian products of the four defining Frobenius spans gives

$$C_P \otimes C_Q \cong C_{P\times Q}.$$

Hence

$$C_P^\circ \otimes C_Q \cong C_{P\times Q}.$$

Applying Step 1 with base $P \times Q$ shows that a right $C_P^\circ \otimes C_Q$-coaction on the fixed underlying set X is represented by the graph of a unique map

$$d_X : X \longrightarrow P \times Q.$$

3. **Expose the two boundary coordinates.** Write

$$d_X = (\ell_X, r_X).$$

The biduality of the P-coordinate is represented in spans by the diagonal evaluation and coevaluation

$$P \times P \xleftarrow{\Delta_P} P \longrightarrow 1, \qquad 1 \longleftarrow P \xrightarrow{\Delta_P} P \times P.$$

Their snake composites are identity spans: the defining pullbacks impose equality of the repeated P-labels.

Taking the mate of the P-coordinate of the graph coaction therefore leaves the carrier X unchanged and turns the two coordinates of d_X into the graph coactions

$$x \longmapsto (\ell_X(x), x), \qquad x \longmapsto (x, r_X(x)).$$

These are precisely the left C_P- and right C_Q-coactions of the boundary diagram

$$P \xleftarrow{\ell_X} X \xrightarrow{r_X} Q.$$

4. **Invert the mate construction.** Applying the opposite snake identity moves the exposed P-coordinate back to the right and recovers the graph

$$d_X = (\ell_X, r_X) : X \longrightarrow P \times Q.$$

Thus, on the fixed underlying set X, the right-comodule and boundary-bicomodule presentations determine one another uniquely. This is the explicit span form of the bidual convention used in Chikhladze's multitensor calculus [5, §3]. $\square$

Proposition 4.16 (Boundary presentations of the arrow object)**.** *The right $C_A^{\circ} \otimes C_A$-comodule underlying the arrow object has boundary map*

$$d_A^R(X) = (\ell_A(X), r_A(X)) = (q, s)$$

for a box with vertices p, q, r, s as in (34)*. Equivalently, its right-mate boundary presentation is*

$$\mathsf{Conf}(A) \xleftarrow{\ \ell_A\ } \mathsf{WH}_{\mathrm{sp}}(A) \xrightarrow{\ r_A\ } \mathsf{Conf}(A). \tag{65}$$

The associated endocomonad on $C_{\mathsf{Conf}(A)\times\mathsf{Conf}(A)}$ retains both boundary maps

$$d_A^L(X) = (p, r), \qquad d_A^R(X) = (q, s). \tag{66}$$

Its comultiplication records horizontal box factorizations, and its boundary counit is the span

$$\mathsf{Boxes}_A \xleftarrow{v \mapsto \mathrm{id}_v^{\mathrm{h}}} \mathsf{Sym}_1^+(A) \xrightarrow{v \mapsto (s(v), t(v))} \mathsf{Conf}(A) \times \mathsf{Conf}(A). \tag{67}$$

Proof.

1. **Identify the right-comodule boundary.** Propositions 4.14 and 4.15 identify the right coaction with $d_A^R(X) = (q, s)$.

2. **Compute the boundary counit.** A horizontal factorization with both factors in the support of (67) consists uniquely of two copies of the same horizontal identity. Thus this span preserves the comonoid structure.

3. **Recover both endocomonad boundaries.** Applying the boundary counit to the first horizontal factor of $\Delta(X)$ extracts the left vertical edge; applying it to the second extracts the right vertical edge. The associated endocomonad therefore has boundary diagram

$$\mathsf{Conf}(A)^2 \xleftarrow{X \mapsto (\mathrm{nw}(X), \mathrm{sw}(X))} \mathsf{Boxes}_A \xrightarrow{X \mapsto (\mathrm{ne}(X), \mathrm{se}(X))} \mathsf{Conf}(A)^2.$$

4. **Verify the comonad laws.** The relative comultiplication records horizontal factorizations. Their shared edge supplies the matching boundary, and the displayed counit selects horizontal identity factors. Associativity and the identity laws of the horizontal box groupoid give the comonad equations. $\square$

Put

$$\mathsf{Comp}_A := \{(X, Y) \in \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A) \mid r_A(X) = \ell_A(Y)\}. \tag{68}$$

Equivalently, Comp_A is the boundary pullback

$$\begin{tikzcd}
\mathsf{Comp}_A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathsf{WH}_{\mathrm{sp}}(A) \arrow[d, "\ell_A"] \\
\mathsf{WH}_{\mathrm{sp}}(A) \arrow[r, "r_A"'] & \mathsf{Conf}(A).
\end{tikzcd}$$

Let

$$j_A : \mathsf{Comp}_A \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A)$$

be the graph of the inclusion, and let p_A be its converse span. Put

$$a_A := j_A p_A.$$

Thus a_A is the partial identity on $\mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A)$ supported on the boundary equation

$$r_A(X) = \ell_A(Y),$$

and

$$p_A j_A = 1_{\mathsf{Comp}_A}, \qquad j_A p_A = a_A.$$

Lemma 4.17 (Local exactness for the game cotensor)**.** *The object* Comp_A *is an explicit representative of the cotensor*

$$\mathsf{WH}_{\mathrm{sp}}(A) \boxtimes_{C_A} \mathsf{WH}_{\mathrm{sp}}(A) \cong \mathsf{Comp}_A. \tag{69}$$

Moreover:

1. *the graph inclusion*
$$j_A : \mathsf{Comp}_A \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A)$$
is a split, hence absolute, equalizer of the two boundary-coaction maps;
2. *the Pastro–Street cotensor idempotent is* $a_A = j_A p_A$*;*
3. *the induced boundaries on* Comp_A *are*
$$\ell_A^{(2)}(X,Y) = \ell_A(X), \qquad r_A^{(2)}(X,Y) = r_A(Y);$$
4. *every tensoring of this equalizer occurring in the forward Pastro–Street construction is again an equalizer;*
5. *for every* $n \geq 2$*, the iterated cotensor is represented explicitly by*
$$\mathsf{Comp}_A^{(n)} := \{(X_1, \ldots, X_n) \in \mathsf{WH}_{\mathrm{sp}}(A)^n \mid r_A(X_i) = \ell_A(X_{i+1}) \text{ for } 1 \leq i < n\}. \tag{70}$$
with exterior boundaries
$$\ell_A^{(n)} = \ell_A \pi_1, \qquad r_A^{(n)} = r_A \pi_n.$$
If
$$j_A^{(n)} : \mathsf{Comp}_A^{(n)} \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A)^n$$
is the graph inclusion and $p_A^{(n)}$ *its converse, then*
$$p_A^{(n)} j_A^{(n)} = 1_{\mathsf{Comp}_A^{(n)}}, \qquad j_A^{(n)} p_A^{(n)} = a_A^{(n)},$$
where $a_A^{(n)}$ *is the partial identity supported on all adjacent boundary equations.*

Proof.

1. **Write the two cotensor maps explicitly.** By Proposition 4.14, the right boundary coaction of the first box records $r_A(X)$, while the left boundary coaction of the second records $\ell_A(Y)$. Hence the defining cotensor equalizer compares the graph maps

$$f_A, g_A : \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A) \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{Conf}(A) \times \mathsf{WH}_{\mathrm{sp}}(A)$$

given by

$$f_A(X,Y) = \big(X, r_A(X), Y\big), \qquad g_A(X,Y) = \big(X, \ell_A(Y), Y\big).$$

They agree precisely on

$$r_A(X) = \ell_A(Y),$$

so their equalizing locus is Comp_A.

The Pastro–Street idempotent on the ambient tensor product is

$$e_A^{\mathrm{PS}} := (1 \otimes \epsilon_A^C \otimes 1)(1 \otimes \mu_A^C \otimes 1)(\gamma_r \otimes \gamma_l). \tag{71}$$

The coactions record $(X, r_A(X), \ell_A(Y), Y)$; the canonical Frobenius multiplication imposes equality of the two middle labels and the counit forgets their common value. Hence

$$e_A^{\mathrm{PS}} = j_A p_A = a_A. \tag{72}$$

2. **Construct the split equalizer.** Let

$$d_A : \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{Conf}(A) \times \mathsf{WH}_{\mathrm{sp}}(A) \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A)$$

be the span whose apex is $\mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A)$, with left leg

$$(X,Y) \longmapsto \big(X, r_A(X), Y\big)$$

and right leg

$$(X,Y) \longmapsto (X,Y).$$

Then

$$d_A f_A = 1_{\mathsf{WH}_{\mathrm{sp}}(A)\times\mathsf{WH}_{\mathrm{sp}}(A)}, \qquad d_A g_A = a_A,$$

while

$$f_A a_A = g_A a_A.$$

3. **Verify the equalizer universal property.** Let

$$z : Z \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A)$$

satisfy

$$f_A z = g_A z.$$

Then

$$a_A z = d_A g_A z = d_A f_A z = z.$$

Since $a_A = j_A p_A$, this gives

$$z = j_A p_A z.$$

The factorization through j_A is unique because $p_A j_A = 1_{\mathsf{Comp}_A}$. Thus j_A is the required equalizer.

4. **Conclude absoluteness.** The equations
$$d_A f_A = 1, \qquad d_A g_A = a_A, \qquad f_A a_A = g_A a_A$$
exhibit the equalizer as split. Split equalizers are absolute, hence preserved by every functor. In particular, every left or right tensoring of j_A occurring in the Pastro–Street construction is again an equalizer.

5. **Identify the induced boundaries.** Restricting the tensor-product coactions along j_A leaves the left boundary of the first factor and the right boundary of the second, so
$$\ell_A^{(2)}(X,Y) = \ell_A(X), \qquad r_A^{(2)}(X,Y) = r_A(Y).$$

6. **Construct the threefold cotensor.** Applying the same boundary-pullback calculation once more gives
$$\mathsf{Comp}_A^{(3)} = \{(X_1,X_2,X_3) \mid r_A(X_1) = \ell_A(X_2),\ r_A(X_2) = \ell_A(X_3)\}.$$
This represents either parenthesization of the threefold cotensor, and the associativity comparison is induced by the identity on this simultaneous boundary pullback.

7. **Iterate the construction.** Repeating the same argument gives
$$\mathsf{Comp}_A^{(n)} = \{(X_1,\dots,X_n) \mid r_A(X_i) = \ell_A(X_{i+1}) \text{ for } 1 \le i < n\}$$
for every $n \ge 2$, with exterior boundaries $\ell_A\pi_1$ and $r_A\pi_n$. Its graph inclusion $j_A^{(n)}$ has converse $p_A^{(n)}$, so
$$p_A^{(n)} j_A^{(n)} = 1_{\mathsf{Comp}_A^{(n)}},$$
while
$$a_A^{(n)} := j_A^{(n)} p_A^{(n)}$$
is the partial identity imposing all adjacent boundary equations. Hence every finite cotensor and every splitting needed below is represented explicitly in **Sp**. ☐

Proposition 4.18 (Transport of the Pastro–Street retracts)**.** *Let*
$$J : \mathbf{Sp} \longrightarrow \mathrm{Kar}(\mathbf{Sp}), \qquad X \longmapsto (X, 1_X).$$
Then the Pastro–Street target object
$$\big(\mathsf{WH}_{\mathrm{sp}}(A), t_A^{\mathrm{PS}}\big)$$
is isomorphic to $J(\mathsf{Conf}(A))$ through i_s, r_s; its Frobenius structure transports to $J(C_A)$; and its cotensor Karoubi object is the split image of
$$a_A = j_A p_A,$$
hence isomorphic to $J(\mathsf{Comp}_A)$.

The splitting of the composition idempotent is
$$\begin{tikzcd} \mathsf{Comp}_A \arrow[rr, bend left, "j_A"] && \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A) \arrow[ll, bend left, "p_A"] \arrow[loop right, "a_A"] \end{tikzcd}$$
with
$$p_A j_A = 1_{\mathsf{Comp}_A}, \qquad j_A p_A = a_A.$$

Proof.

1. **Split the target retract.** Proposition 4.12 gives

$$r_s i_s = 1_{\mathsf{Conf}(A)}, \qquad i_s r_s = t_A^{\mathrm{PS}}.$$

   Hence i_s, r_s exhibit $\big(\mathsf{WH}_{\mathrm{sp}}(A), t_A^{\mathrm{PS}}\big)$ as isomorphic to $J(\mathsf{Conf}(A))$ in $\mathrm{Kar}(\mathbf{Sp})$.

2. **Transport the Frobenius structure.** Proposition 4.13 identifies the Frobenius structure transported through this retract with the canonical structure C_A. Hence the target Frobenius object is identified with $J(C_A)$.

3. **Identify the cotensor idempotent.** The two boundary coactions record

$$r_A(X) \qquad \text{and} \qquad \ell_A(Y).$$

   Therefore the Pastro–Street cotensor idempotent is the partial identity requiring

$$r_A(X) = \ell_A(Y).$$

   By Lemma 4.17, this is precisely

$$a_A = j_A p_A,$$

   whose split image is Comp_A. □

Restrict weak multiplication along the split cotensor inclusion:

$$\mu_A := m\, j_A : \mathsf{Comp}_A \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A), \tag{73}$$

$$u_A := i_s : C_A \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A). \tag{74}$$

The composition span has domain Comp_A and support the pairs whose full intermediate horizontal edges agree. On this support its value is vertical box composition.

Proposition 4.19 (Quantum-category structure)**.** *The arrow comonoid $\mathsf{WH}_{\mathrm{sp}}(A)$, its boundary coactions γ_l, γ_r, the composition μ_A, and the unit u_A form the Pastro–Street realization of a Day–Street quantum category in* $\mathbf{Sp}$.

Proof.

1. **Recall the forward Pastro–Street construction.** The forward direction of [16, Thm. 6.1 and §6.1] constructs the quantum category in the Karoubi completion from the target separable-Frobenius retract and the required relative tensor products. For the present weak Hopf monoid, their defining equalizers and tensor preservation are supplied explicitly by Lemma 4.17.

2. **Identify the target base.** Proposition 4.18 identifies the target object with $J(\mathsf{Conf}(A))$, while Proposition 4.13 identifies its transported separable Frobenius structure with $J(C_A)$.

3. **Identify the relative composition object.** Proposition 4.14 identifies the relevant coactions with the boundary maps r_A and ℓ_A. Lemma 4.17 therefore identifies the cotensor with

$$\mathsf{Comp}_A = \{(X, Y) \mid r_A(X) = \ell_A(Y)\},$$

   and identifies the corresponding Karoubi idempotent as

$$a_A = j_A p_A.$$

4. **Supply the required exactness locally.** Lemma 4.17 shows that j_A is a split absolute equalizer and constructs every finite iterated cotensor as the simultaneous boundary pullback

$$\mathsf{Comp}_A^{(n)} = \{(X_1, \dots, X_n) \mid r_A(X_i) = \ell_A(X_{i+1}) \text{ for } 1 \le i < n\}\,.$$

   Hence all tensorings of the equalizers used in the forward construction are preserved.

5. **Identify composition and unit.** On these split representatives the Pastro–Street composition is the restriction of weak multiplication,

$$\mu_A = m j_A : \mathsf{Comp}_A \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A),$$

   and the unit is

$$u_A = i_s : C_A \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A).$$

   Hence all the quantum-category structure maps have representatives in **Sp**.

6. **Reflect the quantum-category equations to spans.** The forward Pastro–Street construction gives the required equations in Kar(**Sp**). Since

$$J : \mathbf{Sp} \longrightarrow \mathrm{Kar}(\mathbf{Sp})$$

   is fully faithful, they reflect to the displayed spans in **Sp**. □

Proposition 4.20 (Nullary mate of the unit). *The nullary multitensor unit corresponding to $u_A = i_s$ is*

$$\eta_A^T = \left[\mathsf{Conf}(A) \times \mathsf{Conf}(A) \xleftarrow{h \mapsto (s(h),t(h))} \mathsf{Sym}_1^-(A) \xrightarrow{h \mapsto \mathrm{id}_h^{\mathrm{v}}} \mathsf{Boxes}_A\right]. \tag{75}$$

It is a map of comonads from $T_0(C_A)$ to the arrow endocomonad.

Proof.

1. **Match the boundaries.** At $h : p \to q$, the vertical identity box has source (p,p) and target (q,q) by (66). These are the boundaries of $(p,q) \in T_0(C_A)$ from Proposition 3.12.

2. **Take the nullary mate.** The direct unit has apex $\mathsf{Sym}_1^-(A)$ and left leg $t(h)$. The nullary Kan correspondence records both the incoming coevaluation label $s(h)$ and the outgoing label $t(h)$, giving

$$\begin{tikzcd}[column sep=small] & \mathsf{Sym}_1^-(A) \arrow[dl, "h \mapsto (s(h),t(h))"'] \arrow[dr, "h \mapsto \mathrm{id}_h^{\mathrm{v}}"] & \\ \mathsf{Conf}(A)^2 & & \mathsf{Boxes}_A. \end{tikzcd}$$

3. **Verify comultiplication.** Horizontal factorizations of $\mathrm{id}_h^{\mathrm{v}}$ are precisely the vertical identity boxes of factorizations $h = h_2 \circ h_1$. The matrix comultiplication of $T_0(C_A)$ chooses their intermediate configuration, followed by the two witnesses h_1, h_2. This is a bijection of the two composite apices preserving both exterior legs.

4. **Verify the counit.** The counit selects the identity negative arrows. There is one such witness at each pair (p,p), giving the counit of $T_0(C_A)$. □

4.5 The quantum groupoid of a thin game

We have constructed the quantum-category data. It remains to construct the inversion data of Definition 3.9 and verify equations (G1)–(G3).

Construction 4.21 (Transported inversion data)**.** We transport the Pastro–Street inversion data of $\mathsf{WH}_{\mathrm{sp}}(A)$ to the explicit configuration base and split composition object constructed above.

1. **Transport the object and arrow inversion maps.** Pastro–Street's weak-Hopf construction gives
$$\nu = S : \mathsf{WH}_{\mathrm{sp}}(A)^{\circ} \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A), \qquad \upsilon = t_A^{\mathrm{PS}} S^2 t_A^{\mathrm{PS}}.$$
Since
$$S^2 = 1_{\mathsf{WH}_{\mathrm{sp}}(A)},$$
transport through the target splitting i_s, r_s gives
$$\upsilon_A = 1_{C_A}, \qquad \nu_A(X) = X^{-1}. \tag{76}$$

2. **Identify the Pastro–Street composition comodules.** The remaining inversion datum is a left $\mathsf{WH}_{\mathrm{sp}}(A)^{\otimes 3}$-comodule isomorphism
$$\theta_{\mathrm{PS}} : P_{A,l}^{\mathrm{PS}} \xrightarrow{\cong} P_{A,r}^{\mathrm{PS}}, \tag{77}$$
where $P_{A,l}^{\mathrm{PS}}$ and $P_{A,r}^{\mathrm{PS}}$ are the two Pastro–Street comodule structures on the common composition object; see [16, §5.3, Thm. 6.1 and §6.1].

   By Proposition 4.18, their common underlying Karoubi object is
$$\big(\mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A), a_A\big), \qquad a_A = j_A p_A,$$
with split image $J(\mathsf{Comp}_A)$. The splitting maps
$$j_A : J(\mathsf{Comp}_A) \longrightarrow \big(\mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A), a_A\big),$$
and
$$p_A : \big(\mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A), a_A\big) \longrightarrow J(\mathsf{Comp}_A)$$
satisfy
$$p_A j_A = 1_{\mathsf{Comp}_A}, \qquad j_A p_A = a_A.$$
Hence j_A and p_A are mutually inverse morphisms in Kar(**Sp**).

3. **Transport the two comodule structures.** Transport the comodule structure of $P_{A,l}^{\mathrm{PS}}$ along j_A, p_A, and denote the resulting $\mathsf{WH}_{\mathrm{sp}}(A)^{\otimes 3}$-comodule on Comp_A by
$$\mathsf{Comp}_{A,l}.$$
Likewise, transport the comodule structure of $P_{A,r}^{\mathrm{PS}}$ and denote the result by
$$\mathsf{Comp}_{A,r}.$$
With these typings, the common underlying span j_A gives comodule isomorphisms
$$\kappa_l := j_A : \mathsf{Comp}_{A,l} \xrightarrow{\cong} P_{A,l}^{\mathrm{PS}}, \tag{78}$$
$$\kappa_r := j_A : \mathsf{Comp}_{A,r} \xrightarrow{\cong} P_{A,r}^{\mathrm{PS}}, \tag{79}$$
whose inverses are both represented by p_A, with the corresponding comodule typings.

4. **Transport the inversion isomorphism.** Define

$$\theta_A := p_A\,\theta_{\mathrm{PS}}\,j_A : \mathsf{Comp}_{A,l} \xrightarrow{\cong} \mathsf{Comp}_{A,r}. \tag{80}$$

   In the symmetric category **Sp**, the defining Pastro–Street composite becomes

$$\theta_A = p_A(1 \otimes S)(1 \otimes m)(c \otimes 1)(1 \otimes \Delta)j_A. \tag{81}$$

   Here c exchanges the first box and the left factor of the second box; multiplication pastes the remaining two boxes vertically, and S inverts their paste. This is the symmetric form of the inversion composite in [16, §6, p. 182].

Proposition 4.22 (Box formula for the transported inversion)**.** *For $(X, Y) \in \mathsf{Comp}_A$, let k be the bottom edge of X, and write $Y = h \urcorner v$. There is a unique horizontal factorization $Y = Y_R \circ_{\mathrm{h}} Y_L$ whose right factor has top edge k. With $Z = Y_R \circ_{\mathrm{v}} X$, the map θ_A is the graph of the bijection*

$$\theta_A(X, Y) = (Y_L, Z^{-1}). \tag{82}$$

The two factors are explicitly

$$Y_R = k \urcorner v, \qquad Y_L = (k^{-1} \circ h) \urcorner \mathsf{L}(k, v).$$

Proof.

1. **Construct the required factorization.** Boundary matching gives $t(k) = t(h)$, so $h = k \circ (k^{-1} \circ h)$. The horizontal factorization clause of Lemma 4.4 gives exactly the displayed Y_L, Y_R. Since the top edge of Y_R is the bottom edge of X, their vertical paste Z is defined.

2. **Read the structural composite.** A witness of (81) first factorizes Y horizontally and then requires its right factor to paste below X. This forces that factor's top edge to be k, so the witness just constructed is unique. The output is (Y_L, Z^{-1}). Its boundaries match because $r_A(Y_L) = \mathrm{sw}(Y_R) = \mathrm{sw}(Z) = \ell_A(Z^{-1})$.

3. **Construct an inverse.** Given $(U, V) \in \mathsf{Comp}_A$, put $Z = V^{-1}$. The right edge of U and the bottom edge of Z form a compatible southwest corner: their common endpoint is $\mathrm{se}(U) = \mathrm{ne}(V) = \mathrm{sw}(Z)$. Vacancy gives the unique box Y_R with that left edge and that bottom edge. Vertical groupoid composition determines the unique X with $Z = Y_R \circ_{\mathrm{v}} X$; put $Y = Y_R \circ_{\mathrm{h}} U$. The pair (X, Y) is in Comp_A, since both relevant boundary labels are $\mathrm{ne}(Y_R)$.

4. **Verify the inverse equations.** Starting from (U, V), the right factor selected in Step 1 is the same Y_R, by its top and right edges. Thus the displayed formula returns (U, V). Starting from (X, Y), the southwest corner in Step 3 reconstructs the original Y_R; cancellation then reconstructs X, and horizontal pasting reconstructs Y. Hence the two constructions are inverse. The equality with the structural composite identifies this bijection with the transported Pastro–Street inversion. □

Proposition 4.23 (Transported inversion datum in spans)**.** *The transported comodules $\mathsf{Comp}_{A,l}$ and $\mathsf{Comp}_{A,r}$ are represented by comodule structures in* **Sp**, *and*

$$\theta_A = p_A\theta_{\mathrm{PS}}j_A : \mathsf{Comp}_{A,l} \xrightarrow{\cong} \mathsf{Comp}_{A,r}$$

is represented by an isomorphism of those **Sp***-comodules.*

The fixed split composition object

$$\mathsf{Comp}_A \xrightarrow{j_A} \mathsf{WH}_{\mathrm{sp}}(A) \times \mathsf{WH}_{\mathrm{sp}}(A) \xrightarrow{p_A} \mathsf{Comp}_A$$

and its splitting maps uniquely determine the transport of the Pastro–Street inversion map θ_{PS}*, namely*

$$\theta_A = p_A \theta_{\mathrm{PS}} j_A.$$

Put

$$\varsigma_A := (\bar{s}_A \otimes 1_{C_A} \otimes \bar{t}_A)(\gamma_r \otimes 1_{\mathsf{WH}_{\mathrm{sp}}(A)}) j_A. \tag{83}$$

Equivalently,

$$\varsigma_A = (\bar{s}_A \otimes 1_{C_A} \otimes \bar{t}_A)(1_{\mathsf{WH}_{\mathrm{sp}}(A)} \otimes \gamma_l) j_A, \tag{84}$$

since j_A *equalizes the two boundary coactions.*

Together with

$$\upsilon_A = 1_{C_A}, \qquad \nu_A = S_A,$$

these data satisfy (G1)–(G3)*. In particular,*

$$(1_{C_A} \otimes 1_{C_A} \otimes \upsilon_A) c_{C_A, C_A \otimes C_A} \varsigma_A = \varsigma_A \theta_A. \tag{85}$$

Proof.

1. **Identify the splitting in the Karoubi completion.** The relations

$$p_A j_A = 1_{\mathsf{Comp}_A}, \qquad j_A p_A = a_A$$

say precisely that j_A and p_A are inverse morphisms between $J(\mathsf{Comp}_A)$ and the Pastro–Street composition idempotent in Kar(**Sp**).

2. **Transport the two comodule structures.** Transport of a comodule structure along an isomorphism is uniquely determined by the fixed inverse pair j_A, p_A. Hence the two Pastro–Street comodule structures transport uniquely to

$$\mathsf{Comp}_{A,l}, \qquad \mathsf{Comp}_{A,r},$$

and j_A, with its two respective typings, becomes the comodule isomorphisms

$$\kappa_l : \mathsf{Comp}_{A,l} \xrightarrow{\cong} P^{\mathrm{PS}}_{A,l}, \qquad \kappa_r : \mathsf{Comp}_{A,r} \xrightarrow{\cong} P^{\mathrm{PS}}_{A,r}.$$

3. **Transport the Pastro–Street inversion map.** The Pastro–Street morphism

$$\theta_{\mathrm{PS}} : P^{\mathrm{PS}}_{A,l} \xrightarrow{\cong} P^{\mathrm{PS}}_{A,r}$$

is a comodule isomorphism. Therefore its conjugate

$$\theta_A = p_A \theta_{\mathrm{PS}} j_A$$

is a comodule morphism

$$\mathsf{Comp}_{A,l} \longrightarrow \mathsf{Comp}_{A,r}.$$

Moreover,

$$\begin{aligned} j_A\theta_A &= j_A p_A \theta_{\mathrm{PS}} j_A \\ &= a_A \theta_{\mathrm{PS}} j_A \\ &= \theta_{\mathrm{PS}} j_A, \end{aligned} \tag{86}$$

because θ_{PS} is a morphism between the corresponding Karoubi objects and hence is fixed by the relevant composition idempotents.

4. **Construct the inverse of the transported map.** Define

$$\theta_A^{-1} := p_A \theta_{\mathrm{PS}}^{-1} j_A.$$

Using $j_A p_A = a_A$, together with the fact that θ_{PS} is a morphism of the corresponding Karoubi objects, one obtains

$$\begin{aligned} \theta_A^{-1}\theta_A &= \big(p_A \theta_{\mathrm{PS}}^{-1} j_A\big)\big(p_A \theta_{\mathrm{PS}} j_A\big) \\ &= p_A \theta_{\mathrm{PS}}^{-1} a_A \theta_{\mathrm{PS}} j_A \\ &= p_A j_A = 1_{\mathsf{Comp}_A}. \end{aligned}$$

The same calculation in the opposite order gives

$$\theta_A \theta_A^{-1} = 1_{\mathsf{Comp}_A}.$$

Hence θ_A is a comodule isomorphism.

5. **Reflect the transported data to spans.** The source and target of every transported coaction and of θ_A lie in the image of the fully faithful embedding

$$J : \mathbf{Sp} \longrightarrow \mathrm{Kar}(\mathbf{Sp}).$$

Therefore each transported morphism is represented uniquely by a morphism in **Sp**.

6. **Identify the transported ς-map.** Let

$$\varsigma_{\mathrm{PS}}$$

denote the map appearing in the Pastro–Street (G3) equation before splitting the target and composition idempotents. By definition, ς_{PS} is obtained from the source and target maps and either of the two boundary coactions on the composition object.

Transport the target object through

$$C_A \xrightarrow{i_s} \big(\mathsf{WH}_{\mathrm{sp}}(A), t_A^{\mathrm{PS}}\big) \xrightarrow{r_s} C_A$$

and the composition object through j_A, p_A. Since the transported base-valued source and target maps are

$$\bar{s}_A = r_t, \qquad \bar{t}_A = r_s,$$

and the transported boundary coactions are γ_l, γ_r, the conjugate of ς_{PS} is precisely

$$\varsigma_A = (\bar{s}_A \otimes 1_{C_A} \otimes \bar{t}_A)(\gamma_r \otimes 1) j_A.$$

Because j_A is the cotensor equalizer,

$$(\gamma_r \otimes 1)j_A = (1 \otimes \gamma_l)j_A,$$

which also gives

$$\varsigma_A = (\bar{s}_A \otimes 1_{C_A} \otimes \bar{t}_A)(1 \otimes \gamma_l)j_A.$$

Equivalently, in Kar(**Sp**),

$$\varsigma_{\mathrm{PS}} j_A = (i_s \otimes i_s \otimes i_s)\varsigma_A. \tag{87}$$

Indeed, the left-hand side lands in the tensor cube of the target retract, while

$$i_s r_s = t_A^{\mathrm{PS}}.$$

7. **Verify (G1)–(G2).** By Proposition 4.12,

$$\bar{s}_A \nu_A = r_t S_A = r_s = \bar{t}_A,$$

giving (G1). Likewise,

$$\bar{t}_A \nu_A = r_s S_A = r_t = \upsilon_A \bar{s}_A,$$

since

$$\upsilon_A = 1_{C_A}.$$

This gives (G2).

8. **Verify (G3) explicitly.** Put

$$I_A := i_s \otimes i_s \otimes i_s.$$

Since

$$r_s i_s = 1_{C_A},$$

the morphism I_A is split monic.

By the transport of υ_{PS} to υ_A, naturality of the braiding, and (87),

$$\begin{aligned} I_A(1_{C_A} \otimes 1_{C_A} \otimes \upsilon_A)c_{C_A, C_A \otimes C_A}\varsigma_A \\ = (1 \otimes 1 \otimes \upsilon_{\mathrm{PS}})c\,\varsigma_{\mathrm{PS}} j_A. \end{aligned}$$

The Pastro–Street (G3) equation gives

$$(1 \otimes 1 \otimes \upsilon_{\mathrm{PS}})c\,\varsigma_{\mathrm{PS}} j_A = \varsigma_{\mathrm{PS}} \theta_{\mathrm{PS}} j_A.$$

Using (86) and then (87),

$$\begin{aligned} \varsigma_{\mathrm{PS}} \theta_{\mathrm{PS}} j_A &= \varsigma_{\mathrm{PS}} j_A \theta_A \\ &= I_A \varsigma_A \theta_A. \end{aligned}$$

Hence

$$I_A(1_{C_A} \otimes 1_{C_A} \otimes \upsilon_A)c_{C_A, C_A \otimes C_A}\varsigma_A = I_A \varsigma_A \theta_A.$$

Since I_A is split monic, cancellation gives

$$(1_{C_A} \otimes 1_{C_A} \otimes \upsilon_A)c_{C_A, C_A \otimes C_A}\varsigma_A = \varsigma_A \theta_A.$$

This is precisely (G3).

9. **Conclude the construction in Sp.** Thus the transported comodule structures

$$\mathsf{Comp}_{A,l}, \qquad \mathsf{Comp}_{A,r},$$

the isomorphism

$$\theta_A : \mathsf{Comp}_{A,l} \xrightarrow{\cong} \mathsf{Comp}_{A,r},$$

and the map ς_A are represented in **Sp**. Together with

$$\upsilon_A = 1_{C_A}, \qquad \nu_A = S_A,$$

they satisfy (G1)–(G3). □

Definition 4.24 (Quantum groupoid associated to a thin game)**.** Let $\mathsf{QG}_{\mathrm{sp}}(A)$ be the Day–Street quantum groupoid with object-of-objects C_A, arrow object $\mathsf{WH}_{\mathrm{sp}}(A)$, boundary coactions γ_l, γ_r, composition object Comp_A, composition μ_A, unit u_A, and inversion data

$$\upsilon_A : C_A^{\circ\circ} \to C_A, \qquad \nu_A : \mathsf{WH}_{\mathrm{sp}}(A)^\circ \to \mathsf{WH}_{\mathrm{sp}}(A), \qquad \theta_A : \mathsf{Comp}_{A,l} \xrightarrow{\cong} \mathsf{Comp}_{A,r}.$$

Theorem 4.25 (Canonical quantum groupoid)**.** *Every thin concurrent game canonically determines* $\mathsf{QG}_{\mathrm{sp}}(A)$*. Its object-of-objects is the configuration set with its canonical separable commutative Frobenius structure, and its arrow inversion is total box inversion.*

Proof.

1. **Pass from symmetry to boxes.** Polarized decomposition gives the vacant double groupoid.
2. **Construct the weak Hopf monoid.** Theorem 4.9 gives the incidence weak Hopf monoid.
3. **Identify the Pastro–Street base and composition object.** Propositions 4.13 and 4.18 identify the target base and cotensor object, and Proposition 4.19 supplies the quantum category.
4. **Transport inversion.** Equation (76) gives the object and arrow inversion maps, and Proposition 4.23 supplies θ_A and verifies (G1)–(G3) in **Sp**. □

4.6 Tensor and negation

Here tensor product of quantum groupoids is taken componentwise, and $\cong$ denotes an isomorphism of the complete quantum-groupoid data.

Theorem 4.26 (Monoidal compatibility)**.** *For thin games A, B,*

$$\mathsf{WH}_{\mathrm{sp}}(A \otimes B) \cong \mathsf{WH}_{\mathrm{sp}}(A) \otimes \mathsf{WH}_{\mathrm{sp}}(B), \tag{88}$$
$$\mathsf{QG}_{\mathrm{sp}}(A \otimes B) \cong \mathsf{QG}_{\mathrm{sp}}(A) \otimes \mathsf{QG}_{\mathrm{sp}}(B). \tag{89}$$

Proof.

1. **Factor boxes componentwise.** By Proposition 4.5,

$$T(A \otimes B) \cong T(A) \times T(B).$$

Hence a polarized box of $A \otimes B$ is canonically a pair consisting of one polarized box of A and one polarized box of B.

2. **Factor the incidence witness objects.** Vertical composability, horizontal factorization, vertical and horizontal identity boxes, and total box inversion are all computed componentwise under this identification. Therefore the witness objects defining

$$m_{A\otimes B}, \qquad \eta_{A\otimes B}, \qquad \Delta_{A\otimes B}, \qquad \varepsilon_{A\otimes B}, \qquad S_{A\otimes B}$$

are Cartesian products of the corresponding witness objects for A and B.

3. **Identify the weak Hopf structure.** Since the tensor product in **Sp** is Cartesian product, the preceding componentwise identifications give

$$\mathsf{WH}_{\mathrm{sp}}(A\otimes B) \cong \mathsf{WH}_{\mathrm{sp}}(A) \otimes \mathsf{WH}_{\mathrm{sp}}(B)$$

as weak Hopf monoids. This proves (88).

4. **Factor the configuration base.** By (3),

$$\mathsf{Conf}(A\otimes B) \cong \mathsf{Conf}(A) \times \mathsf{Conf}(B),$$

and the canonical Frobenius structure on this product is the tensor product of the canonical Frobenius structures on $\mathsf{Conf}(A)$ and $\mathsf{Conf}(B)$. Hence

$$C_{A\otimes B} \cong C_A \otimes C_B.$$

5. **Factor the boundary and composition data.** The boundary maps, boundary coactions, split composition object, composition, and unit are all obtained componentwise. In particular,

$$\mathsf{Comp}_{A\otimes B} \cong \mathsf{Comp}_A \times \mathsf{Comp}_B,$$

since the boundary-matching equation for $A\otimes B$ is precisely the pair of boundary-matching equations for A and B.

6. **Factor the inversion data.** The target splitting, arrow inversion, transported composition comodules, and transported Pastro–Street map θ are likewise products of the corresponding structures for A and B.

7. **Conclude monoidal compatibility.** Thus every datum of the game quantum groupoid is computed componentwise under tensor product, giving

$$\mathsf{QG}_{\mathrm{sp}}(A\otimes B) \cong \mathsf{QG}_{\mathrm{sp}}(A) \otimes \mathsf{QG}_{\mathrm{sp}}(B).$$

This proves (89). □

Proposition 4.27 (Compact self-duality of spans)**.** *Every set X is canonically self-dual in* **Sp***, with evaluation and coevaluation*

$$\begin{tikzcd} & X \arrow[dl, "\Delta_X"'] \arrow[dr] & & & X \arrow[dl] \arrow[dr, "\Delta_X"] & \\ X\times X & & 1 & 1 & & X\times X. \end{tikzcd}$$

Hence **Sp** *is compact closed.*

Proof.

1. **Compute the first snake composite.** Compose the coevaluation and evaluation spans in the first snake identity. The defining pullbacks impose equality of the repeated X-coordinates, so the resulting apex is canonically X.

2. **Identify the first exterior legs.** Under this canonical apex identification, both exterior legs are 1_X. Hence the first snake composite is the identity span on X.

3. **Compute the second snake composite.** The reflected pullback calculation again has apex canonically X, with both exterior legs equal to 1_X. Hence the second snake composite is also the identity span.

4. **Conclude compact self-duality.** The displayed evaluation and coevaluation therefore exhibit X as its own left and right dual. Since this construction is available for every set, **Sp** is compact closed. ☐

If

$$K = (K, m_K, \eta_K, \Delta_K, \varepsilon_K, S_K)$$

is a weak Hopf monoid in **Sp**, write $K^\vee$ for the compactly transposed weak Hopf structure

$$\begin{aligned} m_{K^\vee} &:= \Delta_K^\vee, & \eta_{K^\vee} &:= \varepsilon_K^\vee, \\ \Delta_{K^\vee} &:= m_K^\vee, & \varepsilon_{K^\vee} &:= \eta_K^\vee, \\ S_{K^\vee} &:= S_K^\vee. \end{aligned}$$

If Q is the Pastro–Street quantum groupoid obtained from a weak Hopf monoid K in **Sp**, write

$$Q^\vee$$

for the Pastro–Street quantum groupoid obtained from the compactly transposed weak Hopf monoid $K^\vee$ in $\mathrm{Kar}(\mathbf{Sp})$.

Thus the target object, the two boundary coactions, the composition idempotent, and the inversion data of $Q^\vee$ are those determined by the weak Hopf structure of $K^\vee$ itself. When these target and composition idempotents are equipped with explicit splittings in **Sp**, we use those splittings to present $Q^\vee$ in **Sp**.

Lemma 4.28 (Weak Hopf structure under compact transposition)**.** *Let*

$$K = (K, m, \eta, \Delta, \varepsilon, S)$$

be a weak Hopf monoid in **Sp**. *Under the canonical compact self-duality of* **Sp***, the transposed structure*

$$K^\vee := \big(K, \Delta^\vee, \varepsilon^\vee, m^\vee, \eta^\vee, S^\vee\big)$$

is again a weak Hopf monoid.

If $S^2 = 1_K$, then

$$(S^\vee)^2 = 1_K.$$

Moreover,

$$(K^\vee)^\vee \cong K$$

canonically as weak Hopf monoids.

Proof.

1. **Fix the tensor-order convention for transposition.** Under the canonical compact self-duality of Proposition 4.27, every object X is identified with its dual. For a span morphism

$$f : X \longrightarrow Y,$$

write

$$f^\vee : Y \longrightarrow X$$

for its compact transpose.

Compact transposition is contravariant:

$$(g \circ f)^\vee = f^\vee \circ g^\vee, \qquad 1_X^\vee = 1_X.$$

The canonical compact dual of a tensor product reverses tensor order. Throughout this lemma we compose that canonical identification with the symmetry of **Sp**, and therefore use the coherent identification

$$(X \otimes Y)^\vee \cong X^\vee \otimes Y^\vee.$$

With this convention,

$$(f \otimes g)^\vee \cong f^\vee \otimes g^\vee,$$

and the symmetry is fixed by transposition.

2. **Transpose the monoid and comonoid structures.** Since

$$\Delta : K \longrightarrow K \otimes K,$$

its transpose has type

$$\Delta^\vee : K \otimes K \longrightarrow K$$

under the preceding tensor-order identification. Likewise,

$$\varepsilon^\vee : I \longrightarrow K.$$

Thus

$$m_{K^\vee} := \Delta^\vee, \qquad \eta_{K^\vee} := \varepsilon^\vee$$

have the types of multiplication and unit.

Similarly,

$$m^\vee : K \longrightarrow K \otimes K, \qquad \eta^\vee : K \longrightarrow I,$$

so

$$\Delta_{K^\vee} := m^\vee, \qquad \varepsilon_{K^\vee} := \eta^\vee$$

have the types of comultiplication and counit.

3. **Transpose associativity, coassociativity, unit, and counit.** Transposing the coassociativity equation for Δ gives the associativity equation for $\Delta^\vee$. Transposing the counit equations for ε gives the two unit equations for $\varepsilon^\vee$.

Conversely, transposing associativity of m gives coassociativity of $m^\vee$, and transposing the unit equations for η gives the counit equations for $\eta^\vee$. Hence the displayed maps define a monoid and a comonoid on $K^\vee$.

4. **Transpose multiplicativity.** The multiplicativity equation

$$\Delta m = (m \otimes m)(1 \otimes c \otimes 1)(\Delta \otimes \Delta)$$

transposes to

$$m^\vee \Delta^\vee = (\Delta^\vee \otimes \Delta^\vee)(1 \otimes c \otimes 1)(m^\vee \otimes m^\vee),$$

after the coherent tensor-order identifications of Step 1. This is precisely the multiplicativity equation for the multiplication $\Delta^\vee$ and comultiplication $m^\vee$ of $K^\vee$.

5. **Transpose the weak unit and weak counit equations.** Compact transposition exchanges the weak-unit and weak-counit equations (45)–(46). The symmetry occurring in those equations is preserved by transposition under the convention of Step 1. Hence

$$\left(K, \Delta^\vee, \varepsilon^\vee, m^\vee, \eta^\vee\right)$$

is a weak bimonoid.

6. **Transpose the counital antipode equations.** Transposing the defining composites of the canonical counital maps gives

$$\varepsilon_t^{K^\vee} = (\varepsilon_t^K)^\vee, \qquad \varepsilon_s^{K^\vee} = (\varepsilon_s^K)^\vee.$$

Hence the two equations

$$m(1 \otimes S)\Delta = \varepsilon_t, \qquad m(S \otimes 1)\Delta = \varepsilon_s$$

transpose to the corresponding counital convolution equations for $S^\vee$ in the transposed weak bimonoid. Thus the first two antipode equations hold for $S^\vee$.

7. **Transpose the third antipode equation.** Transposing

$$m^{(3)}(S \otimes 1 \otimes S)\Delta^{(3)} = S$$

gives

$$m^{(3)}_{K^\vee}(S^\vee \otimes 1 \otimes S^\vee)\Delta^{(3)}_{K^\vee} = S^\vee.$$

Hence $S^\vee$ is an antipode and $K^\vee$ is a weak Hopf monoid.

8. **Transpose involutivity.** If

$$S^2 = 1_K,$$

then

$$1_K = (S^2)^\vee = S^\vee S^\vee,$$

so

$$(S^\vee)^2 = 1_K.$$

9. **Transpose twice.** The two snake identities identify the double compact transpose of every span with the original span. These identifications are compatible with tensor product and symmetry, so they identify every structure map of $(K^\vee)^\vee$ with the corresponding structure map of K. Therefore

$$(K^\vee)^\vee \cong K$$

canonically as weak Hopf monoids. □

Theorem 4.29 (Negation–duality)**.** *For every thin game A,*

$$\mathsf{WH}_{\mathrm{sp}}(A^{\perp}) \cong \mathsf{WH}_{\mathrm{sp}}(A)^{\vee}, \tag{90}$$

$$\mathsf{QG}_{\mathrm{sp}}(A^{\perp}) \cong \mathsf{QG}_{\mathrm{sp}}(A)^{\vee}. \tag{91}$$

Proof.

1. **Calculate the compact transpose of a span.** Let

   $$X \xleftarrow{\ell} S \xrightarrow{r} Y$$

   represent a morphism in **Sp**. Composing with the canonical diagonal coevaluation and evaluation spans leaves the apex canonically S, with the two exterior legs exchanged. Hence its compact transpose is represented by

   $$Y \xleftarrow{r} S \xrightarrow{\ell} X.$$

2. **Transpose the polarized double groupoid.** By Proposition 4.6,

   $$T(A^{\perp}) \cong T(A)^{\mathsf{t}}.$$

   Under this identification, the horizontal and vertical edge groupoids are exchanged. Consequently vertical box composition for $A^{\perp}$ is horizontal box composition for A, while horizontal box factorization for $A^{\perp}$ is vertical box factorization for A.

3. **Transpose the incidence structure maps.** Since compact transposition of a span exchanges its two legs, the preceding identification gives

   $$m_{A^{\perp}} \cong \Delta_A^{\vee}, \qquad \eta_{A^{\perp}} \cong \varepsilon_A^{\vee},$$

   and

   $$\Delta_{A^{\perp}} \cong m_A^{\vee}, \qquad \varepsilon_{A^{\perp}} \cong \eta_A^{\vee}.$$

   Total box inversion commutes with transposition:

   $$(X^{\mathsf{t}})^{-1} = (X^{-1})^{\mathsf{t}},$$

   so

   $$S_{A^{\perp}} \cong S_A^{\vee}.$$

   By Lemma 4.28, these identifications assemble to an isomorphism of weak Hopf monoids

   $$\mathsf{WH}_{\mathrm{sp}}(A^{\perp}) \cong \mathsf{WH}_{\mathrm{sp}}(A)^{\vee}.$$

   This proves (90).

4. **Apply the Pastro–Street construction to the transposed weak Hopf monoid.** By definition,

   $$\mathsf{QG}_{\mathrm{sp}}(A)^{\vee}$$

   is the Pastro–Street quantum groupoid obtained from

   $$\mathsf{WH}_{\mathrm{sp}}(A)^{\vee}.$$

   The weak-Hopf isomorphism of Step 3 therefore identifies this Pastro–Street quantum groupoid in $\mathrm{Kar}(\mathbf{Sp})$ with the one obtained from

   $$\mathsf{WH}_{\mathrm{sp}}(A^{\perp}).$$

5. **Identify the target base.** The Pastro–Street target idempotent is determined functorially by the weak-bimonoid structure. Hence the weak-Hopf isomorphism of Step 3 carries the target idempotent of $\mathsf{WH}_{\mathrm{sp}}(A^{\perp})$ to the target idempotent of $\mathsf{WH}_{\mathrm{sp}}(A)^{\vee}$.

   For $A^{\perp}$, the explicit splitting constructed above is

$$\mathsf{Conf}(A^{\perp}) \xrightarrow{i_s} \mathsf{WH}_{\mathrm{sp}}(A^{\perp}) \xrightarrow{r_s} \mathsf{Conf}(A^{\perp}).$$

   Since

$$\mathsf{Conf}(A^{\perp}) = \mathsf{Conf}(A),$$

   Proposition 4.13 identifies the corresponding object-of-objects with the same canonical separable commutative Frobenius monoid

$$C_A.$$

6. **Identify the composition object from the transposed coactions.** The Pastro–Street composition object for $\mathsf{WH}_{\mathrm{sp}}(A)^{\vee}$ is, by definition, the split equalizer of the two boundary coactions determined by that transposed weak Hopf structure.

   Under the weak-Hopf isomorphism of Step 3 these are precisely the boundary coactions of $\mathsf{WH}_{\mathrm{sp}}(A^{\perp})$. Therefore Lemma 4.17, applied to $A^{\perp}$, supplies the explicit split representative

$$\mathsf{Comp}_{A^{\perp}} = \left\{(X,Y) \in \mathsf{WH}_{\mathrm{sp}}(A^{\perp})^2 \;\middle|\; r_{A^{\perp}}(X) = \ell_{A^{\perp}}(Y)\right\}.$$

   Its inclusion and converse split the composition idempotent calculated from the transposed weak Hopf structure.

7. **Identify composition and unit.** On this split composition object, the Pastro–Street composition is the restriction of the multiplication of $\mathsf{WH}_{\mathrm{sp}}(A^{\perp})$, namely

$$\mu_{A^{\perp}} = m_{A^{\perp}} j_{A^{\perp}},$$

   and the unit is the target-base inclusion

$$u_{A^{\perp}} = i_s.$$

   Under the weak-Hopf isomorphism of Step 3 these are exactly the composition and unit obtained by applying the Pastro–Street construction to $\mathsf{WH}_{\mathrm{sp}}(A)^{\vee}$.

8. **Transport the inversion data.** The Pastro–Street inversion maps are constructed from the weak Hopf antipode, the target idempotent, and the two comodule structures on the Pastro–Street composition object. The weak-Hopf isomorphism of Step 3 therefore transports the inversion data of $\mathsf{WH}_{\mathrm{sp}}(A)^{\vee}$ to those of $\mathsf{WH}_{\mathrm{sp}}(A^{\perp})$.

   In the explicit split presentation for $A^{\perp}$, the arrow inversion is total box inversion and the composition-comodule isomorphism is the transported map

$$\theta_{A^{\perp}} : \mathsf{Comp}_{A^{\perp},l} \xrightarrow{\cong} \mathsf{Comp}_{A^{\perp},r}.$$

9. **Conclude quantum-groupoid duality.** Thus the Pastro–Street quantum groupoid associated to the compactly transposed weak Hopf monoid $\mathsf{WH}_{\mathrm{sp}}(A)^{\vee}$ is presented in **Sp** by the explicit game quantum groupoid of $A^{\perp}$. Hence

$$\mathsf{QG}_{\mathrm{sp}}(A^{\perp}) \cong \mathsf{QG}_{\mathrm{sp}}(A)^{\vee}.$$

   This proves (91). □

5 Thin strategies as quantum modules

We construct the module associated to a distributor using Definition 3.10. The configuration Frobenius bases express boundary coactions as graphs, and the module equations become identifications of spans.

5.1 Configuration-boundary span calculus

Theorem 5.1 (Configuration comodules and boundary spans)**.** *There is a symmetric monoidal bicategory* **BndSp** *whose objects are sets and whose* 1*-cells* $P \rightsquigarrow Q$ *are diagrams*

$$\begin{tikzcd} & X \arrow[dl, "\ell_X"'] \arrow[dr, "r_X"] & \\ P & & Q. \end{tikzcd}$$

The identity 1*-cell on* P *is*

$$P \xleftarrow{1_P} P \xrightarrow{1_P} P.$$

Its 2*-cells are isomorphism classes of spans* $X \xleftarrow{u} S \xrightarrow{v} Y$ *satisfying*

$$\ell_X u = \ell_Y v, \qquad r_X u = r_Y v.$$

Horizontal composition of 1*-cells is boundary pullback. Vertical composition of* 2*-cells is pullback over the intermediate carrier, horizontal composition is induced by pullback over the common boundary, and identity* 2*-cells are identity spans. The assignment* $P \mapsto C_P$*, with graph coactions on* 1*-cells, gives a symmetric monoidal biequivalence with the configuration-base fragment of* **Comod**(**Sp**)*. For* $X : P \rightsquigarrow Q$ *and* $Y : Q \rightsquigarrow R$*, the cotensor inclusion is the graph*

$$j : X \times_Q Y \hookrightarrow X \times Y$$

and is a split absolute equalizer.

Proof.

1. **Identify coactions and 2-cells.** Proposition 4.15 identifies the coactions with the graphs of ℓ_X, r_X. Intertwining them is precisely preservation of the two boundary functions by the legs of a 2-span. The stated pullback formulas for vertical and horizontal composition agree with composition of comodule maps; associativity, units, and interchange are the canonical comparison isomorphisms for iterated pullbacks.

2. **Construct the cotensor.** Put $K = X \times_Q Y$, with its defining pullback

$$\begin{tikzcd} K \arrow[r, "\pi_Y"] \arrow[d, "\pi_X"'] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "\ell_Y"] \\ X \arrow[r, "r_X"'] & Q. \end{tikzcd}$$

   The two cotensor maps are the graphs

$$f(x, y) = (x, r_X(x), y), \qquad g(x, y) = (x, \ell_Y(y), y).$$

Let $p : X \times Y \to K$ be the converse of j. Define $h : X \times Q \times Y \to X \times Y$ to be the partial span supported on $q = r_X(x)$, with value (x, y). Then

$$pj = 1_K, \qquad hf = 1_{X\times Y}, \qquad hg = jp.$$

If a span $z : Z \to X \times Y$ satisfies $fz = gz$, these equations give $z = hfz = hgz = jpz$. Thus pz factors z through j, uniquely because $pj = 1_K$.

3. **Exhibit the splitting.** The preceding equations display the cotensor as the split equalizer

$$\begin{tikzcd} K \arrow[r, bend left, "j"] & X \times Y \arrow[l, bend left, "p"] \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & X \times Q \times Y. \arrow[l, bend left, "h"] \end{tikzcd}$$

Split equalizers are absolute, so tensoring preserves this cotensor.

4. **Construct the bicategorical and tensor comparisons.** Iterated composites are simultaneous boundary pullbacks. Rebracketing their tuples gives the associators, and removing a repeated boundary gives the unitors. Cartesian product gives $C_P \otimes C_{P'} \cong C_{P\times P'}$ and the bijection

$$(X \times_Q Y) \times (X' \times_{Q'} Y') \longrightarrow (X \times X') \times_{Q\times Q'} (Y \times Y'),$$
$$((x, y), (x', y')) \longmapsto ((x, x'), (y, y')).$$

Each coherence equation identifies the same rebracketing or permutation of tuples, giving the symmetric monoidal biequivalence. □

We calculate in **BndSp**. For sets P, Q, write $E_{P,Q} = P \times Q$. The configuration biduality is represented by

$$e_P : \; P^2 \xleftarrow{\Delta_P} P \longrightarrow 1, \qquad n_P : \; 1 \longleftarrow P \xrightarrow{\Delta_P} P^2.$$

The snake composites identify the repeated labels and give identity spans. Put

$$L_{P,Q} = E_{P,P} \times E_{P,Q} \times E_{Q,Q}.$$

The canonical action has boundary span

$$\begin{tikzcd} & P \times P \times Q \times Q \arrow[dl, "\iota_{\mathsf{act}}"'] \arrow[dr, "\tau_{\mathsf{act}}"] & \\ L_{P,Q} & & E_{P,Q}, \end{tikzcd}$$

where

$$\iota_{\mathsf{act}}(a', a, b, b') = ((a', a), (a, b), (b, b')), \tag{92}$$
$$\tau_{\mathsf{act}}(a', a, b, b') = (a', b'). \tag{93}$$

This is $\mathsf{act}_{P,Q}$ from (24).

Proposition 5.2 (Configuration-base right-mate quantum modules). *For canonical configuration bases $C_{\mathsf{Conf}(A)}, C_{\mathsf{Conf}(B)}$, Chikhladze's formal module data are equivalently represented in* **BndSp** *by an endocomonad M on $E_{\mathsf{Conf}(A),\mathsf{Conf}(B)} = \mathsf{Conf}(A) \times \mathsf{Conf}(B)$ and an action*

$$\kappa_M : \mathsf{act}_{\mathsf{Conf}(A),\mathsf{Conf}(B)}(A_1 \otimes M \otimes B_1) \Longrightarrow M\mathsf{act}_{\mathsf{Conf}(A),\mathsf{Conf}(B)}$$

satisfying (26)–(29). *Module maps correspond to the right-mate comonad transformations intertwining these actions.*

Proof.

1. **Transport the comonad data.** Proposition 4.15 and Theorem 5.1 identify the relevant configuration-base comodule hom-categories with boundary-span hom-categories. The mutually inverse mate operations preserve identities and pullback composition by the snake identities, hence transport comonads and their transformations.

2. **Transport the action.** Under the same correspondence the bidual evaluations give the canonical boundary action (93). Thus the action cell, its four equations, and the intertwining equation for module maps are transported bijectively. □

5.2 The distributor comonad and the direct action

Let $F : \mathsf{Sym}(A)^{\mathrm{op}} \times \mathsf{Sym}(B) \to \mathbf{Set}$ be a distributor. Set

$$W_F = \coprod_{a \in \mathsf{Conf}(A),\, b \in \mathsf{Conf}(B)} F(a,b), \qquad d_F(w) = (a,b) \quad (w \in F(a,b)). \tag{94}$$

For $\Omega_A : a' \to a$, $\Omega_B : b \to b'$, write

$$\Omega_{B*}\Omega_A^* w = F(\Omega_A, \Omega_B)(w).$$

The endpoint transports commute and are functorial.

Proposition 5.3 (Diagonal distributor comonad)**.** *The endo-*1*-cell*

$$\begin{tikzcd} & W_F \arrow[dl, "d_F"'] \arrow[dr, "d_F"] & \\ E_{\mathsf{Conf}(A),\mathsf{Conf}(B)} & & E_{\mathsf{Conf}(A),\mathsf{Conf}(B)} \end{tikzcd}$$

is a comonad M_F, with comultiplication $w \mapsto (w,w)$ and counit $w \mapsto d_F(w)$. Its carrier comonoid in **Sp** *is the diagonal comonoid on W_F.*

Proof.

1. **Construct the comonad cells.** The diagonal lands in $W_F \times_{\mathsf{Conf}(A)\times\mathsf{Conf}(B)} W_F$ and preserves the two boundary maps. The graph of d_F is a 2-cell from M_F to the identity 1-cell on $E_{\mathsf{Conf}(A),\mathsf{Conf}(B)}$.

2. **Verify the equations.** Both iterated comultiplications send w to (w,w,w). Either counit removes one repeated copy and returns w. Composing with the cotensor inclusion gives the carrier diagonal; composing the boundary counit with the terminal map gives $W_F \to 1$. □

By Proposition 4.16, a box X of A has endocomonad boundaries

$$d_A^L(X) = (\mathrm{nw}(X), \mathrm{sw}(X)), \qquad d_A^R(X) = (\mathrm{ne}(X), \mathrm{se}(X)).$$

Put $\mathsf{act} = \mathsf{act}_{\mathsf{Conf}(A),\mathsf{Conf}(B)}$ and $D_F = \mathsf{WH}_{\mathrm{sp}}(A) \otimes M_F \otimes \mathsf{WH}_{\mathrm{sp}}(B)$. The carrier of $\mathsf{act}D_F$ is

$$K_F = \{(X, w, Y) : w \in F(a,b),\ \mathrm{se}(X) = a,\ \mathrm{ne}(Y) = b\}, \tag{95}$$

with boundary maps

$$\begin{aligned}
d^L_{K_F}(X,w,Y) &= ((\mathrm{nw}(X),\mathrm{sw}(X)),(a,b),(\mathrm{nw}(Y),\mathrm{sw}(Y))),\\
d^R_{K_F}(X,w,Y) &= (\mathrm{ne}(X),\mathrm{se}(Y)).
\end{aligned} \tag{96}$$

The carrier of $M_F\mathsf{act}$ is

$$J_F = \mathsf{Conf}(A)\times\mathsf{Conf}(B)\times W_F.$$

For $w'\in F(a',b')$, its boundaries are

$$\begin{aligned}
d^L_{J_F}(a,b,w') &= ((a',a),(a,b),(b,b')),\\
d^R_{J_F}(a,b,w') &= (a',b').
\end{aligned} \tag{97}$$

Both descriptions follow by composing with (93); the labels a,b are the contraction witnesses in J_F.

Define the action apex

$$\mathrm{Act}_F = \left\{ (v,w,u) \;\middle|\; \begin{array}{l} v: a'\to a \text{ in } \mathsf{Sym}^+(A),\\ w\in F(a,b),\\ u: b\to b' \text{ in } \mathsf{Sym}^+(B) \end{array} \right\}. \tag{98}$$

The direct action is the span

$$\begin{tikzcd}
 & \mathrm{Act}_F \arrow[dl, "\iota_F"'] \arrow[dr, "\tau_F"] & \\
K_F & & J_F,
\end{tikzcd} \qquad \kappa_F = [\iota_F,\tau_F], \tag{99}$$

with legs

$$\iota_F(v,w,u) = (\mathrm{id}^{\mathrm{h}}_v, w, \mathrm{id}^{\mathrm{h}}_u), \qquad \tau_F(v,w,u) = (a,b,u_*v^*w). \tag{100}$$

Its two source boundaries are $((a',a),(a,b),(b,b'))$, and its two target boundaries are (a',b'). Thus it defines $\kappa_F : \mathsf{act}D_F \Rightarrow M_F\mathsf{act}$. The endpoint transport appears as

$$\begin{tikzcd}
F(a,b) \arrow[r, "v^*"] \arrow[d, "u_*"'] & F(a',b) \arrow[d, "u_*"] \\
F(a,b') \arrow[r, "v^*"'] & F(a',b').
\end{tikzcd}$$

Proposition 5.4 (The direct action and its multitensor mate)**.** *The multitensor* $T_3(\mathsf{WH}_{\mathrm{sp}}(A), M_F, \mathsf{WH}_{\mathrm{sp}}(B))$ *has carrier*

$$\{(X,w,Y) : \mathrm{sw}(X)=\mathrm{se}(X)=a(w),\ \mathrm{nw}(Y)=\mathrm{ne}(Y)=b(w)\}.$$

The Kan-extension mate of κ_F *has apex* Act_F *and legs*

$$(v,w,u)\longmapsto (\mathrm{id}^{\mathrm{h}}_v, w, \mathrm{id}^{\mathrm{h}}_u), \qquad (v,w,u)\longmapsto u_*v^*w.$$

Proof.

1. **Identify the matching locus.** Equality of the source boundaries of K_F and J_F imposes $\mathrm{sw}(X)=a$ and $\mathrm{nw}(Y)=b$, in addition to $\mathrm{se}(X)=a$ and $\mathrm{ne}(Y)=b$. These are exactly the two-boundary equations of Proposition 3.12.

2. **Construct the correspondence of 2-cells.** On this locus, a boundary-preserving span to $M_F\mathsf{act}$ determines a span to M_F by forgetting the uniquely determined contraction labels a, b. Inserting those labels is inverse to this operation. Applying this correspondence to (100) gives the displayed mate. $\square$

5.3 The quantum-module equations

Theorem 5.5 (Distributor-induced quantum module). *Every distributor* $F : \mathsf{Sym}(A)^{\mathrm{op}} \times \mathsf{Sym}(B) \to$ **Set** *canonically determines a Chikhladze quantum module*

$$Q_F^{\mathrm{sp}} : \mathsf{QG}_{\mathrm{sp}}(A) \nrightarrow \mathsf{QG}_{\mathrm{sp}}(B),$$

whose right-mate presentation has comonad M_F *and direct action* κ_F*.*

Proof.

1. **Verify the counit equation.** The comonad counit equation is

$$\begin{tikzcd} \mathsf{act}D_F \arrow[r, "\kappa_F"] \arrow[d, "\mathsf{act}\epsilon_{D_F}"'] & M_F\mathsf{act} \arrow[dl, "\epsilon_{M_F}\mathsf{act}"] \\ \mathsf{act}. & \end{tikzcd}$$

On the direct route the two arrow counits select horizontal identity boxes, and the distributor counit is defined on every w. Its apex is therefore Act_F. The route through κ_F has the same apex. Both spans take (v, w, u) to the contraction witness (a', a, b, b') and have the same input $(\mathrm{id}_v^{\mathrm{h}}, w, \mathrm{id}_u^{\mathrm{h}})$.

2. **Verify the comultiplication equation.** The diagram to be checked is

$$\begin{tikzcd} \mathsf{act}D_F \arrow[r, "\mathsf{act}\delta_{D_F}"] \arrow[d, "\kappa_F"'] & \mathsf{act}D_F^2 \arrow[r, "\kappa_F D_F"] & M_F\mathsf{act}D_F \arrow[d, "M_F\kappa_F"] \\ M_F\mathsf{act} \arrow[rr, "\delta_{M_F}\mathsf{act}"'] & & M_F^2\mathsf{act}. \end{tikzcd}$$

Equivalently,

$$(\delta_{M_F}\mathsf{act})\kappa_F = (M_F\kappa_F)(\kappa_F D_F)(\mathsf{act}\delta_{D_F}). \tag{101}$$

A witness on the upper route chooses horizontal factorizations of X, Y and duplicates w. The two actions require all four box factors to be horizontal identities. Horizontally composable horizontal identities lie on the same vertical edge, giving the unique surviving factorizations

$$X = \mathrm{id}_v^{\mathrm{h}} = \mathrm{id}_v^{\mathrm{h}} \circ_{\mathrm{h}} \mathrm{id}_v^{\mathrm{h}}, \qquad Y = \mathrm{id}_u^{\mathrm{h}} = \mathrm{id}_u^{\mathrm{h}} \circ_{\mathrm{h}} \mathrm{id}_u^{\mathrm{h}}.$$

Both routes therefore have apex Act_F. Their output consists of two copies of $u_* v^* w$ and the contraction labels a, b, with identical exterior legs.

3. **Verify associativity of the action.** Use π, λ from Definition 3.10 and put

$$\chi = \mathsf{act}\pi \cong \mathsf{act}\lambda, \qquad D_F^{(2)} = \mathsf{WH}_{\mathrm{sp}}(A) \otimes \mathsf{WH}_{\mathrm{sp}}(A) \otimes M_F \otimes \mathsf{WH}_{\mathrm{sp}}(B) \otimes \mathsf{WH}_{\mathrm{sp}}(B).$$

Write

$$L_F = \mathsf{act}(\phi_A \otimes 1_{M_F} \otimes \phi_B), \qquad R_F = \mathsf{act}(1_{\mathsf{WH}_{\mathrm{sp}}(A)} \otimes \kappa_F \otimes 1_{\mathsf{WH}_{\mathrm{sp}}(B)}).$$

The action equation is the commutativity of

$$\begin{tikzcd}
\chi D_F^{(2)} \arrow[r, "L_F"] \arrow[d, "R_F"'] & \mathsf{act} D_F \pi \arrow[d, "\kappa_F \pi"] \\
\mathsf{act} D_F \lambda \arrow[r, "\kappa_F \lambda"'] & M_F \chi.
\end{tikzcd}$$

On the multiplication-first route, the action selects horizontal identity vertical pastes. By Lemma 4.4, every vertical factor of such a paste is a horizontal identity. The action-first route has the same support. The common apex is the set of tuples (v_1, v_2, w, u_1, u_2) with

$$v_1 : a_0 \to a_1, \quad v_2 : a_1 \to a_2, \quad w \in F(a_2, b_0),$$
$$u_1 : b_0 \to b_1, \quad u_2 : b_1 \to b_2,$$

where the four arrows are positive. Both exterior input legs give $(\mathrm{id}^{\mathrm{h}}_{v_1}, \mathrm{id}^{\mathrm{h}}_{v_2}, w, \mathrm{id}^{\mathrm{h}}_{u_1}, \mathrm{id}^{\mathrm{h}}_{u_2})$. The outputs are

$$(u_2 \circ u_1)_*(v_2 \circ v_1)^* w = u_{2*} v_1^* u_{1*} v_2^* w$$

by functoriality and commuting endpoint actions. Both output spans retain the same contraction labels a_1, a_2, b_0, b_1. Thus the complete spans agree.

4. **Verify the unit equation.** Let $\jmath$ be the canonical unit insertion and put $U_F = \mathsf{act}(\phi_A^0 \otimes 1_{M_F} \otimes \phi_B^0)$. Under $\mathsf{act}\jmath \cong 1$, the unit equation is

$$\begin{tikzcd}
M_F \arrow[r, "U_F"] \arrow[dr, equal] & \mathsf{act} D_F \jmath \arrow[d, "\kappa_F \jmath"] \\
 & M_F.
\end{tikzcd}$$

The quantum-category units select vertical identity boxes. Intersecting their support with the horizontal identities selected by the action gives the double identities:

$$\{\mathrm{id}^{\mathrm{v}}_h : h \in \mathsf{Sym}_1^-(A)\} \cap \{\mathrm{id}^{\mathrm{h}}_v : v \in \mathsf{Sym}_1^+(A)\} = \{\Theta_a : a \in \mathsf{Conf}(A)\},$$

and similarly for B. Over $w \in F(a, b)$ there is exactly one surviving witness, $(\mathrm{id}_a, w, \mathrm{id}_b)$, and its output is w. Both exterior legs are identities. The four equations establish the required right-mate module action; Proposition 5.2 transports it to the formal Chikhladze quantum module. $\square$

5.4 Naturality and tensor product

Proposition 5.6 (Functoriality and tensor compatibility)**.** *For fixed A, B, a natural transformation $\eta : F \Rightarrow F'$ induces a quantum-module map $Q^{\mathrm{sp}}_\eta : Q^{\mathrm{sp}}_F \to Q^{\mathrm{sp}}_{F'}$, functorially. External products give canonical isomorphisms*

$$Q^{\mathrm{sp}}_{F \otimes_{\mathrm{ext}} F'} \cong Q^{\mathrm{sp}}_F \otimes Q^{\mathrm{sp}}_{F'}, \tag{102}$$

over the object tensor comparisons of Theorem 4.26. They are natural and satisfy associativity, unit, and symmetry coherence.

Proof.

1. **Construct the comonad transformation.** Define $\widehat{\eta}(w) = \eta_{a,b}(w)$ for $w \in F(a,b)$. Its graph preserves d_F, the diagonal, and the counit. Hence it gives a comonad transformation $M_F \Rightarrow M_{F'}$.

2. **Intertwine the actions.** Naturality gives $\widehat{\eta}(u_* v^* w) = u_* v^* \widehat{\eta}(w)$. Sending (v, w, u) to $(v, \widehat{\eta}(w), u)$ identifies the apices of the two composites in

$$\begin{tikzcd} \mathsf{act}D_F \arrow[r, "\kappa_F"] \arrow[d, "\mathsf{act}(1\otimes\widehat{\eta}\otimes 1)"'] & M_F\mathsf{act} \arrow[d, "\widehat{\eta}\mathsf{act}"] \\ \mathsf{act}D_{F'} \arrow[r, "\kappa_{F'}"'] & M_{F'}\mathsf{act}. \end{tikzcd}$$

   Both exterior legs agree by the displayed naturality equation. Graphs preserve identity maps and composition, proving functoriality.

3. **Construct the tensor comparison.** The external product has carrier $W_{F\otimes_{\mathrm{ext}}F'} \cong W_F \times W_{F'}$. The configuration and box product identifications give $\mathrm{Act}_{F\otimes_{\mathrm{ext}}F'} \cong \mathrm{Act}_F \times \mathrm{Act}_{F'}$. Under these bijections, the two action legs, the comonad diagonal, and its counit are componentwise products. This gives (102).

4. **Verify tensor coherence.** The associativity and unit comparisons rebracket tuples or remove the singleton coordinate of the tensor unit. The symmetry comparison exchanges the two factors. Both sides of each coherence equation have the same action apex and the same permutation of its coordinates. Naturality follows by applying the component maps to these coordinates. $\square$

Proposition 5.7 (Positive restriction and quantum-module maps)**.** *For a distributor $F : \mathsf{Sym}(A)^{\mathrm{op}} \times \mathsf{Sym}(B) \to \mathbf{Set}$, put*

$$F^+ = F\big|_{\mathsf{Sym}^+(A)^{\mathrm{op}}\times\mathsf{Sym}^+(B)}.$$

The quantum module Q_F^{sp} depends only on F^+. For fixed games A, B, there is a canonical bijection

$$\mathrm{Hom}_{\mathrm{qmod}}(Q_F^{\mathrm{sp}}, Q_{F'}^{\mathrm{sp}}) \cong \mathrm{Nat}(F^+, F'^+), \tag{103}$$

where the left-hand side consists of the fixed-endpoint quantum-module maps of Definition 3.10.

Proof.

1. **Identify the data used in the construction.** The positive symmetry subgroupoids are wide, so restriction retains every fibre $F(a,b)$ and the endpoint map d_F. It therefore retains the diagonal comonad. The action in (100) uses only $u_* v^*$ for positive u, v, and is determined by F^+.

2. **Recover a function from a module map.** Represent its underlying comonad transformation by a boundary span $W_F \xleftarrow{a} S \xrightarrow{b} W_{F'}$. Preservation of the boundary counit says that

$$[W_F \xleftarrow{a} S \xrightarrow{d_{F'}\circ b} \mathsf{Conf}(A) \times \mathsf{Conf}(B)] = [W_F \xleftarrow{1} W_F \xrightarrow{d_F} \mathsf{Conf}(A) \times \mathsf{Conf}(B)].$$

   An apex isomorphism therefore identifies a with a bijection. The module map is the graph of the unique function $f = b \circ a^{-1}$, and $d_{F'} \circ f = d_F$. Such a graph automatically preserves the diagonal comultiplication.

3. **Identify action compatibility.** The two composite action spans have the same possible inputs $(\mathrm{id}^{\mathrm{h}}_v, w, \mathrm{id}^{\mathrm{h}}_u)$ and one witness for each such input. They agree exactly when

$$f(u_* v^* w) = u_* v^* f(w)$$

for all positive endpoint arrows. Together with endpoint preservation, these equations are precisely naturality of $F^+ \Rightarrow F'^+$.

4. **Construct the inverse correspondence.** A natural transformation of the restrictions gives an endpoint-preserving function with these equations. Its graph preserves the two comonad maps and intertwines the actions, hence is a quantum-module map. The two constructions are inverse and preserve identities and composition. □

Corollary 5.8 (Classical distributor specialization)**.** *If the negative symmetry groupoids of A, B are discrete, Q^{sp}_F is the ordinary span realization of F.*

Proof.

1. **Identify the arrow structures.** Polarized decomposition gives

$$\mathsf{Sym}^+(A) = \mathsf{Sym}(A), \qquad \mathsf{Sym}^+(B) = \mathsf{Sym}(B).$$

Every box is a horizontal identity, identified with its positive edge. Vertical multiplication is groupoid composition and horizontal comultiplication is diagonal.

2. **Identify the action.** The action span is the ordinary distributor action $(v, w, u) \mapsto u_* v^* w$, over the endpoint-indexed carrier W_F. □

5.5 Thin strategies and copycat

Theorem 5.9 (Game-generated quantum module)**.** *Every thin strategy $\sigma : A \to B$ canonically determines*

$$Q^{\mathrm{sp}}_\sigma = Q^{\mathrm{sp}}_{\|\sigma\|} : \mathsf{QG}_{\mathrm{sp}}(A) \nrightarrow \mathsf{QG}_{\mathrm{sp}}(B).$$

Globular positive morphisms induce quantum-module maps, and tensor products of strategies induce tensor products of these modules up to the canonical isomorphisms.

Proof.

1. **Construct the module.** Apply Theorem 5.5 to the positive-witness distributor of Theorem 2.13.

2. **Transport morphisms and tensor products.** The collapse sends globular positive morphisms to natural transformations and supplies its symmetric monoidal comparisons. Proposition 5.6 carries these to the stated module maps and tensor isomorphisms. □

Proposition 5.10 (The quantum module of copycat)**.** *For every thin game A, composition of the polarized frames gives a natural isomorphism*

$$\zeta_A : \|\mathsf{cc}_A\| \xrightarrow{\cong} \mathsf{Sym}(A)(-,-), \qquad (\theta^-, \mathsf{cc}_A x, \theta^+) \longmapsto \theta^+ \circ \theta^-. \tag{104}$$

Its quantum image is the canonical isomorphism

$$Q^{\mathrm{sp}}_{\zeta_A} : Q^{\mathrm{sp}}_{\mathsf{cc}_A} \xrightarrow{\cong} Q^{\mathrm{sp}}_{\mathsf{Sym}(A)(-,-)}.$$

Proof.

1. **Identify the copycat witnesses.** The $+$-covered configurations of copycat are $\mathsf{cc}_A x$, with display $x \otimes x$. A witness from a to b is therefore the factorization triangle

$$\begin{tikzcd} a \arrow[rr, "\zeta_A(w)"] \arrow[dr, "\theta^-"'] & & b \\ & x \arrow[ur, "\theta^+"'] & \end{tikzcd}.$$

2. **Construct the inverse.** For $\omega : a \to b$, Theorem 2.4 gives the unique factorization $\omega = \omega^+ \circ \omega^-$, through x. Send ω to $(\omega^-, \mathsf{cc}_A x, \omega^+)$. Uniqueness makes this the inverse of (104).

3. **Verify naturality.** For endpoint symmetries $\Omega_A : a' \to a$, $\Omega_B : b \to b'$, write the transported witness as $(\vartheta^-, \mathsf{cc}_A y, \vartheta^+)$. The two components of the copycat symmetry are the same map $\varphi : x \to y$. Endpoint transport gives

$$\vartheta^- = \varphi \circ \theta^- \circ \Omega_A, \qquad \vartheta^+ \circ \varphi = \Omega_B \circ \theta^+.$$

   Multiplying these equations proves commutativity of

$$\begin{tikzcd} \mathsf{Wit}^+_{\mathsf{cc}_A}(a,b) \arrow[r, "\zeta_A"] \arrow[d, "\Omega_{B*}\Omega_A^*"'] & \mathsf{Sym}(A)(a,b) \arrow[d, "\omega \mapsto \Omega_B \circ \omega \circ \Omega_A"] \\ \mathsf{Wit}^+_{\mathsf{cc}_A}(a',b') \arrow[r, "\zeta_A"'] & \mathsf{Sym}(A)(a',b'). \end{tikzcd}$$

   This is the copycat comparison of [7, Proposition 5.12].

4. **Identify the quantum-module isomorphism.** The graph of ζ_A preserves the endpoint map, diagonal, and counit. For positive $v : a' \to a$, $u : b \to b'$, naturality gives

$$\zeta_A(u_* v^* w) = u \circ \zeta_A(w) \circ v.$$

   The two action spans therefore intertwine under this graph. Its inverse gives the inverse module map. Tensor decomposition identifies $\zeta_{A\otimes B}$ with the product of ζ_A, ζ_B, since composition of symmetry arrows is componentwise. $\square$

6 Quantum realization as a symmetric monoidal double functor

We assemble the game quantum groupoids and distributor-induced modules into a double category. A horizontal arrow consists of a distributor together with its quantum module, and horizontal composition is induced by the distributor coend. Copycat gives the generated horizontal identity, and companion distributors supply the vertical arrows and squares.

6.1 The generated horizontal bicategory

For a thin game A, define

$$\mathbb{Q}_A = (A, \mathsf{Sym}(A), \mathsf{QG}_{\mathrm{sp}}(A)). \tag{105}$$

For $F : \mathsf{Sym}(A)^{\mathrm{op}} \times \mathsf{Sym}(B) \to \mathbf{Set}$, define

$$\mathbb{M}_F = (F, Q_F^{\mathrm{sp}}) : \mathbb{Q}_A \nrightarrow \mathbb{Q}_B. \tag{106}$$

A natural transformation $\eta : F \Rightarrow F'$ gives the globular cell $\mathbb{M}_\eta = (\eta, Q_\eta^{\mathrm{sp}})$. These quantum components are supplied by Theorem 5.5 and Proposition 5.6.

For composable distributors $F : \mathsf{Sym}(A) \nrightarrow \mathsf{Sym}(B)$ and $G : \mathsf{Sym}(B) \nrightarrow \mathsf{Sym}(C)$, put

$$\mathbb{M}_G \star \mathbb{M}_F = \mathbb{M}_{G\circ F}, \qquad (G \circ F)(a, c) = \int^{b\in\mathsf{Sym}(B)} F(a, b) \times G(b, c). \tag{107}$$

For fixed a, c, this coend has the coequalizer presentation in sets

$$\coprod_{r:b\to b'} F(a, b) \times G(b', c) \underset{d_1}{\overset{d_0}{\rightrightarrows}} \coprod_{b} F(a, b) \times G(b, c) \xrightarrow{\quad q \quad} (G \circ F)(a, c),$$

where $d_0(r, x, y) = (r_*x, y)$, $d_1(r, x, y) = (x, r^*y)$. Thus

$$[r_*x, y] = [x, r^*y], \qquad r : b \to b', \quad x \in F(a, b), \quad y \in G(b', c). \tag{108}$$

The exterior transports

$$h^*[x, y] = [h^*x, y], \qquad k_*[x, y] = [x, k_*y]$$

commute with these relations and make the coend a distributor. The generated horizontal identity is

$$\mathbb{I}_A^{\mathrm{gen}} = \mathbb{M}_{\mathsf{Sym}(A)(-,-)}. \tag{109}$$

Proposition 5.10 identifies its quantum component with $Q^{\mathrm{sp}}_{\mathsf{cc}_A}$.

Proposition 6.1 (Generated horizontal bicategory)**.** *The objects $\mathbb{Q}_A$, arrows $\mathbb{M}_F$, and cells $\mathbb{M}_\eta$ form a bicategory with composition $\star$ and identities $\mathbb{I}_A^{\mathrm{gen}}$.*

Proof.

1. **Construct composites of arrows and cells.** The full coend and the hom-distributor are distributors, so Theorem 5.5 supplies their quantum modules. Horizontal composition of natural transformations is

   $$[x, y] \longmapsto [\eta(x), \zeta(y)].$$

   Naturality identifies the images of the two representatives in (108). Applying $Q^{\mathrm{sp}}_{(-)}$ gives the quantum component.

2. **Construct the associator.** For F, G, H, the two iterated coends are quotients of triples by the same two middle-symmetry relations. The bijection

   $$[x, [y, z]] \longmapsto [[x, y], z]$$

   therefore induces

   $$\mathfrak{a}_{H,G,F} : (\mathbb{M}_H \star \mathbb{M}_G) \star \mathbb{M}_F \xrightarrow{\cong} \mathbb{M}_H \star (\mathbb{M}_G \star \mathbb{M}_F).$$

   The inverse rebrackets in the opposite direction. Both are natural, and their quantum images are inverse module maps.

3. **Construct the unitors.** The co-Yoneda maps and their inverses are

$$\begin{aligned} [h,w] &\longmapsto h^*w, \quad w \longmapsto [\mathrm{id}_a, w], \\ [w,k] &\longmapsto k_*w, \quad w \longmapsto [w, \mathrm{id}_b], \end{aligned} \qquad w \in F(a,b).$$

   The two inverse equations are instances of (108). Their quantum images give

$$\begin{tikzcd} \mathbb{I}^{\mathrm{gen}}_B \star \mathbb{M}_F \arrow[dr, "\lambda_F"'] & & \mathbb{M}_F \star \mathbb{I}^{\mathrm{gen}}_A \arrow[dl, "\rho_F"] \\ & \mathbb{M}_F. & \end{tikzcd}$$

4. **Verify coherence and interchange.** Every pentagon path sends a quadruple representative to the same rebracketed quadruple. Both triangle paths remove the identity factor by the same endpoint transport. The interchange equation applies the component natural transformations to the same pair (x, y). These are equalities of distributor maps; functoriality of $Q^{\mathrm{sp}}_{(-)}$ gives the corresponding equalities of quantum-module maps. □

Let $\mathbb{R}^{\mathrm{reg}}_A$ denote the regular quantum module of $\mathsf{QG}_{\mathrm{sp}}(A)$, with carrier $\mathsf{WH}_{\mathrm{sp}}(A)$ and action induced by multiplication. Its comparison with $\mathbb{I}^{\mathrm{gen}}_A$ is a comparison with the quantum component $Q^{\mathrm{sp}}_{\mathsf{Sym}(A)(-,-)}$.

Proposition 6.2 (Copycat and the regular quantum module)**.** *The quantum component $Q^{\mathrm{sp}}_{\mathsf{Sym}(A)(-,-)}$ of the generated horizontal identity and the regular quantum module $\mathbb{R}^{\mathrm{reg}}_A$ are isomorphic if and only if*

$$\mathsf{Sym}^{-}(A) = \mathsf{Disc}(\mathsf{Conf}(A)).$$

Through $Q^{\mathrm{sp}}_{\zeta_A}$, this also characterizes when the quantum module of copycat is isomorphic to the regular quantum module.

Proof.

1. **Identify the generated carrier.** Its carrier is $\coprod_{a,b} \mathsf{Sym}(A)(a,b)$, with diagonal comultiplication and terminal counit. An invertible morphism of **Sp** is represented by a bijection: composing with its inverse gives identity spans, forcing a unique witness over each exterior element.

2. **Derive discreteness from an identity comparison.** A quantum-module isomorphism induces an isomorphism of the underlying endocomonads. By the argument of Proposition 5.7, its underlying span is therefore the graph of a boundary-preserving bijection

$$f : \coprod_{a,b} \mathsf{Sym}(A)(a,b) \xrightarrow{\cong} \mathsf{WH}_{\mathrm{sp}}(A).$$

   The boundary counit of the distributor comonad is defined on every element of its carrier, whereas the boundary counit of the arrow endocomonad is

$$\mathsf{Boxes}_A \xleftarrow{v \mapsto \mathrm{id}^{\mathrm{h}}_v} \mathsf{Sym}^{+}_1(A) \xrightarrow{v \mapsto (s(v), t(v))} \mathsf{Conf}(A) \times \mathsf{Conf}(A)$$

   by Proposition 4.16. Preservation of the boundary counit therefore implies that every value of f is a horizontal identity box. Since f is surjective, every box in Boxes_A is a horizontal identity.

In particular, for every $h \in \mathsf{Sym}_1^-(A)$, the vertical identity $\mathrm{id}_h^{\mathrm{v}}$ is also a horizontal identity. A box which is both a vertical identity and a horizontal identity is a double identity, so h is an identity arrow. Hence

$$\mathsf{Sym}^-(A) = \mathsf{Disc}(\mathsf{Conf}(A)).$$

3. **Construct the comparison under discreteness.** Polarized decomposition gives $\mathsf{Sym}(A) = \mathsf{Sym}^+(A)$, and every box is $\mathrm{id}_v^{\mathrm{h}}$ for a unique $v \in \mathsf{Sym}(A)$. The bijection

$$\coprod_{a,b} \mathsf{Sym}(A)(a,b) \longrightarrow \mathsf{WH}_{\mathrm{sp}}(A), \qquad v \longmapsto \mathrm{id}_v^{\mathrm{h}}$$

preserves both endocomonad boundaries. Horizontal comultiplication is the diagonal and its counit is terminal. The regular and distributor actions both compose the positive edges, so this bijection is a quantum-module isomorphism. Composing with $Q^{\mathrm{sp}}_{\zeta_A}$ gives the stated comparison with copycat. $\square$

Proposition 6.3 (Full symmetry transport in composition)**.** *There are thin games I, B, distributors $F_0, F_1 : \mathsf{Sym}(I) \nrightarrow \mathsf{Sym}(B)$, and $N : \mathsf{Sym}(B) \nrightarrow \mathsf{Sym}(I)$, such that*

$$Q^{\mathrm{sp}}_{F_0} = Q^{\mathrm{sp}}_{F_1}, \qquad |(N \circ F_0)(*,*)| = 2, \qquad |(N \circ F_1)(*,*)| = 1.$$

Proof.

1. **Construct the game and distributors.** Let I be the empty game and let B have two concurrent negative events, with all selected bijections. Put $b = \{e_1, e_2\}$. Its automorphism group is C_2, and $\mathsf{Sym}^+(B)$ is discrete. Define F_0, F_1 to have fibre $\{0,1\}$ at b and empty fibres elsewhere, with respectively the trivial and swapping C_2-actions. Define N to have singleton fibre at b, trivial action, and empty fibres elsewhere. Symmetries preserve configuration cardinality, so these prescriptions define distributors.

2. **Identify their quantum modules.** The two carriers and endpoint maps agree. Their diagonals and counits agree as well. The action in (100) uses positive endpoint transport, which is the identity transport in this example. Hence the two comonads and action spans coincide.

3. **Compute the composites.** Formula (108) gives the orbit sets

$$(N \circ F_i)(*,*) = F_i(*,b)/C_2.$$

The trivial action has orbits $\{0\}, \{1\}$; the swapping action has the single orbit $\{0,1\}$. This gives the two displayed cardinalities. Proposition 5.7 identifies the common restriction retained by their quantum modules. The retained distributor in a generated arrow specifies the additional symmetry transport used by this full coend. $\square$

6.2 Representable vertical arrows and squares

A functor $u : \mathsf{Sym}(A) \to \mathsf{Sym}(C)$ has companion and conjoint distributors

$$u_*(a,c) = \mathsf{Sym}(C)(u(a),c), \qquad u^*(c,a) = \mathsf{Sym}(C)(c,u(a)). \tag{110}$$

Applying Theorem 5.5 gives the generated arrows

$$\begin{tikzcd} \mathbb{Q}_A \arrow[r, Rightarrow, bend left, "\mathbb{M}_{u_*}"] & \mathbb{Q}_C \arrow[l, Rightarrow, bend left, "\mathbb{M}_{u^*}"] \end{tikzcd}.$$

Proposition 6.4 (Representable adjunction and composition)**.** *There is a canonical adjunction* $\mathbb{M}_{u_*} \dashv \mathbb{M}_{u^*}$ *in the generated bicategory. For composable* u, v*, there are coherent canonical isomorphisms*

$$\mathbb{M}_{v_*} \star \mathbb{M}_{u_*} \cong \mathbb{M}_{(v\circ u)_*}, \qquad \mathbb{M}_{u^*} \star \mathbb{M}_{v^*} \cong \mathbb{M}_{(v\circ u)^*}.$$

Proof.

1. **Construct the unit and counit.** At distributor level the unit and counit are

$$\begin{aligned} \eta_u : \mathsf{Sym}(A)(-,-) &\longrightarrow u^* \circ u_*, & h : a \to a' &\longmapsto [\mathrm{id}_{u(a)}, u(h)], \\ \epsilon_u : u_* \circ u^* &\longrightarrow \mathsf{Sym}(C)(-,-), & [\alpha, \beta] &\longmapsto \beta \circ \alpha, \end{aligned}$$

   where $\alpha : c \to u(a)$, $\beta : u(a) \to c'$. Coend relations and functoriality of u give their well-definedness and naturality.

2. **Verify the adjunction identities.** The two triangle composites are

$$u_* \xrightarrow{u_*\eta_u} u_* \circ u^* \circ u_* \xrightarrow{\epsilon_u u_*} u_*,$$

$$u^* \xrightarrow{\eta_u u^*} u^* \circ u_* \circ u^* \xrightarrow{u^*\epsilon_u} u^*.$$

   For $\alpha : u(a) \to c$, the first inserts identity arrows and then composes to α. For $\beta : c \to u(a)$, the second similarly returns β. Applying the module-map functor proves the generated adjunction.

3. **Construct composition comparisons.** The companion and conjoint comparisons are respectively

$$\begin{aligned} &[\alpha : u(a) \to c,\ \beta : v(c) \to d] \longmapsto \beta \circ v(\alpha), \\ &[\alpha : d \to v(c),\ \beta : c \to u(a)] \longmapsto v(\beta) \circ \alpha. \end{aligned}$$

   Their inverses insert $c = u(a)$ and the corresponding identity arrow. Coend relations give both inverse equations.

4. **Verify coherence.** For three functors u, v, w, both companion comparison paths send $[\alpha, \beta, \gamma]$ to $\gamma \circ w(\beta) \circ (w \circ v)(\alpha)$. Both conjoint paths give the corresponding reversed-endpoint composite. Identity functors insert identity arrows. These equations carry to the quantum components by functoriality. ☐

A distributor square consists of functors $u : \mathsf{Sym}(A) \to \mathsf{Sym}(C)$, $v : \mathsf{Sym}(B) \to \mathsf{Sym}(D)$ and a natural family

$$\alpha_{a,b} : F(a, b) \longrightarrow F'(u(a), v(b)). \tag{111}$$

It is displayed by

$$\begin{tikzcd} \mathsf{Sym}(A) \arrow[r, Rightarrow, "F"] \arrow[d, "u"'] \arrow[dr, phantom, "\alpha"] & \mathsf{Sym}(B) \arrow[d, "v"] \\ \mathsf{Sym}(C) \arrow[r, Rightarrow, "F'"'] & \mathsf{Sym}(D). \end{tikzcd}$$

Its companion mate is

$$\begin{gathered} \bar{\alpha} : v_* \circ F \Longrightarrow F' \circ u_*, \\ [w, \beta : v(b) \to d] \longmapsto [\mathrm{id}_{u(a)}, \beta_* \alpha(w)]. \end{gathered} \tag{112}$$

The associated **representable quantum square** is the pair consisting of α and

$$Q^{\mathrm{sp}}_{\bar{\alpha}} : Q^{\mathrm{sp}}_{v_* \circ F} \longrightarrow Q^{\mathrm{sp}}_{F' \circ u_*}. \tag{113}$$

The companion construction turns the square into a globular cell:

$$\mathbb{M}_{v_*} \star \mathbb{M}_F \xrightarrow{\ \mathbb{M}_{\bar{\alpha}}\ } \mathbb{M}_{F'} \star \mathbb{M}_{u_*}.$$

Both sides have endpoints $\mathbb{Q}_A, \mathbb{Q}_D$. For identity vertical boundaries, the co-Yoneda unitors identify $Q^{\mathrm{sp}}_{\bar{\alpha}}$ with Q^{sp}_{α}.

Lemma 6.5 (Companion mates and pasting)**.** *Formula* (112) *is well defined and natural. Identity squares and both kinds of square pasting correspond to identity and pasting of companion mates.*

Proof.

1. **Verify the coend relation and naturality.** For $r : b \to b'$, the representatives $[r_* w, \beta]$, $[w, \beta \circ v(r)]$ have the same image because $\alpha(r_* w) = v(r)_* \alpha(w)$. For $h : a' \to a$, naturality gives $\alpha(h^* w) = u(h)^* \alpha(w)$; moving $u(h)$ across the target coend proves naturality at the first endpoint. At the last endpoint it is postcomposition of β.

2. **Compute vertical pasting.** Let $\alpha' : F' \Rightarrow F''(u'-, v'-)$ be a second square. The composite is

   $$w \longmapsto \alpha'_{u(a),v(b)}(\alpha_{a,b}(w)).$$

   Its mate sends $[w, \rho : (v' \circ v)(b) \to d]$ to

   $$[\mathrm{id}_{u'(u(a))},\ \rho_* \alpha'_{u(a),v(b)}(\alpha_{a,b}(w))].$$

   Pasting the two mates inserts two companion factors. The comparisons of Proposition 6.4 compose those factors and give exactly the displayed value.

3. **Compute horizontal pasting.** For horizontally adjacent squares, use the diagram

   $$\begin{tikzcd} \mathsf{Sym}(A) \arrow[r, Rightarrow, "F"] \arrow[d, "u"'] \arrow[dr, phantom, "\alpha"] & \mathsf{Sym}(B) \arrow[r, Rightarrow, "G"] \arrow[d, "v"] \arrow[dr, phantom, "\beta"] & \mathsf{Sym}(C) \arrow[d, "t"] \\ \mathsf{Sym}(A') \arrow[r, Rightarrow, "F'"'] & \mathsf{Sym}(B') \arrow[r, Rightarrow, "G'"'] & \mathsf{Sym}(C'). \end{tikzcd}$$

   Their paste sends $[x, y]$ to $[\alpha(x), \beta(y)]$. Its well-definedness is the calculation

   $$\begin{aligned} [\alpha(r_* x), \beta(y)] &= [v(r)_* \alpha(x), \beta(y)] \\ &= [\alpha(x), v(r)^* \beta(y)] \\ &= [\alpha(x), \beta(r^* y)]. \end{aligned}$$

   For $\rho : t(c) \to d$, the mate of this paste sends

   $$[[x, y], \rho] \longmapsto [\mathrm{id}_{u(a)}, [\alpha(x), \rho_* \beta(y)]].$$

   Pasting the two companion mates gives the same formula after rebracketing the coends and eliminating identity companion factors.

4. **Identify identities and quantum components.** For an identity square, the mate is the co-Yoneda identity comparison. The preceding representative calculations therefore establish identities and both pasting laws for distributor mates. Functoriality of $Q^{\mathrm{sp}}_{(-)}$ establishes the same equations for their quantum-module maps. □

Theorem 6.6 (Generated quantum-module double category)**.** *There is a double category $\mathbf{QMod}^{\mathrm{rep}}_{\mathrm{sp}}$ whose objects are $\mathbb{Q}_A$, horizontal arrows are $\mathbb{M}_F$, vertical arrows are functors $u : \mathsf{Sym}(A) \to \mathsf{Sym}(C)$ equipped with their canonical representable quantum adjunctions, and squares are the representable quantum squares. Horizontal composition is $\star$, horizontal identities are $\mathbb{I}^{\mathrm{gen}}_A$, and vertical composition is composition of functors.*

Proof.

1. **Construct the two compositions.** Proposition 6.1 supplies the horizontal bicategory. Proposition 6.4 identifies the representable arrows of composite functors. Define square pasting by pasting the distributor squares and taking the quantum image of the resulting companion mate.

2. **Identify the pasted quantum maps.** Lemma 6.5 identifies this definition with pasting the quantum components, including the horizontal associators, unitors, and companion comparisons. It also identifies the identity squares.

3. **Verify the double-category equations.** Let $\mathbf{Dist}_{\mathrm{TCG}}$ be the distributor double category labelled by games A, with groupoid $\mathsf{Sym}(A)$ at A. Forgetting the quantum components defines

$$U : \mathbf{QMod}^{\mathrm{rep}}_{\mathrm{sp}} \longrightarrow \mathbf{Dist}_{\mathrm{TCG}}.$$

It is faithful on squares because each distributor square determines its quantum component. The identities, associativity, and interchange equations hold in $\mathbf{Dist}_{\mathrm{TCG}}$. The construction of pasting in Steps 1–2 preserves these equations under U, and faithfulness reflects them to the generated squares. □

6.3 Symmetric monoidal structure

Tensor products are defined by

$$\mathbb{Q}_A \otimes \mathbb{Q}_B = \mathbb{Q}_{A\otimes B}, \qquad \mathbb{M}_F \otimes \mathbb{M}_{F'} = \mathbb{M}_{F\otimes_{\mathrm{ext}} F'}.$$

The unit is $\mathbb{Q}_I$. Theorem 4.26 and Proposition 5.6 identify the quantum components with the corresponding tensor products.

Theorem 6.7 (Symmetric monoidal target and realization)**.** *The generated target is a symmetric monoidal double category. The canonical realization*

$$\mathcal{R} : \mathbf{Dist}_{\mathrm{TCG}} \longrightarrow \mathbf{QMod}^{\mathrm{rep}}_{\mathrm{sp}}$$

is a symmetric monoidal double functor satisfying $U \circ \mathcal{R} = 1$. The canonical prescribed components also identify $\mathcal{R} \circ U$ with the identity, so U and $\mathcal{R}$ give a canonical symmetric monoidal equivalence of the generated target with $\mathbf{Dist}_{\mathrm{TCG}}$.

Proof.

1. **Tensor vertical arrows and squares.** Use products of functors. Hom-sets in product groupoids give
$$(u \times v)_* \cong u_* \otimes_{\text{ext}} v_*, \qquad (u \times v)^* \cong u^* \otimes_{\text{ext}} v^*.$$
Tensor distributor squares by external product of their natural transformations, and use their quantum companion mates. The formula for a companion mate is componentwise under these product identifications.

2. **Compare tensor and horizontal composition.** The map
$$\xi : \ (G \otimes_{\text{ext}} G') \circ (F \otimes_{\text{ext}} F') \longrightarrow (G \circ F) \otimes_{\text{ext}} (G' \circ F')$$
sends $[(x, x'), (y, y')]$ to $([x, y], [x', y'])$. The two coends impose independent middle-symmetry relations in the two coordinates. Thus the inverse pairs representatives coordinatewise, and both inverse equations follow from the respective coend relations. Its quantum image is the tensor comparison for composition, by Proposition 5.6.

3. **Verify the tensor equations.** Associators, unitors, and symmetries rebracket product coordinates, remove singleton coordinates, or permute factors. These operations commute with the two independent coend relations and the componentwise action spans. Therefore the tensor comparisons satisfy their coherence equations on distributor squares, and the faithful U reflects those equations to the quantum squares.

4. **Construct the realization functor.** Define $\mathcal{R}$ by adjoining the quantum groupoid and module components and taking companion mates on squares. The definitions of $\star$, identities, and pasting make it a double functor. The comparisons of Steps 1–3 make it symmetric monoidal. Forgetting those components recovers the labelled distributor data, giving $U \circ \mathcal{R} = 1$. Conversely, every object, horizontal arrow, vertical arrow, and square of the generated target already carries precisely the components prescribed by $\mathcal{R}$. Adjoining those components after forgetting them recovers the original data and all the chosen comparisons. Hence $\mathcal{R} \circ U$ is canonically the identity, proving the stated equivalence. □

6.4 The global quantum realization

For a positive morphism $f : \sigma \to \tau$ with vertical boundaries $h : A \to C$, $k : B \to D$, write
$$\alpha_f : \|\sigma\| \Longrightarrow \|\tau\|(\mathsf{Sym}(h)-, \mathsf{Sym}(k)-)$$
for its distributor square. For composable strategies $\sigma : A \to B$, $\tau : B \to C$, interaction and hiding identify a $+$-covered composite configuration with a causally compatible matching pair (x^σ, x^τ). Write its common middle configuration as b. The collapse compositor is
$$\begin{aligned} &\gamma_{\sigma,\tau}(\theta_A^-, x^\tau \odot x^\sigma, \theta_C^+) \\ &\qquad = [(\theta_A^-, x^\sigma, \mathrm{id}_b), (\mathrm{id}_b, x^\tau, \theta_C^+)]. \end{aligned} \tag{114}$$
This is the full-coend construction of [7, Proposition 5.13].

For exterior symmetries $h : a' \to a$, $k : c \to c'$, let $[x, y] = \gamma_{\sigma,\tau}(w)$. The symmetry lifting of the composite restricts to symmetries of its two interacting configurations, with common middle symmetry $r : b \to b'$. Uniqueness of endpoint transport identifies the two component witnesses of the transported composite with $r_* h^* x$ and $(r^{-1})^* k_* y$. Consequently,

$$\begin{aligned} \gamma_{\sigma,\tau}(k_* h^* w) &= [r_* h^* x, (r^{-1})^* k_* y] \\ &= [h^* x, k_* y] = k_* h^* \gamma_{\sigma,\tau}(w). \end{aligned} \tag{115}$$

The middle equality is (108). Thus the naturality square is

$$\begin{tikzcd} \|\tau \odot \sigma\|(a,c) \arrow[r, "\gamma_{\sigma,\tau}"] \arrow[d, "k_* h^*"'] & (\|\tau\| \circ \|\sigma\|)(a,c) \arrow[d, "k_* h^*"] \\ \|\tau \odot \sigma\|(a',c') \arrow[r, "\gamma_{\sigma,\tau}"'] & (\|\tau\| \circ \|\sigma\|)(a',c'). \end{tikzcd}$$

Proposition 6.8 (A two-move interaction)**.** *There are thin strategies $\sigma : I \to B$, $\tau : B \to I$ for which the compositor has the form*

$$\gamma_{\sigma,\tau} : \{*\} \longrightarrow \{[\varnothing, \varnothing], [b, b]\}, \qquad * \longmapsto [\varnothing, \varnothing].$$

Proof.

1. **Construct the game and strategies.** Let B have two concurrent events b^-, b^+, of the indicated polarities, with discrete symmetry. Put $b = \{b^-, b^+\}$. The strategy $\sigma : I \to B$ has the same events and adds the causal order $b^- < b^+$. The strategy $\tau : B \to I$ has display in $B^\perp$ and adds the causal order $b^+ < b^-$. Their display maps are the identity on the two named events:

   $$\sigma : \quad b^- \longrightarrow b^+ \qquad\qquad \tau : \quad b^+ \longrightarrow b^-.$$

   In each strategy the new dependency runs from a negative event to a positive event in its displayed game. Courtesy therefore holds. The sole negative event is available initially and has a unique occurrence, giving receptivity. With discrete symmetry, symmetry-receptivity and thinness follow from the identity maps. Thus both are thin strategies.

2. **Compute the distributor composite.** Each strategy has exactly two +-covered configurations: the empty configuration and the full configuration. All endpoint symmetries are identities. The coend over the discrete groupoid $\mathsf{Sym}(B)$ therefore consists of the two matching pairs $[\varnothing, \varnothing]$ and $[b, b]$.

3. **Compute interaction and hiding.** The full matching pair induces the causal cycle

   $$\begin{tikzcd} b^- \arrow[r, bend left, "\sigma"] & b^+. \arrow[l, bend left, "\tau"] \end{tikzcd}$$

   The empty pair is causally compatible. Hence $\mathsf{Conf}^+(\tau \odot \sigma) = \{\varnothing\}$, and (114) gives the displayed map. Its quantum image is the graph of the same proper inclusion. $\square$

Theorem 6.9 (Symmetric monoidal oplax quantum realization)**.** *The assignments*

$$A \longmapsto \mathbb{Q}_A, \tag{116}$$

$$h : A \to C \longmapsto \mathsf{Sym}(h), \tag{117}$$

$$\sigma : A \to B \longmapsto \mathbb{M}_{\|\sigma\|} = (\|\sigma\|, Q^{\mathrm{sp}}_\sigma), \tag{118}$$

$$f \longmapsto (\alpha_f, Q^{\mathrm{sp}}_{\alpha_f}) \tag{119}$$

extend canonically to a symmetric monoidal oplax double functor

$$\mathfrak{Q} : \mathbf{TCG} \longrightarrow \mathbf{QMod}^{\mathrm{rep}}_{\mathrm{sp}}. \tag{120}$$

The vertical arrow $\mathsf{Sym}(h)$ *carries the representable adjunction*

$$\mathbb{M}_{\mathsf{Sym}(h)_*} \dashv \mathbb{M}_{\mathsf{Sym}(h)^*}.$$

The horizontal identity comparison is the quantum image of

$$\zeta_A : \|\mathsf{cc}_A\| \xrightarrow{\cong} \mathsf{Sym}(A)(-,-). \tag{121}$$

For composable σ, τ*, the compositor has quantum component*

$$\Gamma^{\mathrm{sp}}_{\sigma,\tau} = Q^{\mathrm{sp}}_{\gamma_{\sigma,\tau}} : Q^{\mathrm{sp}}_{\|\tau\odot\sigma\|} \longrightarrow Q^{\mathrm{sp}}_{\|\tau\|\circ\|\sigma\|}. \tag{122}$$

The identity and tensor comparisons are invertible.

Proof.

1. **Compose the two realizations.** Keep the game labels in Theorem 2.13. The resulting factorization is
$$\begin{tikzcd} \mathbf{TCG} \arrow[rr, "\mathfrak{Q}"] \arrow[dr, "\|-\|"'] & & \mathbf{QMod}^{\mathrm{rep}}_{\mathrm{sp}} \\ & \mathbf{Dist}_{\mathrm{TCG}} \arrow[ur, "\mathcal{R}"'] & \end{tikzcd} .$$
Theorem 6.7 constructs $\mathcal{R}$. Their composite has the displayed object, vertical, horizontal, and square assignments.

2. **Construct the identity and composition comparisons.** Proposition 5.10 supplies the identity comparison and its inverse. The graph of (114) gives $Q^{\mathrm{sp}}_{\gamma_{\sigma,\tau}}$; its action compatibility follows from (115) and Proposition 5.6. These are the quantum images of the collapse comparisons.

3. **Verify horizontal coherence.** For three interacting strategy witnesses, both compositor paths produce their class in the threefold full coend, with the parenthesizations identified by $\mathfrak{a}$. The copycat comparison composes the polarized frames, and the co-Yoneda unitors then apply the corresponding endpoint transport. These are the associativity and unit equations of the positive-witness collapse. Applying the module-map functor preserves the equations and gives horizontal coherence.

4. **Verify square and tensor compatibility.** Lemma 6.5 identifies the companion mates of pasted non-globular squares with the pasted quantum components. The collapse is symmetric monoidal, and $\mathcal{R}$ has the tensor comparisons constructed in Theorem 6.7. Their composite therefore satisfies the symmetric monoidal double-functor equations. The identity and tensor comparisons are images of natural isomorphisms and are invertible. □

The realization satisfies $U \circ \mathfrak{Q} = \|-\|$, with game labels retained. It assigns to each game its canonical quantum groupoid, to each strategy its distributor-induced quantum module, and to each positive morphism its representable quantum square.

7 Conclusion

The polarized symmetry of a thin concurrent game determines a quantum groupoid. Unique negative–positive factorization produces the vacant box structure; its two composition directions yield the weak Hopf operations; and the configuration set supplies the explicit Frobenius base. Tensor and negation transport to corresponding operations on this quantum structure. The construction therefore gives an algebraic interpretation of symmetry and polarity within an established model of concurrency.

Thin strategies extend this interpretation through their associated quantum modules. Their positive-restriction characterization identifies the endpoint transport retained by the module construction. Together with their generating distributors, these modules support a symmetric monoidal oplax double-categorical realization. The necessity of the retained distributor and the exact criterion for agreement with the regular quantum module are established by the comparison results above. The identity and composition comparisons explain how the realization relates copycat, symmetry transport, and causal compatibility.

A natural next goal is a higher categorical extension of this construction. The role of unique polarized factorization suggests studying groupoids of factorizations and their comparison symmetries, with contractible uniqueness taking the place of strict uniqueness. This raises the question of how the box construction, its quantum algebraic structure, and its extension to strategies can be developed while retaining these comparisons and their coherence data. Such an extension would clarify the role of thinness in the emergence of quantum structure from concurrent games.

These developments provide concrete objects for further study at the interface of semantics and quantum algebra. Alongside the higher categorical generalization, two questions arise directly: which quantum groupoids can be obtained from thin concurrent games, and which properties of strategies can be expressed through their associated quantum modules?

8 Related work

The semantic input is the theory of thin concurrent games of Castellan, Clairambault, and Winskel [4]. Their construction uses event structures and symmetry to model concurrent computation and its interaction with higher-order behaviour. Clairambault, Olimpieri, and Paquet [7] develop the polarized factorization and positive-witness distributors used here, and establish a symmetric monoidal oplax collapse to distributors. Our strategy realization equips that collapse with the quantum groupoids and quantum modules constructed from its game-semantic data.

The algebraic mechanism is related to the construction of weak Hopf algebras from matched pairs of groupoids and cocycle data by Andruskiewitsch and Natale [2], formulated through double groupoids. We identify the required vacant double groupoid in polarized configuration symmetry and realize its box operations in spans. Pastro and Street [16] establish the passage from weak Hopf monoids in braided monoidal categories to quantum groupoids, developing the categorical setting introduced by Day and Street [9]. We make this passage explicit for games by constructing the configuration Frobenius base, the boundary and composition objects, and the inversion data.

Chikhladze [5] develops quantum modules as quantum analogues of distributors. We use the direct action formulation to construct modules from ordinary distributors between the symmetry groupoids of games. Their diagonal carriers and positive endpoint transports determine explicit action spans. The resulting equations establish the module structure, and natural transformations and external products give its functorial and monoidal behaviour. The positive-restriction characterization specifies exactly which endpoint actions these modules retain, and the composition example shows why the generating distributor is needed for the full coend. Full distributor composition defines horizontal composition in the generated target, which is canonically equivalent to the game-labelled distributor double category equipped with its prescribed quantum data.

A neighbouring connection between logical interaction and quantum topology is developed in Melliès's ribbon tensorial logic [1]. Motivated by the graphical representation of proofs, that work equips tensorial logic with ribbon-group exchange and proves a faithful interpretation of proofs as ribbon tangles. The topological structure guides the design of the calculus. Here the input is the existing model of thin concurrent games, and the construction extracts a quantum groupoid from each game's polarized configuration symmetries. The two developments relate logical interaction to quantum-group-theoretic structures through different constructions: graphical representations of proofs and algebraic structures associated with games.

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